Kronos at Table



The dials of the machine, which register days, thousands of days, millions of days and thousands of millions, set for the year 802,701.



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Chronophagy, Entropy, and the Ethics of the Remainder in H. G. Wells's The Time Machine

Written by Claude Opus 5.5
Assembled by Andrew Van Blair



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Abstract

THIS THESIS READS H. G. Wells's *The Time Machine* (1895) as the founding text of *chronophagy*: the figuration of irreversible time as an alimentary process in which every order sustains itself by consuming another, and in which an uncompensated loss gives the process its direction. Where criticism since Bernard Bergonzi has treated the novella's governing preoccupations — class antagonism, evolutionary degeneration, thermodynamic dissipation — as parallel themes, I argue that they are homologous registers of a single metabolic operator, rendered mutually convertible by the device of the time machine itself. The dinner-table frame, in which a famished Traveller demands mutton before narrating a future in which the descendants of his hosts' class have become meat, binds that operator to the reader's own appetite.

Three original concepts structure the argument. The first, *the speciation of metaphor*, names Wells's characteristic procedure: the governing social figures of Victorian discourse — Disraeli's "two nations" "formed by a different breeding" and "fed by a different food", William Booth's "submerged tenth", George Gissing's "nether world" — are passed through geological time until they harden into biological fact. The time machine is accordingly a rhetorical engine for converting figure into flesh, and this temporal literalization is proposed as the constitutive operation of speculative fiction. The second, *the social second law*, reads the world of 802,701 as a heat engine approaching equilibrium: its Eloi a society of exhausted gradients, its Morlocks the last differential still capable of work. The result is an *entropic pastoral* that inverts Hesiod's descending sequence of ages and resolves the ancient contradiction between Kronos, benevolent king of the Golden Age, and Kronos, devourer of his children; they are one regime, seen from above and from below. The third, *the ethics of the remainder*, recovers the counter-economy the novel inscribes in Weena's flowers, "not unlike very large white mallows", which share the Traveller's pocket with his matches. Reconstructing the mallow's Hesiodic genealogy — the "bribe-swallowing", literally

gift-devouring, lords of *Works and Days* and the maxim that the half is more than the whole — the thesis reads the narrator's closing injunction, "it remains for us to live as though it were not so", as a practice of non-total consumption, gift and testimony that neither denies entropy nor permits it to legislate value.

The argument proceeds by a *chronophagic comparatism* that reads Wells's sources and successors as successive digestions of a shared complex: from Hesiod, Plato's *Statesman*, Ovid's Pythagoras, Milton's Moloch, Swift, Carlyle, Darwin, Huxley, Clausius, Kelvin, Hinton, Morris, Bellamy and Wilde; through Lang's *Metropolis*, Lovecraft, Stapledon, Le Guin, Ishiguro, McCarthy and Miyazaki; to Bong Joon-ho, *The Platform*, *Dark Souls*, *Outer Wilds* and *NieR: Automata*; and in dialogue with Nietzsche, Bergson, Weber, Benjamin, Derrida, Lyotard, Meillassoux, Chakrabarty, Jonas and contemporary longtermism. It concludes that *The Time Machine* furnishes a more adequate ethics of deep time than either catastrophist despair or the optimizing futurisms of the present, and that the novel's most radical artefact is not the machine that crosses eight hundred thousand years but the pressed flower that returns from them, flattened by one dimension, as evidence that something was given rather than consumed.

Note on Texts and Citation

Quotations from *The Time Machine* follow the first English edition (London: William Heinemann, 1895), the copy-text preferred by most recent scholarly editions; because pagination varies widely across editions, quotations are identified by episode rather than by page. Variants from "The Chronic Argonauts" (*Science Schools Journal*, 1888), the *National Observer* sketches (1894), the *New Review* serial (1895) and the first American edition (New York: Henry Holt, 1895) are cited from the apparatus of Harry M. Geduld's *The Definitive Time Machine* (1987) and Nicholas Ruddick's Broadview edition (2001).

Classical texts are cited by standard book, line or Stephanus numbers. Translations from Greek and Latin are my own unless attributed; Hesiod's *Works and Days* is quoted, where noted, in Hugh G. Evelyn-White's Loeb translation (1914). Biblical citations follow the Authorized Version. Secondary works are cited by author, title and date in the body of the text, with full details reserved for the Bibliography that closes the thesis. Films, television, games, comics and musical works are cited by creator, title and year of first release.

"The Traveller" designates Wells's unnamed protagonist; "Eloi" and "Morlocks" follow Wells's capitalization; "802,701" designates the year of the Traveller's first destination, as recorded by the dials of his machine.



Tempus edax rerum.

— Ovid, *Metamorphoses* XV.234

Fools! They know not how much more the half is than the whole, nor what great advantage there is in mallow and asphodel.

— Hesiod, *Works and Days* 40–41, trans. Hugh G. Evelyn-White (1914)

Save me some of that mutton. I'm starving for a bit of meat.

— H. G. Wells, *The Time Machine* (1895)

CHAPTER ONE

1

Kronos at Table

Chronophagy, the Speciation of Metaphor, and the Ethics of the Remainder

1.1 The Mutton and the Red Joint

THE DECISIVE SCENE OF *The Time Machine* is neither the White Sphinx nor the descent into the wells nor the eclipse above the terminal beach. It is a dinner. When the Traveller limps into his dining-room at Richmond — his coat smeared with green, his hair greyer, his face ghastly pale, his feet bound in nothing but tattered and blood-stained socks — his first articulate desire is alimentary: "Save me some of that mutton. I'm starving for a bit of meat." The narrative of eight hundred thousand years is deferred until he has washed, dressed and eaten. It is then delivered post-prandially, over the remains of a joint, to a company that the novel, across its two dinners, designates not by name but by function — the Medical Man, the Psychologist, the Editor, the Journalist, the Provincial Mayor, the Very Young Man, the Silent Man — with the single exception of the argumentative Filby. The story the Traveller tells concerns a world in which, as he understands only belatedly, the descendants of precisely this class have become meat.

The novel plants its answer before its hero can read it. Among the Eloi he learns that "horses, cattle, sheep, dogs, had followed the Ichthyosaurus into extinction", that his hosts are "strict vegetarians", and that he must, "in spite of

some carnal cravings", become "frugivorous also". In the Under-world, by the light of a match, he glimpses a meal laid out and records the discovery with the naïveté of a man who has not yet understood his own plot: "The Morlocks at any rate were carnivorous!" He had wondered, he adds, "what large animal could have survived to furnish the red joint I saw". The red joint is the riddle's solution; the mutton on the Richmond table is its rhyme; and the circuit that joins the two tables, the board of the Under-world and the dinner-table of the late-Victorian professional class, is the first and most consequential structure of the book. The reader is seated at one table and made to contemplate the other. The narrative does not merely describe a predatory future; it is served as the dessert of a carnivorous present.

The structure is Saturnalian in the strict sense. At the Roman festival of Saturn — the god the Greeks called Kronos, ruler of the Golden Age and devourer of his own children — the order of the table was inverted and masters waited upon their slaves. The world of 802,701 completes the inversion the festival merely staged: the descendants of the masters no longer serve at table; they are its fare. That the novel's first dinner already anticipates this is a measure of its design. Before the Traveller departs, his guests speculate on the uses of time travel, and their proposals turn at once to political economy. "One might invest all one's money, leave it to accumulate at interest, and hurry on ahead!" cries the Very Young Man; the narrator counters with the prospect of discovering "a society erected on a strictly communistic basis". Accumulation and distribution, compound interest and communism: the two answers the nineteenth century had given to the question of the future are proposed over the soup. Wells would give the Very Young Man's fantasy a novel of its own in *When the Sleeper Wakes* (1899), whose somnolent hero wakes after two centuries to find that accumulated interest has made him master of half the world. The future of *The Time Machine*, however, answers both guests with a term neither can conceive. The society of 802,701 neither accumulates nor shares. It eats.

The frame also legislates a position for its readers, and the position is that of the spectator at a safe distance from catastrophe. Lucretius opens the second book of *De Rerum Natura* with the pleasure of watching from the shore another's great labour on a wind-torn sea ("Suave, mari magno turbantibus aequora ventis, / e terra magnum alterius spectare laborem", II.1–2), a figure Hans Blumenberg made the paradigm of the modern relation to disaster in *Shipwreck with Spectator* (1979). The Richmond dinner is Lucretius's shore,

upholstered. Its guests form a jury of the present: medicine, psychology, municipal government and the press receive testimony from the future and weigh it, as David Hume had taught them to weigh reports of miracles, against the uniformity of experience. Four years later Joseph Conrad would seat a Director of Companies, a Lawyer and an Accountant on the deck of the *Nellie* to hear Marlow concede that the Thames too "has been one of the dark places of the earth"; Wells's guests receive the complementary revelation that it will be one again. In both frames the comfort of the listeners is the material condition of the tale, and the tale's burden is that the comfort is paid for by someone out of sight.

This thesis contends that the dinner is not an ornament of the frame but its hermeneutic key. *The Time Machine* is the first modern fiction to construct time itself as a metabolic process — the irreversible conversion of one order into the sustenance of another — and its frame binds that process to the reader's appetite. I call this construction *chronophagy*. Three further claims follow from it: that Wells's characteristic procedure is a *speciation of metaphor*, by which the social figures of Victorian discourse are passed through deep time until they become flesh; that the world of 802,701 obeys a *social second law*, under which a society that has abolished its gradients approaches equilibrium as an *entropic pastoral*; and that against chronophagy the novel inscribes a counter-economy, an *ethics of the remainder*, condensed in two pressed flowers that share the Traveller's pocket with his matches. The present chapter establishes the genealogy of the central figure, defines its terms, situates the text and its criticism, and sets out the method and architecture of the argument.

1.2
Tempus Edax Rerum: A Genealogy of the Devouring Hour

The figure of time as a devourer is among the oldest in European letters, and its most famous formulation arrives, significantly, inside a sermon on diet. In the fifteenth book of the *Metamorphoses*, Ovid's Pythagoras opens his discourse on universal flux by forbidding mortals to pollute their bodies with impious feasts ("Parcite, mortales, dapibus temerare nefandis / corpora", XV.75–76) and by recalling an age "to which we have given the name golden", fortunate in the fruits of trees and the herbs of the earth, that "did not pollute its mouth with

blood" ("nec polluit ora cruore", XV.96–98). Only after this dietary preamble does he apostrophize the destroyer: "Tempus edax rerum, tuque, invidiosa vetustas, / omnia destruitis vitiataque dentibus aevi / paulatim lenta consumitis omnia morte" (XV.234–236) — "Time, devourer of things, and you, envious Age, destroy all things, and, spoiling them with the teeth of the years, consume them little by little in a lingering death." The teeth of time and the teeth of the carnivore enter the same speech. The Golden Age is defined by abstention from flesh, and its loss by the pollution of the mouth. The alimentary and the temporal are, from the outset, one topic.

The topic has a theological ground, and the ground is Kronos. Hesiod's *Theogony* narrates how Kronos, warned that he would be overthrown by a son, swallowed each of his children at birth until Rhea deceived him with a stone wrapped in swaddling-clothes (453–506). Kronos devours in order to forestall succession: his appetite is an attempt to arrest time, to prevent the future from arriving. Yet in *Works and Days* the same god presides over paradise. The golden race lived "in the time of Kronos, when he was reigning in heaven" (111), without toil or grief, while the earth bore fruit unbidden, and they died "as though overcome by sleep" (ὥσθ' ὕπνῳ δεδμημένοι, 116). The silver race that followed was distinguished by a grotesquely protracted infancy, each child reared for a hundred years at its mother's side, "a great simpleton" (μέγα νήπιος, 130–131). Classical scholarship has long registered the discrepancy between the benevolent Kronos of the Golden Age and the cannibal Kronos of the succession myth. *The Time Machine* dissolves it. The Eloi combine the defining features of Hesiod's first two races — fruit without labour, the intellect of children, a death that comes upon them in their sleep — and they live under the hidden dominion of a devourer. The benevolent king and the eater of children are a single regime, seen from above and from below.

The etymological conflation that made Kronos the god of time completes the genealogy. Cicero's Stoic spokesman in *De Natura Deorum* explains that Saturn "is called Kronos, which is the same as *chronos*, that is, a span of time" ("Κρόνος enim dicitur, qui est idem χρόνος, id est spatium temporis"), and that he is fabled to have devoured his offspring "because age consumes the spans of time and is insatiably filled with the years gone by" ("ex se enim natos comesse fingitur solitus, quod consumit aetas temporum spatia annisque praeteritis insaturabiliter expletur", II.64). Plato's *Statesman* adds a cosmological dimension that bears with uncanny precision on Wells's design. In the age of Kronos, the Eleatic

Stranger relates, the cosmos turned in the reverse direction: the old grew young, the dead returned from the earth, food sprang up without agriculture, there were neither constitutions nor households, and the god himself tended human beings as a herdsman tends his flock (268d–274e). Whether the nurslings of Kronos were happier than we are, the Stranger concludes, depends on whether they spent their boundless leisure in philosophy or merely ate, drank and told stories (272b–d). In Plato the age of Kronos is an age of reversed time, of pastoral tutelage and of a leisure whose worth is undecided. Wells's Traveller reaches a herded paradise by means of a machine that can run time backwards, and finds the Stranger's question answered in the negative.

Virgil secured the Saturnian complex for the European imagination. The fourth *Eclogue* announces that "the reign of Saturn returns" ("redeunt Saturnia regna", 6), and the second *Georgic* places golden Saturn's life on earth before "an impious race feasted on slaughtered bullocks" ("impia quam caesis gens est epulata iuvencis", II.537). Across Hesiod, Plato, Virgil and Ovid the pattern holds: the Golden Age abstains from flesh, its end coincides with the eating of flesh, and its presiding deity is the god who eats his children. *The Time Machine* restores the suppressed term. Its Golden Age is genuinely frugivorous, but the impious feast has not ended it. The feast is what sustains it.

The figure passed into the English Renaissance intact. Shakespeare's nineteenth sonnet addresses "Devouring Time", which can "make the earth devour her own sweet brood"; in *Troilus and Cressida* Ulysses describes Time carrying a wallet "Wherein he puts alms for oblivion, / A great-sized monster of ingratitudes", whose scraps are good deeds "devour'd / As fast as they are made, forgot as soon / As done" (III.iii). Spenser's Garden of Adonis, the seminary of all generation, has one great enemy, "wicked Time, who with his scyth addrest, / Does mow the flowring herbes and goodly things" (*The Faerie Queene* III.vi.39). The scythe is itself a Saturnian relic. Erwin Panofsky's study of "Father Time" in *Studies in Iconology* (1939) showed how the sickle of Kronos, an agrarian god, migrated to the personification of Time once the two names had fused, yielding the winged, scythe-bearing elder of the emblem books and, at the extremity of the tradition, Goya's *Saturn Devouring His Son* (c. 1820–23). Raymond Klibansky, Panofsky and Fritz Saxl traced the further migration by which Saturn became the planet of melancholy and of scholars in *Saturn and Melancholy* (1964), whose emblem is Dürer's *Melencolia I* (1514): a winged figure sunk in inertia among the instruments of geometry, beneath an hourglass and a bell.

Wells's Traveller, the grey-eyed experimentalist who, the narrator tells us, "thought but cheerlessly of the Advancement of Mankind", is the late-Victorian heir of that figure: a melancholic of measurement who builds an instrument to visit Saturn's kingdom and finds it a larder.

Wells's onomastics complete the complex. "Eloi" echoes the cry from the cross in Mark 15:34, "Eloi, Eloi, lama sabachthani?", the dereliction of a son forsaken by his father. "Morlock" recalls Molech, to whom Israel is forbidden to "let any of thy seed pass through the fire" (Leviticus 18:21), and whom Milton enthroned among the fallen as the "horrid King besmear'd with blood / Of human sacrifice, and parents tears" (*Paradise Lost* I.392–393). The names compose a theology in miniature: forsaken children above, a child-devouring god below. The Saturnian matrix — Golden Age, devoured offspring, time, harvest, melancholy, the inverted table — is less a source Wells consulted than a structure his novel reactivates entire. What that reactivation means, once it is joined to the new sciences of the nineteenth century, is the subject of what follows.

1.3
Chronophagy: A Definition and the Four Dimensions of Appetite

By *chronophagy* I mean the figuration of irreversible temporal process as alimentary conversion: the maintenance of one order through the consumption of another, accompanied by an uncompensated loss — heat, waste, forgetting — that gives the process its direction. In *The Time Machine* time is not the neutral medium within which eating occurs; eating is the form in which time becomes perceptible. The definition has three elements, each indispensable: an asymmetry between consumer and consumed, a conversion that sustains the one at the cost of the other, and a residue that cannot be recovered. Remove the asymmetry and nothing moves; remove the conversion and nothing persists; remove the residue and the process could run backwards, which is to say that it would not be temporal at all.

The Traveller's opening lecture insists that "any real body must have extension in four directions: it must have Length, Breadth, Thickness, and — Duration." Every appetite in the novel likewise has extension in four registers, which I

call the four dimensions of appetite. Table 1.1 sets them out; the chapters that follow elaborate each.

The oikonomic register is the most visible. Marx had described capital as "dead labour, that, vampire-like, only lives by sucking living labour, and lives the more, the more labour it sucks" (*Capital*, vol. I, 1867, ch. 10); the Eloi's soft garments, worn by a people who make nothing, are commodities whose fetish-character conceals the labour of the depths. The zoological register is Darwin's "war of nature", Malthus's arithmetic of subsistence, and E. Ray Lankester's demonstration in *Degeneration: A Chapter in Darwinism* (1880) that natural selection may simplify an organism as readily as it elaborates one; Louis Dollo's law of evolutionary irreversibility (1893) guaranteed that what was lost would not return in the same form. The thermodynamic register is Rudolf Clausius's double axiom of 1865, "Die Energie der Welt ist constant. Die Entropie der Welt strebt einem Maximum zu"; the universal tendency in nature to the dissipation of mechanical energy announced by William Thomson, later Lord Kelvin, in 1852; and the heat death Hermann von Helmholtz projected in 1854. The mediatic register, the least discussed, concerns the consumption of time as spectacle. The Traveller sits in a saddle like a cyclist, watches "the palpitation of night and day" merge "into one continuous greyness" like a spectator before a flickering screen, and returns to sell his tale to men of the press.

What makes these registers a system rather than a list is that the novel renders them convertible. Under acceleration, a figure in one register becomes a fact in another. Class, an oikonomic relation, becomes species, a zoological fact; species becomes an energy budget, the last metabolism of a dying planet; and the whole is converted, at the end, into a narrative consumed at table. The time machine is the converter. It does not merely carry the Traveller across the registers; it makes their equivalence visible, because it alone can supply the quantity of time that each conversion requires.

The machine has a further property, which the novel exploits without naming it. It is reversible. It travels forwards and backwards, and its maker returns to the Thursday from which he set out. The world it traverses is not reversible: the Eloi cannot become Victorians again, the sun cannot unswell, and the books of the Palace of Green Porcelain, reduced to "brown and charred rags", cannot recompose themselves. *The Time Machine* thus stages in narrative form the objection Josef Loschmidt put to Ludwig Boltzmann in 1876: how can a world whose mechanical laws are symmetrical in time exhibit irreversible processes at

REGISTER	Oikonomic *(social)*	Zoological *(biological)*	Thermodynamic *(cosmic)*	Mediatic *(aesthetic)*
WHAT IS CONSUMED	Labour, then bodies	Prey; disused faculties	Free energy; gradients	Duration; spectacle; the tale
CONSUMER	Capital, then labour	Predation; selection; disuse	Dissipation; tidal friction	Traveller, audience, adapters
SITE IN THE NOVEL	The Underworld machinery; the Eloi's garments	Eloi and Morlock bodies; the giant crabs	The swollen red sun; the stopped Earth; the beach	The saddle; the flicker of day and night; the dinner
GOVERNING NINETEENTH-CENTURY DISCOURSE	Disraeli, Carlyle, Engels, Marx, Booth	Darwin, Malthus, Lankester, Dollo	Clausius, Thomson, Helmholtz, George Darwin	Cook's tours, the cinematograph, the New Journalism
RESIDUE	Class hardened into species	Degeneration; extinction	Heat death; "a horror of this great darkness"	Testimony; two pressed flowers

Table 1.1 The four dimensions of appetite in *The Time Machine*.

the scale of experience? The Traveller is reversible; history is not. The frame, moreover, enacts recurrence while the content enacts dissipation. The Traveller returns to the same table and the same company, as though the novel's form had been composed under the sign of the eternal return Nietzsche announced in *The Gay Science* (§341, 1882), or of the recurrence theorem Henri Poincaré proved in 1890 and Ernst Zermelo turned against Boltzmann in 1896. But the circle breaks. The Traveller departs again, with a small camera and a knapsack, and the epilogue records that he "vanished three years ago" and has never returned. The form submits at last to the content: the time traveller is himself consumed by time.

Chronophagy is offered neither as a fourth theme beside class, degeneration and entropy nor as a master-signifier that would absorb them. It names the operator that allows the novel to pass from one to another without loss of logical continuity — and, as Chapter Eight will argue, the operator against which the novel's final image is set.

1.1
The Speciation of Metaphor

Victorian social description imagined the classes as nations, races and species long before Wells imagined them as anything else. Its master-text is the passage in Benjamin Disraeli's *Sybil* (1845) in which a stranger defines Queen Victoria's realm as "Two nations; between whom there is no intercourse and no sympathy; who are as ignorant of each other's habits, thoughts, and feelings, as if they were dwellers in different zones, or inhabitants of different planets; who are formed by a different breeding, are fed by a different food, are ordered by different manners, and are not governed by the same laws." The figures are precise and plural: geographical (zones), astronomical (planets), genealogical (breeding) and alimentary (food). The same half-century produced Friedrich Engels's survey of the Manchester poor in their cellar-dwellings (1845); Henry Mayhew's interviews with the sewer-hunters who worked the tunnels beneath London (1851); Andrew Mearns's *The Bitter Cry of Outcast London* (1883); W.T. Stead's "The Maiden Tribute of Modern Babylon" (1885), which cast the capital's traffic in the daughters of the poor as a tribute fed to the Minotaur in its labyrinth; George Gissing's *The Nether World* (1889); and General William Booth's *In Darkest England and*

the Way Out (1890), whose "submerged tenth" was to be guaranteed, under the "Cab Horse Charter", at least the shelter, food and work that London afforded its cab-horses. The poor were below, submerged, differently fed, differently bred, a race apart.

Wells's procedure is to take these figures at their word and give them eight hundred thousand years. "Different breeding" becomes speciation; "different food" becomes frugivory on one side and cannibalism on the other; the "submerged" become subterranean; the "nether world" becomes the Under-world. The Traveller reasons the conversion through in so many words. "Even now," he asks, "does not an East-end worker live in such artificial conditions as practically to be cut off from the natural surface of the earth?" Invoking the underground railways, subways and workrooms of contemporary London, he concludes that "the Upper-world man had drifted towards his feeble prettiness, and the Under-world to mere mechanical industry." The novel's most celebrated conceit is, in the strict sense, a metaphor allowed to come true.

The drafts show the operation at work. In the unsigned sketches W. E. Henley ran in the *National Observer* in 1894, the Traveller's destination was A.D. 12,203; in 1895 it is 802,701, an extension of the horizon by a factor of more than sixty. The revision is not decorative. Ten thousand years suffice for a metaphor to harden into custom, caste or law; only geological time suffices for it to harden into flesh. The leap from 12,203 to 802,701 is the speciation of metaphor made visible in the history of the text: Wells discovered that the figure required deep time in order to become literal.

Samuel R. Delany supplies the theoretical vocabulary. In "About 5,750 Words" (1968) he argued that science fiction is distinguished by its "subjunctivity", the level of possibility at which its sentences are read, so that a phrase such as "her world exploded", a dead metaphor in realist prose, may in science fiction report a planetary catastrophe. Darko Suvin's account of the "novum", the new thing whose cognitive validation organizes the text (*Metamorphoses of Science Fiction*, 1979), makes a related claim at the level of narrative logic. The speciation of metaphor refines both. The Wellsian novum is not primarily a device or a discovery but a quantity of time, and its function is to literalize. The time machine is therefore a rhetorical engine, a machine for converting figure into flesh; and this temporal literalization, I propose, is the constitutive operation of speculative fiction as a genre. Its characteristic sentence is a social figure run forward until it becomes biology, geography or physics. Fritz Lang's *Metropolis* (1927) runs "the

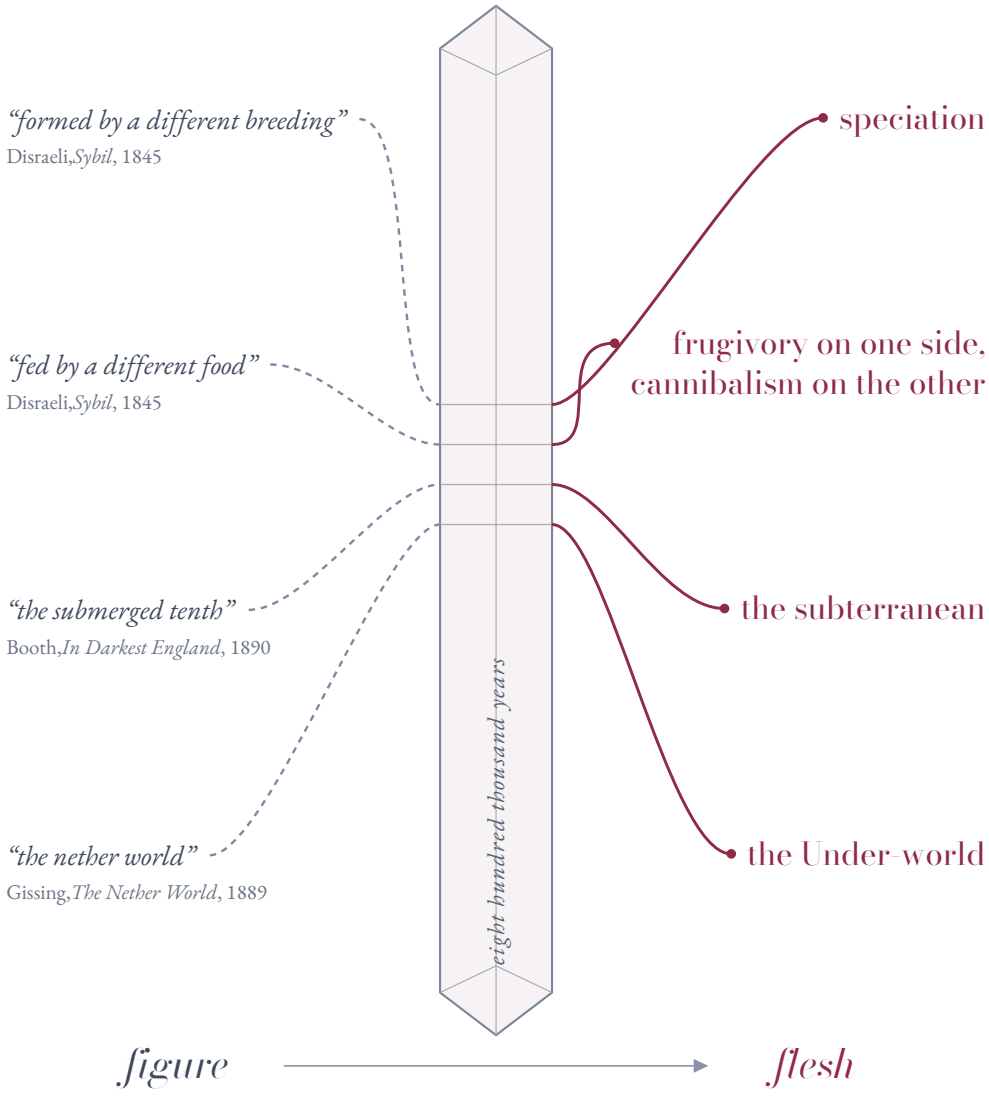


Figure 1.1 The speciation of metaphor. Four figures of Victorian social description, given eight hundred thousand years, harden into biological fact: "different breeding" becomes speciation; "different food" becomes frugivory on one side and cannibalism on the other; the "submerged" become subterranean; the "nether world" becomes the Under-world.

machine devours the workers" forward until the machine becomes Moloch. Bong Joon-ho's *Snowpiercer* (2013) runs "the front and the back of the train of progress" forward until the remnant of humanity rides a single train circling a frozen planet; his *Parasite* (2019) runs "the household below stairs" forward until a man lives for years in a bunker beneath a rich family's house. Galder Gaztelu-Urrutia's *The Platform* (2019) runs "trickle-down" forward until food literally descends through a vertical prison whose lower levels turn to cannibalism. Each is a speciation of metaphor in Wells's sense, and each is legible as a digestion of his text.

Wells had rehearsed the operation in his journalism. "Zoological Retrogression" (*Gentleman's Magazine*, 1891) contested the assumption of continuous evolutionary ascent with Lankester's degenerate forms, warning that nothing in biology guaranteed humanity's permanence. "The Man of the Year Million" (*Pall Mall Budget*, 1893) imagined a cerebral, atrophied descendant retreating underground as the sun cooled, and was promptly parodied in *Punch*. "The Extinction of Man" (*Pall Mall Gazette*, 1894) entertained the succession of humanity by giant crustaceans and intelligent cephalopods. The terminal beach, with its monstrous crabs and its tentacled creature "the size of a football perhaps", "hopping fitfully about" on the shoal, is an essay literalized; the Morlocks are the underground of "The Man of the Year Million" stripped of its intellect.

Wells also had Victorian precedents for converting moral fable into evolutionary narrative. In Charles Kingsley's *The Water-Babies* (1863) the Doasyoulikes, who live without labour in a land of plenty, degenerate across generations into apes, the last of them shot by the gorilla-hunter Paul Du Chaillu; in George MacDonald's *The Princess and the Goblin* (1872) the goblins descend from ordinary people who withdrew underground and altered in the course of generations. Both fables are underwritten by a providential pedagogy: their transformations punish and instruct. Wells removes the providential frame. No author-god guarantees that the transformation means anything; it simply occurs. The comparison with the "noble lie" of Plato's *Republic* (414b–415d) is exact. Socrates proposes to persuade the citizens that the god mixed gold, silver, iron and bronze into their souls, so that class will appear as natural kind. The founders need a lie to naturalize hierarchy; Wells needs only time. *The Time Machine* is the noble lie come true, and its truth is not noble.

1.5
The Social Second Law and the Entropic Pastoral

The Traveller formulates the governing law of 802,701 as an aphorism: "Strength is the outcome of need; security sets a premium on feebleness." Near the end of his narrative he generalizes it: "It is a law of nature we overlook, that intellectual versatility is the compensation for change, danger, and trouble. An animal perfectly in harmony with its environment is a perfect mechanism. … There is no intelligence where there is no change and no need of change." Read beside Sadi Carnot's *Réflexions sur la puissance motrice du feu* (1824), the law is thermodynamic before it is moral. An engine does work only across a difference of temperature; where the difference vanishes, so does the work. The Traveller's sociology restates Carnot's principle in the medium of history: need is a gradient, intelligence is work, and a society that has abolished its gradients has abolished its capacity to do anything at all.

On this reading the Eloi are a society at equilibrium, the social analogue of heat death. They have no war, no disease, no labour and no fear except of the dark; they have also no curiosity, no memory, no fire and no future. The Morlocks are the last gradient, the residual differential that still does work, and the work it does is predation. The world of 802,701 is a heat engine idling towards equilibrium, its final cycle the passage of bodies from the lawn to the larder; the terminal beach is its completion at cosmic scale. I call the principle the *social second law*: in a closed social order, the abolition of need does not abolish asymmetry but concentrates it in a single surviving channel, and that channel becomes the channel of consumption. By the time he reaches the beach, the Traveller has watched the law operate on three scales — the social, the biological and the solar — and found it identical on each.

Wells was not alone in seeking a thermodynamics of history in 1895. Brooks Adams's *The Law of Civilization and Decay*, published in London that year, described civilizations as accumulations of social energy that pass from martial to economic forms before they dissipate; his brother Henry would apply Kelvin's dissipation to historical method in *A Letter to American Teachers of History* (1910). Freud's *Beyond the Pleasure Principle* (1920) posited a drive towards the reduction of tension to zero, adopting Barbara Low's name for it, the "Nirvana

principle"; the languor of the Eloi is its collective form. Thomas Pynchon's story "Entropy" (1960) would restage the dichotomy as two floors of a Washington apartment building, a sealed hothouse above and a disintegrating party below. Wells's priority lies not in the analogy but in its narrative realization: he built a world that runs the analogy forward.

The world he built inverts the oldest schema of historical decline. Hesiod's ages descend from gold through silver, bronze and the heroes to iron; Wells's descend *into* gold. The Golden Age arrives at the end of history rather than at its beginning, not as reward but as exhaustion. I call this inversion the *entropic pastoral*. Its emblem is the inscription Panofsky traced from Guercino to Poussin in "Et in Arcadia Ego: Poussin and the Elegiac Tradition" (1936) — *even in Arcadia, I am there*, the speaker being Death. In 802,701 the speaking "I" lives beneath the lawns. Its pictorial correlate is Thomas Cole's cycle *The Course of Empire* (1833–36), which runs from *The Savage State* through *The Arcadian or Pastoral State* and *The Consummation of Empire* to *Destruction* and *Desolation*. The Eloi inhabit Cole's fifth canvas as though it were his second: shepherds without flocks, idling among the ruins of a consummation they cannot remember.

The pastoral Wells inverts was not only classical. It was also the contemporary socialist pastoral of William Morris's *News from Nowhere* (1890), with its beautiful, long-lived, unhurried citizens and its Houses of Parliament turned into a store for dung, and the aestheticist socialism of Oscar Wilde's "The Soul of Man under Socialism" (1891). Wilde's essay deserves a prominence it has not received in criticism of the novel. It proposes that the drudgery of civilization be transferred to machinery, which "must work for us in coal mines, and do all sanitary services", so that humanity may give itself to cultivated leisure and the making of beautiful things: "Human slavery is wrong, insecure, and demoralising. On mechanical slavery, on the slavery of the machine, the future of the world depends." The world of 802,701 is Wilde's programme realized with a single substitution. The machine-slaves are human; they tend their engines in the dark; and they eat the aesthetes. Bergonzi long ago observed that the Eloi, garlanded, languid and decorative, satirize the aesthetic movement; the link with Wilde's essay converts that resemblance into a direct riposte. The Heinemann edition appeared in May 1895, the month of Wilde's conviction.

The diagnosis would recur in sociology and economics. Max Weber closed *The Protestant Ethic and the Spirit of Capitalism* (1905) by imagining the "last

men" (*letzten Menschen*, after Nietzsche) of a fully rationalized order: "Fachmenschen ohne Geist, Genußmenschen ohne Herz" — specialists without spirit, sensualists without heart. Weber's sentence divides exactly into Morlock and Eloi. John Maynard Keynes, in "Economic Possibilities for our Grandchildren" (1930), predicted that within a century the economic problem would be solved, leaving humanity to face its "permanent problem", the use of leisure, and confessed that he contemplated the readjustment with dread. Keynes's horizon was 2030. Written in 2026, this thesis stands four years short of it, at a moment when the automation of cognitive labour has turned Wilde's machine-slaves and Keynes's problem of leisure into matters of policy rather than speculation.

The entropic pastoral yields the formula that governs Chapters Three to Five: *every Arcadia is a feedlot seen from above*. Hesiod's two Kronoi, the benevolent king and the devourer, are the same god seen from the lawn and from the larder. The formula's modern descendants are numerous and exact: Ursula K. Le Guin's Omelas, whose felicity depends on a child kept in a cellar; the Hailsham of Kazuo Ishiguro's *Never Let Me Go* (2005), where pupils are raised in pastoral tenderness for the harvesting of their organs; the orphanage of *The Promised Neverland* (2016–2020), where children are reared as food. Each will be read as a variation on Wells's discovery that pastoral is a relation of production.

1.6
The Pocket and the Mallow

Against chronophagy the novel inscribes a counter-economy, and it concentrates that economy in an object and a container. The object appears at the climax of the dinner. Pressed by incredulity, the Traveller "put his hand into his pocket, and silently placed two withered flowers, not unlike very large white mallows, upon the little table." The container is the pocket itself, which Weena had repurposed: "My pocket had always puzzled Weena, but at the last she had concluded that it was an eccentric kind of vase for floral decoration. At least she utilized it for that purpose." A man's pocket, the receptacle of tools and coin, has been converted by a creature with the intellect of a child into a vessel for gifts.

The same pocket held matches. The Traveller reached 802,701 with a part-used box, and in the ruined museum he found another, its contents still serviceable after eight thousand centuries. He greets the find in martial terms — "For

now I had a weapon indeed against the horrible creatures we feared" — and celebrates it by performing, on the dust of the gallery floor, "a kind of composite dance, whistling *The Land of the Leal* as cheerfully as I could." The song is Lady Nairne's lament of a dying wife who tells her husband she is bound for "the land o' the leal", the country of the faithful dead; the Traveller whistles it on the eve of the night in which he loses Weena. The matches serve him as lamp and weapon. He uses them to dazzle and repel the Morlocks and to kindle the fire that, spreading through the forest, kills them indiscriminately; in the confusion of that night Weena is lost. The matches are consumed entirely. The flowers remain.

The opposition holds on every axis the novel cares about. Matches are combustion, the rapid oxidation that accelerates entropy; flowers are photosynthesis, the capture of solar energy that Erwin Schrödinger would gloss when he wrote that what an organism feeds upon is "negative entropy" (*What Is Life?*, 1944). Matches are Promethean fire, the gift by which, in Percy Shelley's allegorical reading in *A Vindication of Natural Diet* (1813), humanity first cooked flesh and forfeited its Golden Age; flowers are the food of that age. Matches are weapons; flowers are gifts. And the matches are commodities in a precise historical sense. The match industry of East London was the emblematic sweated trade of the 1880s: Annie Besant's exposé "White Slavery in London" (1888) and the strike of the Bryant and May match-girls in the same year made phosphorus necrosis of the jaw a national scandal. The Traveller's Promethean instrument is congealed labour from the very London the Morlocks literalize. He fights the people of the abyss with fire the people of the abyss had made.

The mallow carries a genealogy that the novel never names and that a chronophagic comparatism can recover. In the opening movement of *Works and Days* Hesiod rebukes his brother Perses for seizing more than his share of their inheritance and for flattering the lords who judge the case, whom he calls δωροφάγοι, literally "gift-eaters", rendered by Evelyn-White as "bribe-swallowing lords". The rebuke turns into a maxim: "Fools! They know not how much more the half is than the whole, nor what great advantage there is in mallow and asphodel" (40–41). Mallow and asphodel were the food of the frugal poor; asphodel is also the flower of the meadows of the dead in the *Odyssey*. Hesiod's maxim is the oldest formulation in European letters of non-total consumption, and he sets it explicitly against an appetite that devours gifts. Whether or not Wells recalled the line, his final image restores it to its context: in a cosmos organized by consumption, a

flower of the mallow kind, given rather than taken, is the sole object that crosses time without being eaten.

The narrator draws the moral in a sentence whose grammar enacts it. The Traveller, he records, "saw in the growing pile of civilization only a foolish heaping that must inevitably fall back upon and destroy its makers in the end. If that is so, it remains for us to live as though it were not so." Two phrases carry the ethic. *Remains* names the ethical subject as a remainder, what is left when the accounting of consumption is complete. *As though* names the mode of action: neither denial of the entropic verdict nor submission to it, but conduct under a fiction knowingly held, in the sense Hans Vaihinger would systematize in *Die Philosophie des Als Ob* (1911), and in the spirit of T. H. Huxley's Romanes Lecture of 1893, which Wells, Huxley's former student, had every reason to know: "Let us understand, once for all, that the ethical progress of society depends, not on imitating the cosmic process, still less in running away from it, but in combating it." The novel then closes on the flowers themselves, "shrivelled now, and brown and flat and brittle — to witness that even when mind and strength had gone, gratitude and a mutual tenderness still lived on in the heart of man." Its last great abstraction, gratitude, answers Shakespeare's Time, that "great-sized monster of ingratitudes". And the word *flat* closes an arc that opened with the lecture on dimensions. A book that begins by adding a dimension to the world ends with an object that has lost one: a pressed flower, a specimen from a herbarium, a Flatland relic of four-dimensional experience.

The *ethics of the remainder* designates the practice that the flowers emblematize. It has five components, each developed in Chapter Eight. It is a practice of non-total consumption, Hesiod's half, which Jacques Derrida would recast as the question not of whether to eat but of how to eat well ("'Il faut bien manger'", 1989). It is a practice of the gift, which Marcel Mauss analysed as the creation of obligation across time (*Essai sur le don*, 1925) and Derrida as that which "gives time" (*Donner le temps*, 1991). It is a practice of negentropy, the local and provisional refusal of equilibrium that every living thing performs. It is a practice of containment rather than conquest — the pocket rather than the spear — in the sense of Le Guin's "The Carrier Bag Theory of Fiction" (1986). And it is a practice of testimony, since the flowers exist "to witness". So defined, the ethics of the remainder stands against two contemporary temptations: the despair of extinction philosophy, which concludes from the eventual death of the sun that meaning is already annulled, and the optimizing futurism of

longtermism, which treats the present as fuel for the maximization of a remote future and so reinstates chronophagy in the name of ethics. Its modern instances are as exact as its ancient source: the seedling WALL-E keeps in an old boot (2008); the untouched panna cotta that becomes a message in *The Platform*; the boy "carrying the fire" in Cormac McCarthy's *The Road* (2006); the robot gardener of Hayao Miyazaki's *Castle in the Sky* (1986); and the closing option of *NieR: Automata* (2017), in which players may erase their entire saved game to help strangers they will never meet.

1.7
The Text in Time: Versions, Variants, Excisions

The Time Machine has a history of composition that is itself a small evolutionary record, complete with ancestors, transitional forms and an excised intermediate. Its earliest ancestor is "The Chronic Argonauts", serialized in three numbers of the *Science Schools Journal* in 1888, a periodical Wells had helped to found as a student at the Normal School of Science. Its protagonist, Dr Moses Nebogipfel, settles in the Welsh village of Llyddwdd, whose inhabitants take him for a warlock and storm his house; he escapes in his "Chronic Argo" with the Reverend Elijah Ulysses Cook. The name invites decomposition into the biblical Nebo, the mountain from which Moses was shown a promised land he would not enter (Deuteronomy 34:1–4), and the German *Gipfel*, summit: the prophet who sees a future he cannot inhabit. The title invokes the Argonautic voyage for the fleece of a sacrificed ram. The villagers' "warlock" offers a phonetic premonition of "Morlock", and the golden fleece, the wool of a slaughtered sheep, finds an ironic fulfilment in a novel whose hero discovers a soft and docile flock and returns home to eat mutton.

The intermediate stages are journalistic. In 1894 Henley published in the *National Observer* a series of unsigned sketches on time-travelling, which set the visit in A.D. 12,203 and broke off when he lost the editorship. In 1895 Henley's *New Review* serialized a fuller version from January to May. That serial contains a passage Wells excised from the book: between 802,701 and the beach of the crabs, the Traveller glimpses grey, hopping creatures resembling rabbits or small kangaroos, possibly the last descendants of humanity, and a monstrous centipede that preys upon them. The book appeared in two forms in 1895: an Amer-

ican edition from Henry Holt, whose title page famously attributed it to "H.S. Wells", and the English edition from William Heinemann in May. Wells revised the text for the Atlantic Edition of his works (1924) and returned to it in later prefaces, each time reading his younger self from a greater distance.

The textual history is chronophagic in its own right. Each version is nourished by its predecessor and discards part of it. The excision of the grey creatures is especially telling. It removes the transitional rung between the Eloi and the crabs, so that the published text leaps from a degraded humanity to a world from which humanity is simply absent. The editorial cut produces a time-jump within the time-jump: extinction becomes abrupt rather than gradual, an absence registered rather than a decline observed. The text, like the species it describes, survives by losing its middle. This thesis takes the Heinemann edition as its base text while drawing on the apparatus of Geduld and Ruddick wherever variants bear on the argument.

1.8 The State of the Field

Criticism of *The Time Machine* falls into three overlapping phases, each of which illuminates one or two of the four registers and none of which unites them. The first, inaugurated by Bernard Bergonzi's essay "The Time Machine: An Ironic Myth" (1960) and his study *The Early H. G. Wells: A Study of the Scientific Romances* (1961), read the novella as an ironic myth of the fin de siècle: a work of cosmic pessimism whose Eloi satirize the aesthetes of the 1890s and whose vision of decline belongs with the mood that Max Nordau would pathologize in *Entartung* (1892–93). Bergonzi established the novel's seriousness and its affinity with the decadence it mocks. He also fixed a reading of Wells's career as a fall from imaginative pessimism into didactic optimism, which subsequent critics have contested.

The second phase was structural and political. Darko Suvin's *Metamorphoses of Science Fiction* (1979) made Wells the turning point of the genre's tradition and the novel a paradigm of "cognitive estrangement", the defamiliarization of the present through a cognitively validated novum. John Huntington's *The Logic of Fantasy: H. G. Wells and Science Fiction* (1982) described the early romances as systems of balanced oppositions — Eloi and Morlock, surface and depth — in

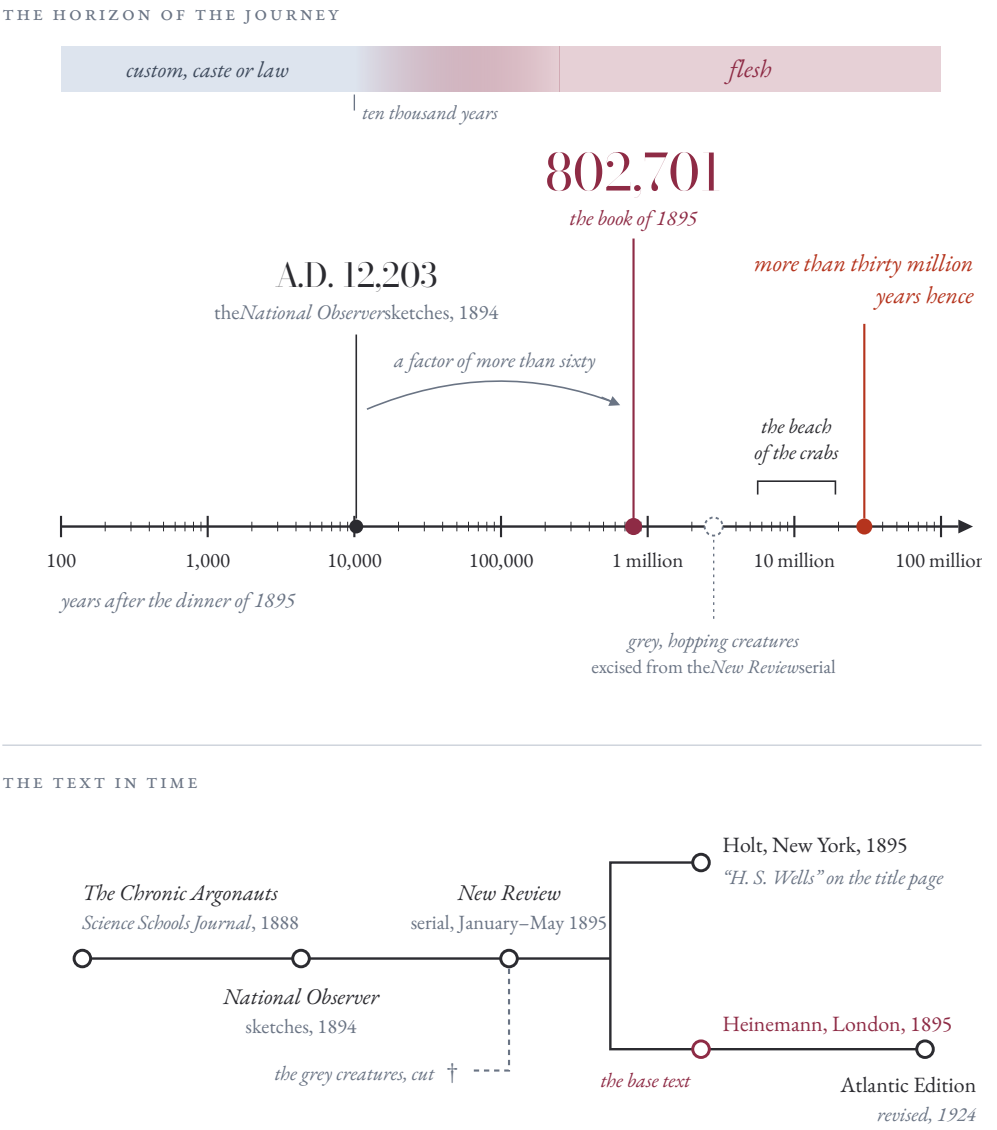


Figure 1.2 The horizon of the journey and the text in time. Above, in years after 1895 on a logarithmic scale: the sketches of 1894 stopped in A.D. 12,203; the book of 1895 went on to 802,701, an extension by a factor of more than sixty, and to a beach "more than thirty million years hence"; the grey, hopping creatures of the *New Review* serial, excised from the book, stood between 802,701 and the beach of the crabs. Below, the versions as a small evolutionary record, with ancestors, transitional forms and an excised intermediate.

which Wells withholds resolution. Frank McConnell's *The Science Fiction of H. G. Wells* (1981) and Patrick Parrinder's *Shadows of the Future: H. G. Wells, Science Fiction and Prophecy* (1995) refined the account of Wells as a prophet whose prophecies are structurally ironic, while Robert M. Philmus and David Y. Hughes's edition of *H. G. Wells: Early Writings in Science and Science Fiction* (1975) recovered the journalism from which the romances grew. Leon Stover's annotated edition (1996) pressed a contentious reading of the Traveller as the vehicle of a Saint-Simonian technocratic elitism that Wells would later espouse openly. Fredric Jameson's essays, from "Progress versus Utopia; or, Can We Imagine the Future?" (1982) to *Archaeologies of the Future* (2005), supplied the period's most influential theory of the genre: science fiction estranges the present by rendering it as the past of a future not yet arrived.

The third phase, the turn to science and culture, has been the most productive. Gillian Beer's *Darwin's Plots* (1983) and *Open Fields* (1996) taught a generation to read evolutionary theory as narrative. Stephen Kern's *The Culture of Time and Space, 1880–1918* (1983) placed the novel within the period's revolution in temporal experience. Daniel Pick's *Faces of Degeneration* (1989) and Kelly Hurley's *The Gothic Body* (1996), with its concept of the "abhuman", situated the Morlocks in the medical and Gothic discourses of the fin de siècle. Thermodynamic readings have deepened steadily, from Bruce Clarke's *Energy Forms* (2001) to Barri J. Gold's *ThermoPoetics* (2010) and Allen MacDuffie's *Victorian Literature, Energy, and the Ecological Imagination* (2014). The editions of Geduld, Ruddick and Roger Luckhurst have consolidated the textual and contextual record. Histories of time travel, from Paul K. Alkon's *Origins of Futuristic Fiction* (1987) and Paul J. Nahin's *Time Machines* (1993) to James Gleick's *Time Travel: A History* (2016), have stated Wells's priority with some precision. He did not invent the time-travelling vehicle, which appears in Enrique Gaspar's *El anacronópete* (1887) and, as a backward-running clock, in Edward Page Mitchell's "The Clock That Went Backward" (1881); he was the first to ground such a vehicle in a geometry of time and to direct it into evolutionary and cosmic time. Recent theory has begun to read the novel at planetary scale. Mark McGurl's "The Posthuman Comedy" (2012) addresses genre fiction's confrontation with inhuman scales of time, and Dipesh Chakrabarty's "The Climate of History: Four Theses" (2009) supplies the frame within which the novel's collapse of human into natural history acquires its contemporary urgency.

The gap this thesis addresses is structural. Existing accounts treat class, degeneration and dissipation as parallel themes, or as allegories of one another, without identifying the operator that renders them convertible. They register the novel's pastoral and its predation without reconstructing the Saturnian matrix in which the two are one. They read the flowers as sentiment rather than as the emblem of a counter-economy. And they catalogue the novel's afterlife as influence rather than reading it as metabolism. Chronophagy, the speciation of metaphor, the social second law and the ethics of the remainder are offered as the concepts these omissions require.

1.9 Method: Chronophagic Comparatism

A thesis about a machine that travels in both directions through time requires a comparative method that does the same. Three twentieth-century theories of literary history supply its elements. T.S. Eliot argued in "Tradition and the Individual Talent" (1919) that the existing monuments of literature form an ideal order which is modified by the arrival of the genuinely new, so that the past is altered by the present as surely as the present is directed by the past. Jorge Luis Borges radicalized the claim in "Kafka and His Precursors" (1951): every writer creates his own precursors, and texts that now seem Kafkaesque were not so before Kafka wrote. Oswald de Andrade's *Manifesto Antropófago* (1928), with its parodic slogan "Tupi or not tupi, that is the question", proposed cultural production as ritual anthropophagy, the devouring and incorporation of the colonizer's culture by the colonized.

Combined, these yield a *chronophagic comparatism*: a method that reads works across media as digestions of a shared complex, attending to what each consumes, what it converts and what it leaves undigested. Forward in time, it reads adaptation and response as metabolism rather than derivation. *Metropolis*, *Snowpiercer* and *The Promised Neverland* are not copies of Wells but bodies built partly from his substance and partly from their own. Backward in time, it reads the Wellsian text as reorganizing its precursors in Borges's sense, so that Hesiod's mallow and Cicero's Saturn become legible as moments of a structure that only *The Time Machine* completes. The method claims influence only

where documentation permits; elsewhere its claims are claims of structural homology across the four registers of Table 1.1.

The method is scalar as well as comparative. Hans Blumenberg's *Lebenszeit und Weltzeit* (1986) described the incommensurability between the span of a human life and the span of the world as a founding scandal of modern consciousness, a gap that myth, history and science each attempt to bridge. *The Time Machine* is the first narrative to cross that gap bodily. Its means of crossing — the acceleration of world-time to the tempo of a life, until days flicker and centuries pass like weather — is the fictional correlate of the "scale framing" that Timothy Clark has diagnosed in environmental thought (*Ecocriticism on the Edge*, 2015). Every object examined in this thesis is therefore asked four questions, corresponding to the elements of the definition in section 1.3 and to its counter-term: what is consumed; by what; with what residue; and what, if anything, remains unconsumed.

The archive is deliberately heterogeneous. It includes epic, drama and lyric; the novel, the essay and the scientific paper; painting, engraving and architecture; opera, symphony and electronic music; film, television, manga and the video game; and philosophy from the Presocratics to speculative realism. The heterogeneity is not ornamental. A complex that operates across the social, the biological, the cosmic and the mediatic can be traced adequately only through the media in which its digestions have occurred. The video game, with its literal economies of resource and consumption and its capacity to make players choose between harvesting and rescuing, is among the most revealing of them.

1.10 The Architecture of the Argument

The thesis proceeds from the machine to the remainder, following the Traveller's journey outward and then reversing it.

Chapter Two, "The Crystal Saddle", examines the machine and its metaphysics. It reconstructs the Victorian fourth dimension from Bernhard Riemann's habilitation lecture of 1854 through Charles Howard Hinton's "What is the Fourth Dimension?" (1880), Edwin A. Abbott's *Flatland* (1884) and the address by Simon Newcomb that the Traveller cites, and follows its consequences into Hermann Minkowski's spacetime and J.M.E. McTaggart's "The Unreality of

Time", both of 1908. Against this spatialization of time it sets Henri Bergson's critique of the cinematographic intellect, and argues that the novel's description of day and night fusing into grey is an account of the flicker-fusion threshold on which cinema depends, written in the year of the first Lumière screenings and of Robert W. Paul's patent application for a time-travel entertainment. The chapter closes with the logic of personal and external time in David Lewis and Kurt Gödel and with the machine's afterlives from Borges to Christopher Nolan's *Interstellar* (2014), whose tesseract borrows Hinton's coinage.

Chapter Three, "Saturn's Arcadia", develops the entropic pastoral through Hesiod's two Kronoi, Plato's *Statesman*, the Victorian utopias of Morris, Edward Bellamy, W. H. Hudson and Edward Bulwer-Lytton, Kingsley's Doasyoulikes, Wilde's machine-slaves, Nietzsche's last men, Weber, Thorstein Veblen, Keynes and Derek Parfit's Repugnant Conclusion, before turning to Weena and the novel's speculative theory of sexual difference. It ends with the feedlot-Arcadias of Le Guin, Ishiguro, *Logan's Run* and *Soylent Green*.

Chapter Four, "The Nether World", reads the Morlocks through the Victorian sociology of the abyss; the underground service passages of Uppark, where Wells's mother was housekeeper; Milton's and Blake's Moloch; Marx's vampire and Lang's *Metropolis*, which Wells reviewed with contempt in 1927; Wagner's Nibelheim as Bernard Shaw allegorized it in *The Perfect Wagnerite* (1898); and the subterranean races of MacDonald and Arthur Machen. It analyses the Traveller's refusal of kinship with the Morlocks through Giorgio Agamben's "anthropological machine", follows the vertical city into *Snowpiercer*, *Parasite*, *Us*, *The Platform* and Moebius and Jodorowsky's *The Incal*, and concludes with the hidden labour of the automated economy.

Chapter Five, "Tempus Edax", is the theoretical centre of the thesis. It reconstructs the alimentary code of the human from Kronos, Goya and Dante's Ugolino through the sacrificial cuisine analysed by Jean-Pierre Vernant and Marcel Detienne, in which humanity is defined by its place between gods nourished by the savour of the altar and beasts that eat raw flesh. It reads the Eloi, the Traveller and the Morlocks as a redistribution of that code; inverts Franco Moretti's "dialectic of fear", in which Frankenstein's creature figured the proletariat and Dracula capital; and tests the theory against the ludic economies of *BioShock*, *Chrono Trigger*, *Horizon Zero Dawn*, *Attack on Titan* and *The Promised Neverland*.

Chapter Six, "The Palace of Green Porcelain", reads the ruined museum as the novel's theory of the archive: South Kensington in ruins; Macaulay's New Zealander sketching the ruins of St Paul's; Volney, Shelley and Horace Smith on the ruins of empire; Hubert Robert's imaginary ruin of the Louvre (1796); the Traveller's grief for his "seventeen papers upon physical optics"; the museum's camphor and matches as an arsenal; and the problem of communicating across deep time, from Walter Benjamin and Michel Foucault's heterotopias to nuclear semiotics, the Voyager record, the Long Now Foundation and William Basinski's *The Disintegration Loops*.

Chapter Seven, "The Terminal Beach", reads the novel's end through Clausius, Kelvin and Helmholtz; George Darwin's theory of tidal friction; Byron's "Darkness" and the tradition of the Last Man; the covenant scene of Genesis 15, whose "horror of great darkness" the Traveller echoes at the moment of eclipse; and the frozen floor of Dante's hell. It confronts the scene with the philosophy of extinction in Jean-François Lyotard, Quentin Meillassoux, Ray Brassier, Eugene Thacker and Claire Colebrook, and with its descendants in William Hope Hodgson, Olaf Stapledon, H.P. Lovecraft, Jack Vance, Gene Wolfe, Stanley Kubrick, *Dark Souls*, *Outer Wilds*, György Ligeti's *Poème symphonique* for one hundred metronomes and Haydn's "Farewell" Symphony.

Chapter Eight, "The Pocket and the Mallow", returns from the beach to the table and develops the ethics of the remainder: the grammar of "it remains"; Hesiod's half and the gift-devouring kings; Mauss and Derrida on the gift that gives time; Schrödinger's negentropy and the thermodynamics of care; Le Guin's carrier bag and the pressed flower; and the remainders of the machine age, from *WALL-E* and *The Road* to Nausicaä's refusal, at the close of Miyazaki's manga, of a purified future bought with the death of the living. It ends by setting the ethics of the remainder against Hans Jonas's heuristics of fear, Derek Parfit's population ethics and the longtermism of Nick Bostrom, Toby Ord and William MacAskill, and argues that the novel's final image unseats Kronos not by defeating time but by refusing to let time's appetite define what is valuable.





Quid est ergo tempus? Si nemo ex me quaerat, scio; si quaerenti explicare velim, nescio.

— Augustine, *Confessions* XI.14

Scientific people … know very well that Time is only a kind of Space.

— H. G. Wells, *The Time Machine* (1895)

Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union of the two will preserve an independent reality.

— Hermann Minkowski, "Space and Time" (1908), trans. W. Perrett and G. B. Jeffery (1923)

CHAPTER TWO

2

The Crystal Saddle

Geometry, Spatialized Time and the Metaphysics of the Machine

THE MACHINE AT THE centre of Wells's novel is two devices in one frame. In the lecture that opens the book it is the corollary of a geometrical theorem: if time is a dimension like the three of space, a vehicle may traverse it as a carriage traverses a road. In the narrative that follows it is a perceptual instrument whose chief effect is to dissolve the spatial order the theorem presupposed, as day and night palpitate into grey and the earth melts and flows beneath its rider. This chapter reconstructs both devices and argues that chronophagy begins at the seam between them. The geometry makes time available for consumption — as a landscape, a reserve and an itinerary — while the perception reveals what is consumed when it is, namely duration itself. The chapter traces the Victorian fourth dimension to its sources and its relativistic sequel (2.1–2.2), sets it against Bergson's critique of spatialized time (2.3), reads the journey as an account of the perceptual threshold on which cinema depends (2.4), analyses the logic of personal and external time (2.5) and the machine's character as a consumer's vehicle (2.6), and closes with the four metaphysical models into which the machine's afterlives divide (2.7).

2.1

"Time Is Only a Kind of Space": Hinton, Abbott, Newcomb and the Victorian Fourth Dimension

The Traveller's lecture is a compact proof in four steps, and its rhetoric is that of the demonstration-room rather than the séance. He begins by denying real existence to the abstractions of school geometry: a mathematical line, a plane, even an "instantaneous cube" cannot exist, because nothing real lacks duration. "Clearly," he proceeds, "any real body must have extension in four directions: it must have Length, Breadth, Thickness, and — Duration." He then diagnoses the error that conceals this truth: "There are really four dimensions, three which we call the three planes of Space, and a fourth, Time. There is, however, a tendency to draw an unreal distinction between the former three dimensions and the latter, because it happens that our consciousness moves intermittently in one direction along the latter from the beginning to the end of our lives." Two illustrations follow. Portraits of a man at eight, fifteen, seventeen and twenty-three are "sections, as it were, Three-Dimensional representations of his Four-Dimensioned being, which is a fixed and unalterable thing"; and a barometric chart records a line traced by the mercury in none of the dimensions of space, which must therefore lie "along the Time-Dimension". The conclusion is practical. If a balloonist can rise against gravity, why should an ingenious man not hope "to stop or accelerate his drift along the Time-Dimension, or even turn about and travel the other way"?

Each step has a lineage. The analytic treatment of time as a coordinate is older than the nineteenth century. Jean le Rond d'Alembert reported in the *Encyclopédie* article "Dimension" (1754) that a clever acquaintance regarded duration as a fourth dimension, and Joseph-Louis Lagrange remarked in the *Théorie des fonctions analytiques* (1797) that mechanics might be regarded as a geometry of four dimensions. Immanuel Kant, in his first publication, *Thoughts on the True Estimation of Living Forces* (1747), had speculated that the three-dimensionality of space might be contingent upon the inverse-square law and that spaces of other dimensions were possible. What the nineteenth century added was the higher-dimensional *imagination*: techniques of analogy by which three-dimensional beings might picture a fourth dimension by contemplating the predica-

ment of two-dimensional ones. Hermann von Helmholtz's lecture on the origin and meaning of geometrical axioms (1870), which imagined intelligent beings confined to the surface of a sphere, supplied the method; Edwin A. Abbott's *Flatland: A Romance of Many Dimensions* (1884) supplied the classic narrative. Abbott's narrator, A. Square, is visited by a Sphere that appears to him as a circle swelling and shrinking as it passes through his plane, and the Traveller's portrait argument is the same analogy rotated into time: as a Flatlander sees successive circular sections of a sphere, so we see successive three-dimensional sections of a four-dimensional person and call their succession a life. Abbott's satire also made dimensionality a figure of social rank — Flatland's women are lines, its labourers isosceles triangles, its priests circles — and in this respect, too, it anticipates a novel in which an added dimension discloses the true geometry of class.

The most direct source is Charles Howard Hinton, whose essay "What is the Fourth Dimension?" appeared in 1880 and again in 1884 as the first of his pamphlet series *Scientific Romances*, a title that would attach itself to Wells's early fiction. Hinton proposed that the changes we experience as time might be the passage of a higher-dimensional reality through our three-dimensional space, so that becoming, seen from a higher vantage, would be the successive sections of a body that simply is. In *A New Era of Thought* (1888) he went further, devising coloured cubes to train the imagination in four-dimensional visualization and coining the word "tesseract" for the four-dimensional analogue of the cube. Hinton's life reads like one of Wells's romances. The son of the surgeon and moral philosopher James Hinton and the husband of George Boole's daughter Mary Ellen, he was convicted of bigamy at the Old Bailey in 1886, emigrated to Japan and then to the United States, and at Princeton built a gunpowder-driven machine for pitching baseballs. Whether Wells read Hinton is undocumented, though he would recall in his autobiography that the fourth dimension had been debated among the students of the Normal School of Science in the 1880s; the conceptual dependence is unmistakable.

The Traveller names one authority only. "Professor Simon Newcomb was expounding this to the New York Mathematical Society only a month or so ago," he tells his guests. Newcomb, the most eminent astronomer in the United States, had delivered his presidential address on "Modern Mathematical Thought" to that society in December 1893, and its text, which entertained the conceivability of four-dimensional space, was printed in *Nature* in February 1894. The citation dates the lecture-scene to the early months of 1894, the season

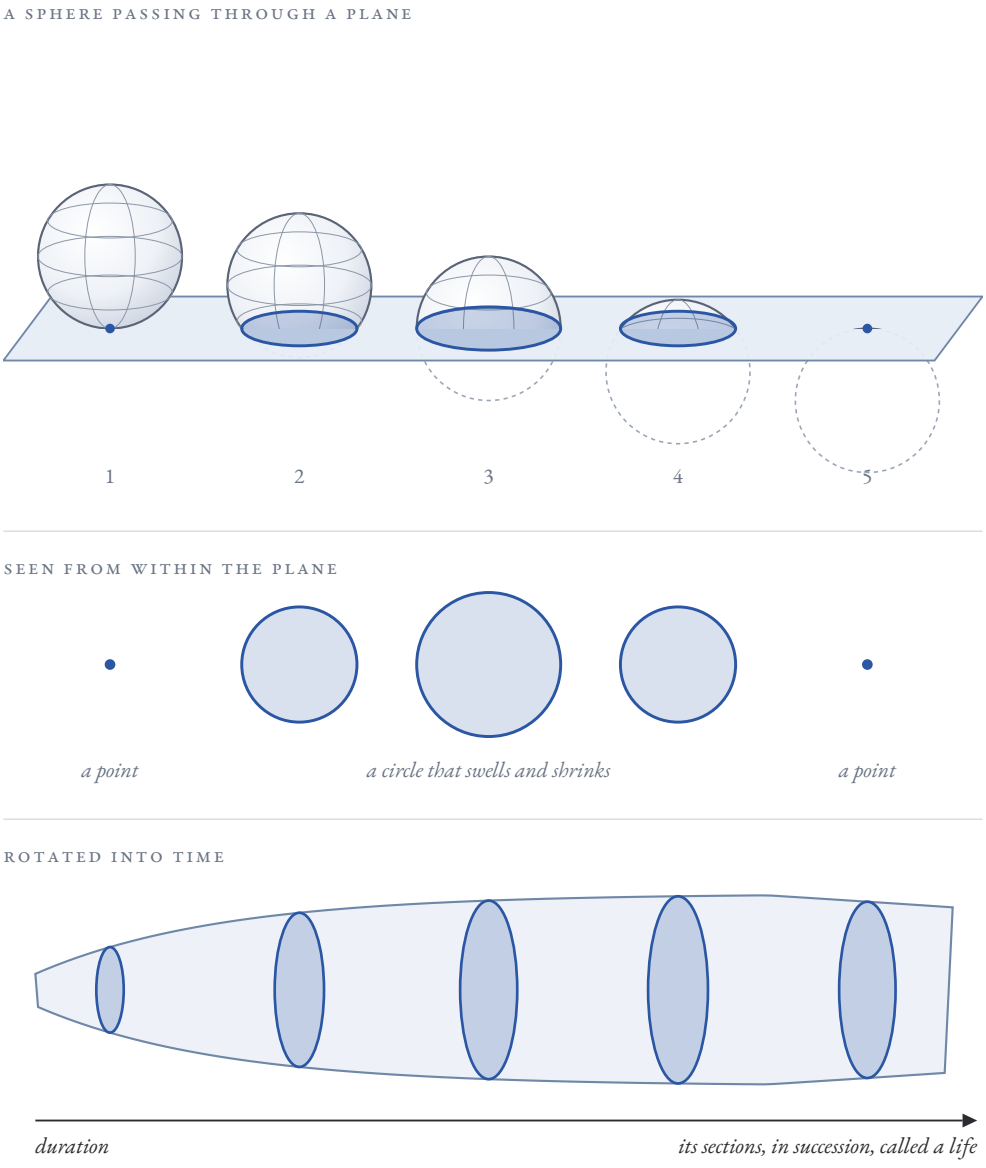


Figure 2.1 Flatland's analogy rotated into time. A sphere passing through a plane appears to the plane's inhabitants as a circle that swells and shrinks; as a Flatlander sees successive circular sections of a sphere, so we see successive three-dimensional sections of a four-dimensional person and call their succession a life.

in which Wells was drafting the *National Observer* sketches, and it performs a precise rhetorical function: it transfers the fourth dimension from the drawing-room of the spiritualists to the lecture-hall of the professionals.

The transfer mattered, because the fourth dimension of the 1870s and 1880s belonged as much to the séance as to the seminar. The Leipzig astrophysicist Johann Karl Friedrich Zöllner had invoked it to explain the feats of the American medium Henry Slade, who claimed to tie knots in cords whose ends were sealed, and Zöllner's *Transcendental Physics* (English translation 1880) made four-dimensional space the refuge of spirits. The Society for Psychical Research was founded in London in 1882. Wells's lecture-scene registers the contamination even as it repudiates it: the Medical Man's suspicion, after the model vanishes, is that the whole business is a trick "like that ghost you showed us last Christmas". The Traveller's achievement, and Wells's, is to secularize the fourth dimension — to move it from the medium's cabinet to the experimentalist's workshop, and to replace the spirit that passes through walls with a machine of nickel, ivory and rock crystal that a man may mount, steer and dismount. The secularization has a cost that the rest of this chapter traces. A dimension that can be ridden can also be owned, mapped, exploited and consumed.

2.2
From Riemann to Minkowski: The Block Universe before Relativity

The mathematical warrant for the Traveller's claim had been laid four decades earlier. Bernhard Riemann's habilitation lecture at Göttingen, "Über die Hypothesen, welche der Geometrie zu Grunde liegen" (delivered 1854, published 1868), generalized geometry to manifolds of any number of dimensions and treated the curvature of space as an empirical question. William Kingdon Clifford translated the lecture for *Nature* in 1873 and speculated, in his brief paper "On the Space-Theory of Matter" (1876), that matter and its motion might be nothing but variations in the curvature of space. The Victorian mathematical public thus possessed, before Wells, the components of a geometrized physics: dimensions beyond three, curvature as a physical property, and motion as geometry. What it lacked was the identification of the further dimension with time, which Hinton approached and Wells asserted.

Thirteen years after the novel, Hermann Minkowski made the assertion canonical. Addressing the Assembly of German Natural Scientists and Physicians at Cologne in September 1908, he recast Einstein's special relativity of 1905 as the geometry of a four-dimensional "world" in which each body's history is a "world-line", and announced that space and time by themselves were doomed to fade into shadows. Minkowski's world-line is the Traveller's "Four-Dimensioned being, which is a fixed and unalterable thing"; the portraits at eight, fifteen, seventeen and twenty-three are its spacelike sections. The congruence has made Wells a commonplace of popular histories of relativity, and it has an exact philosophical name. The Traveller's account of personal identity is perdurantism, the doctrine that persisting things are extended in time as they are in space and are composed of temporal parts, which W.V.O. Quine would defend in "Identity, Ostension, and Hypostasis" (1950) and which has been a leading position in analytic metaphysics since David Lewis. The Victorian lecture compresses a theory of persistence into an anecdote about a family photograph album.

The Traveller's metaphysics is not, however, quite the static block of later physics, and the difference matters. William James had used the image of "an iron block" to name the deterministic whole he rejected in "The Dilemma of Determinism" (1884); Hermann Weyl would give the eternalist picture its most lyrical statement in *Philosophie der Mathematik und Naturwissenschaft* (1927), writing that the objective world simply is, does not happen, and comes to life only for a consciousness crawling along the world-line of its body. Wells's lecture contains such a consciousness and gives it a curious adverb: our consciousness, the Traveller says, "moves intermittently in one direction" along the fourth dimension. A fixed manifold with a moving zone of awareness is what C.D. Broad would describe in *Scientific Thought* (1923) as a present that travels along the series of events like the beam of a policeman's lantern moving along the fronts of the houses in a street — the view philosophers now call the moving spotlight. The time machine is a device for detaching that spotlight from its ordinary drift and steering it. Its premise is that the world is a block; its promise is that consciousness need not be bound to the block's customary order of illumination.

That promise collides with the novel's ethics. If the four-dimensional world is "fixed and unalterable", the future the Traveller visits is not a warning but a fate, and the narrator's injunction to "live as though it were not so" is either incoherent or tragic. The novel does not resolve the antinomy; it inhabits it, in a manner

that recalls the solution Kant offered to his Third Antinomy and restated in the *Groundwork of the Metaphysics of Morals* (1785): theoretical reason may describe a world of exceptionless causal order, but a rational agent must act "under the idea of freedom". Pierre-Simon Laplace had imagined, in his *Essai philosophique sur les probabilités* (1814), an intellect acquainted with every force and position in nature, for which nothing would be uncertain and "l'avenir, comme le passé, serait présent à ses yeux" — the future, like the past, would be present to its eyes. The time machine is Laplace's intellect fitted with a saddle. Yet the Traveller possesses none of its omniscience. He forms four successive hypotheses about 802,701, each overturned by the next; he sees three moments of the remote future and understands none completely; and the narrator insists that the future remains "black and blank — is a vast ignorance, lit at a few casual places by the memory of his story". Epistemically, the block is never available as a whole. The *as though* of the conclusion is not a denial of the block but a recognition that one always lives inside it, illuminated in patches.

Physics has since carried the block in the direction the lecture pointed. In 1949 Kurt Gödel published a solution of Einstein's field equations describing a rotating universe that contains closed timelike curves, paths along which a sufficiently powerful rocket could return into its own past, and drew the philosophical moral in the volume of the Library of Living Philosophers devoted to Einstein: the objective lapse of time, a universal succession of nows, could not exist. Since Gödel, time travel has been a respectable topic of general relativity. Michael Morris, Kip Thorne and Ulvi Yurtsever described traversable wormholes as possible time machines in 1988, and Stephen Hawking proposed in 1992 a "chronology protection conjecture", according to which the laws of physics conspire to forbid them. Section 2.5 returns to the logic of such journeys. The point here is historical. The lecture delivered to a sceptical company in Richmond contained both the eternalist metaphysics that relativity would make orthodox and the paradox that physics has spent a century trying either to exploit or to forbid.

2.3
Bergson's Objection: Durée, the Cinematograph and the Spatialization of Time

In 1889, the year after "The Chronic Argonauts" appeared, Henri Bergson published his *Essai sur les données immédiates de la conscience*, which reached English readers only in 1910 as *Time and Free Will*. It contains the most searching objection ever lodged against the Traveller's lecture, though neither writer knew of the other. Lived time, for Bergson, is *durée*: a qualitative multiplicity in which successive states interpenetrate like the notes of a melody, each coloured by those before it, so that it cannot be divided without being altered. What physics and common sense call time is a homogeneous medium of juxtaposed and interchangeable moments, laid out like points on a line — a time, that is, contaminated by space. The spatialization of time is not an innocent convenience. It is, Bergson argues, the source of the false problem of free will, since a mind that pictures its past and future as positions on a line will picture its choices as already drawn there, waiting to be reached.

The Traveller's lecture is the purest literary statement of the position Bergson attacks. Time is "only a kind of Space"; a life is a line; the portraits are points along it; and the vehicle that moves along it presupposes that every moment already exists, laid out in advance like stations on a railway. Yet the narrative of the journey is something else entirely. The Traveller cannot describe his experience in the vocabulary of his lecture. "I am afraid I cannot convey the peculiar sensations of time travelling," he tells his guests. "They are excessively unpleasant. There is a feeling exactly like that one has upon a switchback — of a helpless headlong motion!" The landscape does not slide past like scenery from a carriage window; it liquefies, and "the whole surface of the earth seemed changed — melting and flowing under my eyes". Stopping brings the fear not of collision but of interpenetration, the risk that his atoms will be forced into "such intimate contact" with whatever occupies the place of his arrival that "a profound chemical reaction" will follow. The experience the lecture promised as the traversal of a line proves to be immersion in a flux of interpenetrating states. The novel stages, within its own two halves, the quarrel between the geometers and Bergson.

Bergson's later work made the quarrel explicitly technological. In *L'Évolution créatrice* (1907) he named the habitual operation of the intellect "the cinematographical mechanism of thought": like the new apparatus, the intellect takes immobile snapshots of a passing reality and restores the appearance of movement by running them through a uniform mechanism, so that becoming is reconstructed out of states and the becoming itself is lost. Wells's lecture had already given the apparatus an operator. Our consciousness, the Traveller says, moves along the fourth dimension "intermittently" — and intermittent movement is the technical name for the mechanism by which a projector holds each frame still before the lens and then advances the film, the device whose perfection made projected cinema possible in the year of the novel's publication. The Traveller's phenomenology of ordinary time is, in its key term, a description of projection.

The quarrel became a public confrontation in 1922. Bergson, who had answered relativity in *Durée et simultanéité*, met Einstein at the Société française de philosophie in Paris on 6 April, and Einstein replied that there was no time of the philosophers distinct from the time of the physicists — only a psychological time different from physical time. Jimena Canales has reconstructed the encounter as the moment at which the twentieth century divided time between physics and the humanities (*The Physicist and the Philosopher*, 2015). *The Time Machine* refuses that division before it was made. Its lecture is Einsteinian before Einstein and its journey Bergsonian before Bergson was read in English; and its account of the end of the world is a third thing, the thermodynamic arrow, which neither Einstein's time-symmetric equations nor Bergson's creative duration could accommodate.

Two further theories of experienced time clarify what the machine does to perception. In lectures delivered in 1905 and published in 1928, Edmund Husserl described the living present as a primal impression bordered by a "retention" of what has just elapsed and a "protention" of what is about to come, a structure that allows a melody to be heard as a melody rather than as a sequence of disconnected tones. The machine overruns that structure. When day and night alternate faster than retention can hold them, they cease to be perceived as alternation at all and fuse into "one continuous greyness": the phenomenological present loses its form. William James, borrowing E.R. Clay's term, called the experienced present "the specious present" and estimated its span in seconds (*The Principles of Psychology*, 1890). In the same chapter he recounted the speculation

of the embryologist Karl Ernst von Baer, who had asked in 1860 how nature would appear to beings whose life-processes ran a thousand times slower than ours: to them, the seasons would pass like quarters of an hour, annual plants would rise and fall like boiling springs, and the sun would cross the sky like a meteor trailing fire. The Traveller, accelerating, sees "the sun hopping swiftly across the sky", then "a streak of fire, a brilliant arch, in space", and "trees growing and changing like puffs of vapour, now brown, now green"; "the slowest snail that ever crawled dashed by too fast for me." Whether or not Wells had read James, he gives to a machine what von Baer gave to a hypothetical physiology: a change in the scale of the specious present, which converts the stable objects of ordinary perception into flows.

One more theory of time travel by perception inverts the novel's alimentary logic with precision. In the overture to *Du côté de chez Swann* (1913), Marcel Proust's narrator raises to his lips a spoonful of tea in which he has softened a piece of madeleine, and at the taste an entire lost past returns; in *Le Temps retrouvé* (1927) such moments yield "un peu de temps à l'état pur", a little time in its pure state. Proust's is the anti-Morlock meal. In Wells, eating consumes time and bodies; in Proust, eating restores them. The Bergsonian tradition that Proust extended and Deleuze would systematize thus supplies the first counter-model to chronophagy in the modern novel: an alimentary anamnesis in which the mouth is an organ of return rather than of destruction. Chapter Eight will set Weena's flowers beside the madeleine as the two great remainders of the modern fiction of time.

2.1
Flicker Fusion: Robert W. Paul's Patent, Time-Lapse and the Crystal-Image

The descriptive core of the journey is an account of the perceptual threshold on which cinema depends. The Traveller pulls the lever and "the night came like the turning out of a lamp, and in another moment came to-morrow". His housekeeper, Mrs Watchett, crossing the laboratory, seems "to shoot across the room like a rocket". As he accelerates, "the twinkling succession of darkness and light was excessively painful to the eye", until "the palpitation of night and day merged into one continuous greyness", the sun became "a streak of fire, a bril-

liant arch, in space" and the moon "a fainter fluctuating band". The sequence passes through three regimes — discrete alternation, painful flicker, fusion — and it is a precise phenomenology of what psychophysics calls the critical flicker-fusion threshold, the rate above which an intermittent light is perceived as steady, and of the related illusions of apparent motion on which projected moving pictures depend. Joseph Plateau had studied the persistence of retinal impressions in 1829 and built his phenakistiscope in 1832; Eadweard Muybridge had decomposed the horse's gallop in 1878; Étienne-Jules Marey had named his own method, with a precision this thesis cannot improve upon, *chronophotographie*. The Traveller's journey runs the chronophotographic analysis in reverse. Instead of decomposing a movement into instants, it fuses instants until the world becomes a smear of light.

The coincidence of dates is exact. *The Time Machine* appeared in book form in May 1895. In October of the same year the London instrument-maker Robert W. Paul, who had already built kinetoscopes and would shortly build the projector that founded the British film industry, lodged a provisional patent for "a novel form of exhibition or entertainment", conceived in conversation with Wells: spectators seated on platforms that could be rocked and moved through a darkened hall would watch projected scenes, still and moving, and seem to travel into the past or the future. On 28 December the Lumière brothers held their first public paid screening at the Grand Café in Paris. Terry Ramsaye, who reprinted Paul's specification in *A Million and One Nights* (1926), drew the conclusion that the medium and the time-travel story had been born together. The patent was never completed, but its idea was realized piecemeal across the following century: in Hale's Tours (from 1905), whose audiences sat in a mock railway carriage that swayed while travel films ran beyond its windows; in the motion simulator; and, with a fidelity that would have gratified Paul, in Universal Studios' *Back to the Future: The Ride* (1991), which seated its riders in a time-travelling car before a domed screen.

The Traveller himself is legible as a showman of optical entertainments. The Medical Man's suspicion of a trick "like that ghost you showed us last Christmas" places him in the tradition of the phantasmagoria that Étienne-Gaspard Robertson staged in Paris in the 1790s and of the stage illusion of a ghost reflected in angled glass that John Henry Pepper popularized at the Royal Polytechnic Institution in 1862. The Psychologist explains the model's invisibility in the vocabulary of the new psychophysics, as "presentation below the threshold ...

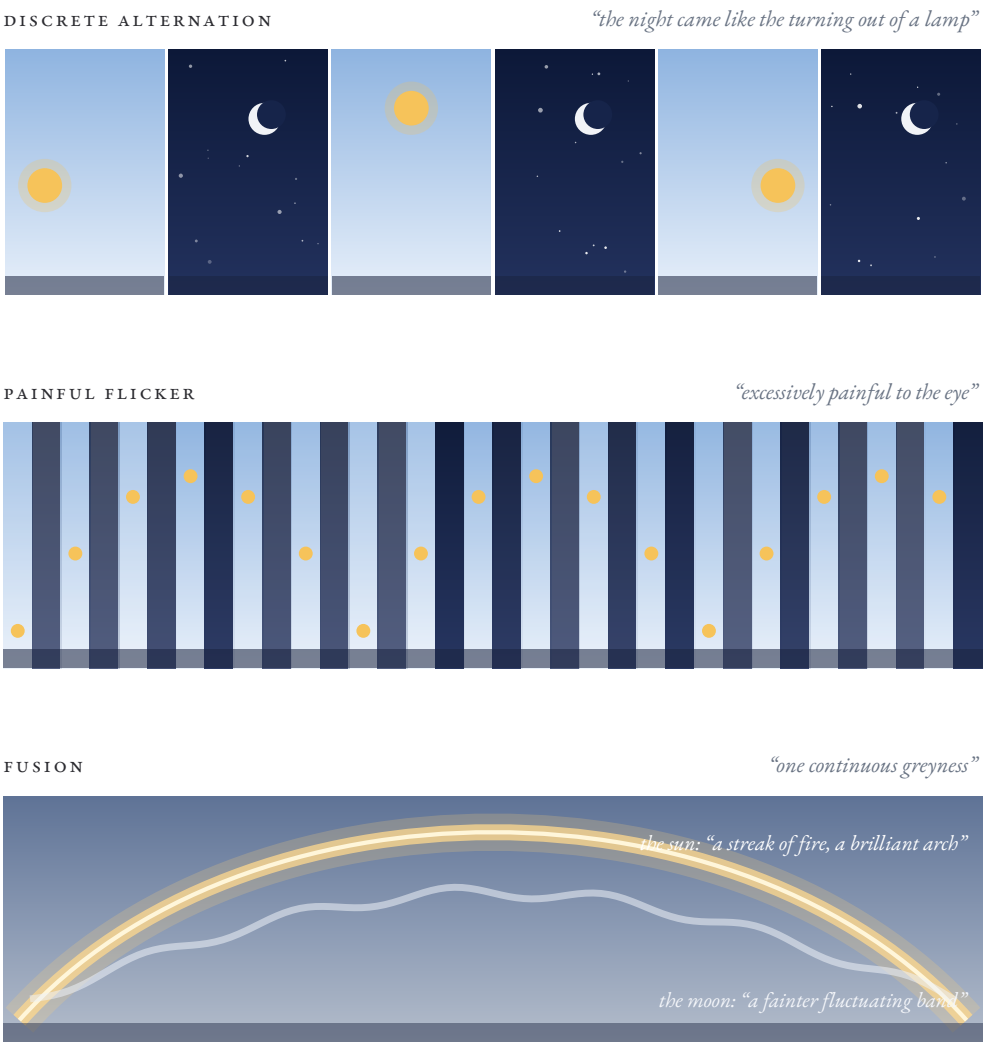


Figure 2.2 The three regimes of the Traveller's acceleration: discrete alternation, painful flicker and fusion, when "the palpitation of night and day merged into one continuous greyness", the sun became "a streak of fire, a brilliant arch, in space" and the moon "a fainter fluctuating band".

diluted presentation": a thing that passes through time faster than we do makes an impression too faint to register, like the spoke of a spinning wheel or a bullet in flight. The lecture-scene is set among the instruments and explanations that were about to converge into cinema, and the journey that follows is the first narrative written, as it were, from inside the projector.

The century of cinema developed the Traveller's vision into a genre of its own. Time-lapse cinematography found its first popular subject in plants: F. Percy Smith's *The Birth of a Flower* (1910), made for Charles Urban, showed blossoms unfolding at an impossible tempo, and its successors in nature film turned vegetal life into choreography. The Traveller had seen precisely this, trees "growing and changing like puffs of vapour", and it is worth noting that the first object to which the machinic acceleration of vision was publicly applied was the flower, the object with which *The Time Machine* ends. Godfrey Reggio's *Koyaanisqatsi* (1982), photographed by Ron Fricke to a score by Philip Glass, turned time-lapse upon the city and found in its accelerated flows of traffic, crowds and production lines a planetary metabolism, cutting between the mechanized packaging of food and commuters borne on escalators as though both were moments of one digestion. Reggio's Hopi title is glossed as "life out of balance": it is the Traveller's accelerated vision turned upon the present, and what it discloses is chronophagy in real time. At the opposite pole stands Christian Marclay's *The Clock* (2010), a twenty-four-hour montage of film clips showing clocks and watches, synchronized to the local time of its exhibition. Where the time machine dissociates the spectator's time from the world's, *The Clock* fuses them absolutely: it is cinema as the abolition of time travel.

The machine's material composition carries the cinematic reading to a final, almost uncanny point. Its parts, the narrator observes of the model, were of nickel and ivory, and some "had certainly been filed or sawn out of rock crystal"; on the Traveller's return the full-sized machine is "a thing of brass, ebony, ivory, and translucent glimmering quartz". Rock crystal is quartz, and quartz is the material in which modern time is kept. Jacques and Pierre Curie discovered its piezoelectricity in 1880; Walter Cady built the first quartz-crystal oscillator in 1921; Warren Marrison and J. W. Horton completed the first quartz clock at Bell Telephone Laboratories in 1927; and the quartz wristwatch, from Seiko's Astron of 1969, put a crystal chronometer on every wrist. Wells could not have intended the prolepsis, but a comparatist may register it: the first fictional time machine is a crystal machine, and the device that would standardize the time of the twenti-

eth century is a crystal resonating under electric charge. Gilles Deleuze, in *Cinéma 2: L'Image-temps* (1985), called the image in which actual and virtual coalesce, and in which time is seen splitting into a present that passes and a past that is preserved, the *crystal-image*, and titled his account of it "The Crystals of Time". The saddle of Wells's machine is mounted, quite literally, on such crystals. What the Traveller sees from it — a present that will not hold and a world that persists only as residue — is the crystal-image avant la lettre: time made visible in the act of splitting.

2.5
Personal Time and External Time: Lewis, Gödel and the Logic of the Journey

The most influential philosophical analysis of time travel, David Lewis's "The Paradoxes of Time Travel" (*American Philosophical Quarterly*, 1976), begins from a distinction Wells dramatizes without naming it. Time travel, Lewis argued, involves a discrepancy between the traveller's *personal time* — the time registered by his watch, his ageing, his hunger and his memories — and *external time*, the time of the world. A journey of an hour in personal time may span a millennium of external time, forwards or backwards. Lewis's purpose was to show that such travel is logically consistent provided nothing the traveller does contradicts what has already happened. His Tim, who journeys back to kill his grandfather, *can* do so in the sense that he has the rifle, the skill and the opportunity, and *cannot* in the sense that his doing so is not compossible with the facts; some trivial circumstance will intervene.

Wells's novel is built on exactly this discrepancy, and it equips the Traveller with two clocks to measure it. External time is recorded by the machine's dials, which register days, thousands of days, millions of days and thousands of millions. Personal time is recorded by the Traveller's body: his famished appetite, his bruises and limp, the hair the narrator thinks greyer, and the eight days he claims to have lived between ten o'clock in the morning and the dinner-hour of the same Thursday. The two clocks are kept in deliberate tension. When he halts in 802,701, his dials tell him the year and nothing of what has happened; his body, hungry and afraid, tells him everything that matters about the world he has entered. The dinner-table frame is where the two times are reconciled: the guests,

for whom the whole expedition has occupied less than a day, receive the report of a man whose personal days contain eight hundred millennia.

Lewis's analysis also clarifies an asymmetry that Wells observes and physics has since ratified. Travel into the future requires only that personal time run slower than external time, and special relativity permits exactly that. Paul Langevin's thought-experiment of 1911, the voyage later known as the twin paradox, showed that a traveller moving close to the speed of light would return to find the earth centuries older; Christopher Nolan's *Interstellar* (2014) dramatizes the gravitational version on a planet where an hour costs seven earthly years. Travel into the past is another matter. It requires closed timelike curves of the kind Gödel found in 1949 and Morris, Thorne and Yurtsever engineered on paper in 1988, and Hawking's conjecture that nature forbids them has not been refuted. The forward journey of *The Time Machine* has become physics; the return remains fiction. Yet the novel depends on the return rather than the outward journey, because the return alone makes testimony possible.

The dependence discloses the narrative structure of testimony as such. Gérard Genette distinguished in *Discours du récit* (1972) between the time of the story and the time of its narration, and observed that the classic narrator speaks from after the events he recounts — from their future. Every retrospective narrative is in this sense a return from the future of its events to an audience in their past. *The Time Machine* literalizes the narrating instance: its narrator has been to the future of his story and come back to a dinner-table to tell it. Paul Ricoeur argued in *Temps et récit* (1983–85) that narrative is the means by which human beings mediate between the phenomenological time of the soul and the cosmological time of the world; Frank Kermode argued in *The Sense of an Ending* (1967) that fiction organizes mere successive time, *chronos*, into significant time, *kairos*, by binding it to an end, as the mind hears the tick of a clock as "tick-tock". Wells's novel tests both theories at their limit. It narrates a cosmological time so vast that it annihilates the soul's, and it supplies the ultimate "tock", the end of the world, only to find that the end confers no meaning. What meaning there is must be carried back to the beginning, in a pocket.

The logic of the journey yields a last consideration, concerning evidence. The Traveller returns with no photographs, no instruments and no specimens except the two flowers, and the only empirical test performed in the frame is botanical. The Medical Man takes them to the lamp, examines them and pronounces: "The gynaeceum's odd." The proof of the future lies in the female reproductive organ

of a flower, the part on which Linnaean botany had built its orders; futurity is legible in the anatomy of generation. That the Traveller's single specimen is a gift rather than a trophy, and that its evidential force lies in its sexuality, will matter in Chapter Eight. Here it marks the limit of Lewis's distinction. External time leaves nothing on the dials that can be shown at table, and personal time nothing on the body that can be proved. Only an object belonging to both — a flower grown in external time and carried in personal time — can cross the threshold of the frame.

2.6
The Saddle and the Tourist: Bicycles, Thomas Cook and the Consumption of Duration

The machine is a personal vehicle with a saddle, and its form belongs to the decade of the bicycle. J.K. Starley's Rover safety bicycle of 1885 and J.B. Dunlop's pneumatic tyre of 1888 had made cycling practicable for the lower-middle classes, and the boom of the mid-1890s turned it into a mass enthusiasm. Wells was among the enthusiasts. His comic novel *The Wheels of Chance* (1896) sends a draper's assistant, Hoopdriver, on a cycling holiday through the south of England, and its pleasures are those of a new mobility available to a class that had never before travelled for its own sake. The great vehicles of the extraordinary voyage before Wells were collective: Verne's *Nautilus* and his lunar projectile carried crews, and Gaspar's *anacronópete* was a flying ark. Wells's machine carries one rider. The genre's founding vehicle is private, and its rider is an individual consumer of experience.

The saddle carries a further irony. The Traveller rides a mechanical horse into a world from which, he discovers, "horses, cattle, sheep, dogs" have vanished; the vestigial seat of the equestrian age is fitted to a vehicle that will outlast the animal it imitates. In 1895 the horse was still the engine of the city — the omnibus, the cab, the dray — and the cab-horse was, in General Booth's charter, the standard of minimum welfare to which the submerged tenth might aspire. A London that ran on horses, and fed and stabled many of them better than it fed and housed many of its poor, provides the template for a vehicle that runs on nothing visible at all.

What the vehicle makes possible is tourism in time. Thomas Cook had organized his first excursion, a temperance outing by rail from Leicester to Loughborough, in 1841; by the 1870s his firm conducted parties round the world and up the Nile, past temples and sphinxes that its clients had come to consume as spectacle. The Traveller's itinerary in 802,701 follows the pattern with comic fidelity. He arrives, ascends a hill for the view, acquires a local guide in Weena, tours the monuments, visits the museum, collects a souvenir and, on his second departure, takes a camera — the instrument with which George Eastman's Kodak (1888), sold under the slogan "You press the button, we do the rest", had made every tourist a documentarian. Dean MacCannell's *The Tourist* (1976) analysed sightseeing as a ritual of modernity in which the past and the elsewhere are staged for consumption, and Guy Debord's *La Société du spectacle* (1967) gave the operation its name: the time of the spectacle is *temps consommable*, consumable time, a pseudo-cyclical time of packaged moments whose model is the holiday. The Traveller is the first tourist of consumable time. He does not inhabit the future; he visits it, samples it and returns to recount it before the Editor and the Journalist.

The tourist's vehicle is made of the materials of empire. The machine is "a thing of brass, ebony, ivory, and translucent glimmering quartz", with fittings of nickel. In the 1890s ivory meant above all the Congo Free State of Leopold II, where the trade Joseph Conrad would anatomize in *Heart of Darkness* was at its height; ebony came from the forests of West Africa and Ceylon; and the world's nickel was then mined chiefly in New Caledonia, a French penal colony, by convicts and indentured labourers. The vehicle that carries the bourgeois experimentalist through time is assembled from the tusks of slaughtered elephants and the labour of the unfree. The commodity, Marx had written, conceals the social relations that produced it beneath the appearance of a relation between things; here the concealment is literal, for the machine glitters. The mediatic register of chronophagy rests upon the oikonomic. The Traveller consumes time as spectacle from a seat made of consumed bodies, and the novel's first image of the machine — the model that vanishes "as an eddy of faintly glittering brass and ivory" — is an image of imperial matter disappearing into the future it will purchase.

The politics of standard time make the same point from the other side. In 1884 the International Meridian Conference at Washington fixed the prime meridian at Greenwich and laid the foundations of world time; British railways

had kept Greenwich time since the 1840s, and Parliament made it the legal time of Great Britain in 1880. On 15 February 1894, in the months in which Wells was drafting the *National Observer* sketches, the French anarchist Martial Bourdin was killed by his own bomb in Greenwich Park, below the Royal Observatory — the episode Conrad would refashion in *The Secret Agent* (1907) as a plot against "the first meridian", the fetish of a bourgeoisie that worships science. The anarchist attacks public time; the Traveller escapes it. Both responses register a single fact: by the 1890s time had become an infrastructure, administered from an observatory, sold by the railway company and the excursion agent, and owned. The time machine is the private appropriation of that infrastructure, the bourgeois dream of owning one's time outright. That the dream discloses a future in which time has eaten everything is the novel's judgement upon it.

2.7
Afterlives of the Machine: Borges, Vonnegut, Marker, Moore, Chiang, Nolan, Carruth

The machine's afterlives are best sorted not by medium or decade but by the metaphysics of time they adopt, since each metaphysics redistributes chronophagy differently. Four models recur: the block, the branching garden, the loop and the arrow.

The block, which is Wells's lecture taken at its word, promises that nothing is ever consumed, because every moment exists for ever. Its best-known fictional statement is Kurt Vonnegut's *Slaughterhouse-Five* (1969), whose Billy Pilgrim has "come unstuck in time" and learns from the Tralfamadorians, who see in four dimensions, that the dead are merely in bad condition at one moment and alive at many others. The consolation is explicitly metaphysical and implicitly quietist: a novel about the fire-bombing of Dresden recommends attention to the pleasant moments. It also contains a joke that is an exact image of chronophagy. The Tralfamadorians know how the universe ends: a test pilot of theirs, experimenting with a new fuel for flying saucers, presses a starter button and the whole universe disappears. They do not prevent it, because the moment is structured that way. The block can absorb even the consumption of the universe by a fuel. Alan Moore and Dave Gibbons gave the block its most austere figure in *Watchmen* (1986–87), whose Dr Manhattan, a watchmaker's son, perceives all his moments

at once and narrates past and future alike in the present tense. Ted Chiang gave it its most moving figure in "Story of Your Life" (1998), filmed by Denis Villeneuve as *Arrival* (2016), in which a linguist who learns an alien script written all at once comes to experience her own life simultaneously and chooses, knowing its end, to have the daughter who will die. Chiang also gave the block its most frightening form, in "What's Expected of Us" (2005), a story of barely a page in which a device whose light flashes one second before its button is pressed proves to its users that their choices are fixed, and many of them simply stop acting. Wells's narrator had answered the story a century in advance: knowing the end, "it remains for us to live as though it were not so."

The branching garden is Borges's. In "El jardín de senderos que se bifurcan" (1941) the labyrinthine novel of Ts'ui Pên turns out to be a theory of time as an infinite, growing net of diverging, converging and parallel times in which every alternative is realized; Hugh Everett's relative-state interpretation of quantum mechanics (1957) would supply the physics. The garden abolishes chronophagy by multiplication: nothing is lost because everything also happens. Borges, however, also wrote the most exact confession of chronophagy in modern prose. "Nueva refutación del tiempo" (1947) assembles idealist arguments against the reality of time only to concede them in its final paragraph, where time is a river that carries the writer away and a tiger that tears him apart — and "es un fuego que me consume, pero yo soy el fuego". In that clause the devourer and the devoured are one. Borges reaches the limit Wells approached in the Morlocks: the eater of humanity is humanity.

The loop is the form of recurrence, and its emblem is the ouroboros, the serpent that eats its own tail. Chris Marker's *La Jetée* (1962) is its masterpiece, and it is also a meditation on the cinematographic mechanism of thought, being composed almost entirely of still photographs. After a third world war the survivors live in the galleries beneath the Palais de Chaillot — here the underground belongs to the experimenters — and they send a prisoner haunted by an image from his childhood into the past and then into the future, whose inhabitants give him the means to set the industry of his own era running again. He chooses instead to return to the woman of the image, and dies on the jetty at Orly, understanding that the death he witnessed as a child was his own. In the whole film there is a single moving shot: a few seconds in which the woman, asleep, opens her eyes. Against a technology that fixes time into stills, the film opens one breath of *durée*, and the breath is a woman waking. Terry Gilliam reworked the

story as *12 Monkeys* (1995); Shane Carruth's *Primer* (2004) reinvented it as an engineering problem. Its inventors build their machine in a suburban garage, the Richmond laboratory's descendant, and use it, as the Very Young Man proposed, to make money on the stock market, until the loops multiply beyond the point at which anyone can tell which version of a man is speaking. In the German series *Dark* (2017–20) the loop becomes cosmic, a knot of recurrences around a cave in a small town that can be undone only by unmaking its origin.

The arrow is Wells's own model, and its afterlives are those that keep the second law in view. Nolan's *Interstellar* sets its time-dilated voyage against a dying earth choked by blight and dust, and resolves it inside a tesseract — Hinton's word — where a father can reach his daughter only by perturbing the second hand of the watch he once gave her: a gift become a channel across time. Nolan's *Tenet* (2020) makes the arrow its subject. Its technology "inverts" the entropy of objects and persons so that they travel backwards in time, and its antagonists are humanity's descendants, who, inheriting a world their ancestors ruined, seek to reverse the entropy of the whole world and annihilate the past. *Tenet* is the most explicit chronophagy in recent cinema: the future seeks to devour the generations that devoured it. Wells's Morlocks eat their contemporaries; Nolan's posterity would eat its ancestors. Robert Wilson and Philip Glass's opera *Einstein on the Beach* (1976), whose title recalls the last shore of Nevil Shute's *On the Beach* (1957), names the destination towards which the arrow points; Chapter Seven follows it there.

The four models can now be read as four answers to the problem Chapter One called the remainder. The block guarantees remainders absolutely, since every moment persists, and thereby makes them worthless, since nothing can be lost. The garden guarantees them by profusion and cheapens them in the same way. The loop recycles them until they are exhausted. The arrow alone makes remainders precious, because it alone makes them rare. Wells's originality lies in having combined the first model and the last in a single book: a lecture that asserts the block and a journey that suffers the arrow. The flowers lie at the seam between them. In the block they would be one moment among infinitely many, as permanent as the Morlocks' larder; in the arrow they are what survives. The ethics of the remainder is what the block universe becomes when it is lived inside

the arrow: a decision to treat what persists as though it might be lost, and what perishes as though it mattered for ever.





When they died, it was as though they were overcome with sleep, and they had all good things; for the fruitful earth unforced bare them fruit abundantly and without stint.

— Hesiod, *Works and Days* 116–118, trans. Hugh G. Evelyn-White (1914)

This has ever been the fate of energy in security; it takes to art and to eroticism, and then come languor and decay.

— H. G. Wells, *The Time Machine* (1895)

„Wir haben das Glück erfunden“ — sagen die letzten Menschen und blinzeln.

— Friedrich Nietzsche, *Also sprach Zarathustra*, “Vorrede” §5 (1883)

CHAPTER THREE

3

Saturn's Arcadia

The Golden Age, the Entropic Pastoral and the Eloi

THE WORLD OF 802,701 is a Golden Age that arrives at the end of history, and this chapter argues that its every amenity is a form of equilibrium sustained by a hidden act of consumption. Chapter One proposed the formula *every Arcadia is a feedlot seen from above*; here the formula is tested against the classical sources of the Golden Age (3.1–3.2), the Victorian utopias Wells inverted (3.3), the aesthetic movement whose ethos the Eloi embody (3.4), the theorists of the last men (3.5), the novel's speculative biology of sex and its single figure of care, Weena (3.6), and the modern fictions that have rediscovered the feedlot beneath the pastoral (3.7). Two new claims emerge. The first is that the Eloi are figured, from their first appearance, as *consumptive* — beautiful in the manner of the wasting disease the century called consumption — so that the aesthetics of the Golden Age is from the outset an aesthetics of being eaten. The second is that the novel splits care in two, giving its practical half to the Morlocks, who clothe and breed the Eloi, and its affective half to Weena, who can give nothing but flowers. The flower, on this reading, is care stripped of every function but regard.

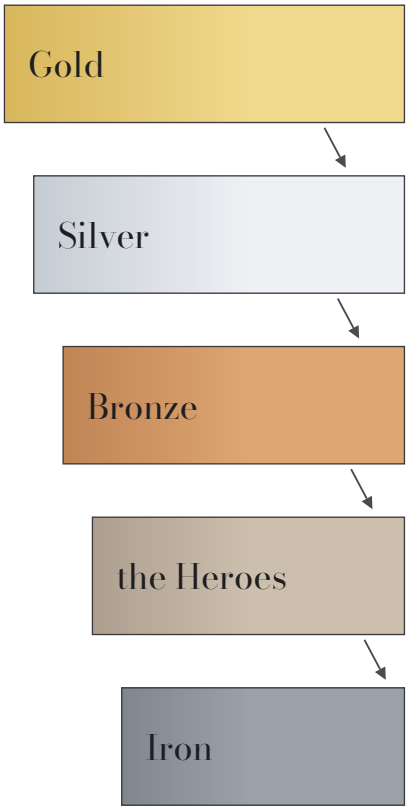
3.1
Hesiod's Two Kronoi

Greek religion never reconciled its two images of Kronos, and the irreconciliation was itself ritualized. In *Works and Days* he is the king of the golden race, under whom human beings "lived like gods without sorrow of heart, remote and free from toil and grief" (112–113, trans. Evelyn-White); in the *Theogony* he is the father who swallows his children. The Athenians celebrated his festival, the Kronia, at the season of harvest, and on that day, according to testimony preserved by Macrobius, heads of households ate together with the slaves who had shared their labour in the fields, as Romans would at the Saturnalia, in memory of the age before servitude. H.S. Versnel's study of Kronos and the Kronia (1993) showed that the god of reversal presides over both utopia and atrocity because both are suspensions of ordinary order: the festival restores for a day the world without masters, and the myth recalls the world in which a father ate his heirs. Pindar placed the Isles of the Blessed by "the tower of Kronos" (*Olympian* 2.70), so that the god who ruled the first age also governed the last, the paradise of the dead. Kronos is the lord of the beginning and the end of time, and of the feast.

Wells's Eloi are drawn, feature by feature, from Hesiod's first two races. Like the golden race they live without toil among fruits that the earth seems to provide unbidden — the Traveller makes his staple of "a floury thing in a three-sided husk" — and they die, as the golden race died, in their sleep, taken from the communal halls in which they huddle together after dark. Like the silver race they are prolonged children. The Traveller, who had "always anticipated that the people of the year Eight Hundred and Two Thousand odd would be incredibly in front of us in knowledge, art, everything", is disabused when one of them asks "if I had come from the sun in a thunderstorm", a question that places its speaker "on the intellectual level of one of our five-year-old children". Hesiod's silver children were reared for a hundred years at their mothers' sides, "a great simpleton" each; the Eloi have no mothers in particular, since the household has vanished with the cottage, but they remain children for life. Even their speech is a child's. Their language, the Traveller finds, is "excessively simple — almost exclusively composed of concrete substantives and verbs", with "few, if any, abstract terms, or little use of figurative language".

HESIOD

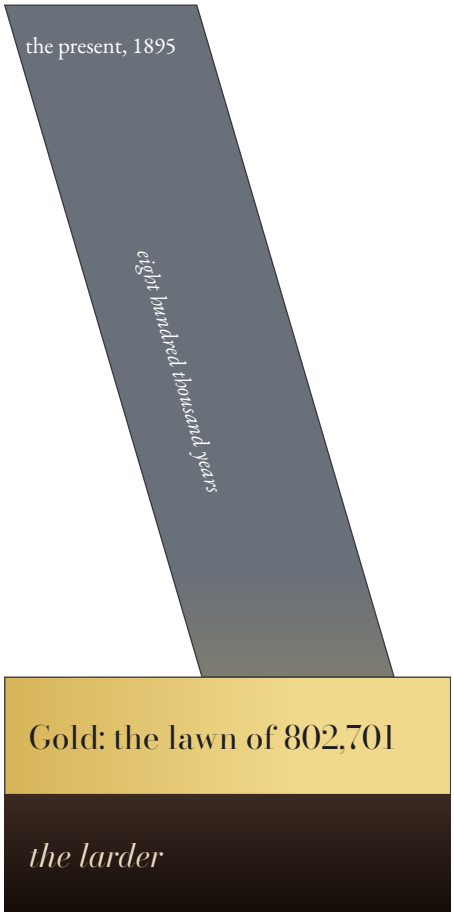
Works and Days



from gold through silver, bronze and the heroes to iron

WELLS

The Time Machine



Kronos the king, seen from the lawn; Kronos the devourer, seen from below

Hesiod's ages descend to iron; Wells's descend into gold

Figure 3.1 Two descents. Hesiod's ages descend from gold through silver, bronze and the heroes to iron; Wells's descend *into* gold. Beneath the golden age of 802,701 lies the larder: the benevolent king and the devourer are one regime, seen from the lawn and from below.

That last observation deserves more weight than criticism has given it. A society produced by the speciation of metaphor has no use for metaphor. The Eloi do not say that they are fed upon by those below; they are fed upon. Figure has become fact, and the faculty of figuration has atrophied with every other faculty for which there is no need. The Golden Age of 802,701 is post-rhetorical: it is the world in which the metaphors of the nineteenth century have been fulfilled so completely that no one remains who could speak them.

The Traveller's first sight of an Eloi supplies the chapter's governing image. The creature is "perhaps four feet high", clad in a purple tunic, "a very beautiful and graceful creature, but indescribably frail", and "his flushed face reminded me of the more beautiful kind of consumptive — that hectic beauty of which we used to hear so much." Consumption, the century's name for pulmonary tuberculosis, was the disease the nineteenth century aestheticized above all others: the wasting that refined, the flush that signified passion, the beauty of Keats and of the Lady of the Camellias. Susan Sontag analysed the mythology in *Illness as Metaphor* (1978), showing how tuberculosis was imagined as a disease of sensibility, of the artist and the aristocrat, a fire that consumed the body from within. Wells, who had suffered pulmonary haemorrhages in 1887 and again in 1893 and believed himself tubercular, gives the inhabitants of the Golden Age the complexion of the disease that bore the name of this thesis's master-figure. The Eloi look consumptive because they are being consumed. Their beauty is the visible surface of a wasting whose agent is invisible, and the novel's first description of the future contains, in a pun the century had allowed to die, its last secret.

The Traveller's own theory of the Golden Age is phrased, appropriately, as a harvest. Contemplating the ruined palaces at sunset, he recalls the slow improvements of the present — "now a new and better peach, now a seedless grape, now a sweeter and larger flower, now a more convenient breed of cattle" — and concludes that the work of amelioration "had gone steadily on to a climax": "And the harvest was what I saw!" The sentence is truer than he knows. The harvest of human improvement is a population bred for convenience, and the more convenient breed of cattle that closes his list of achievements is the one the Morlocks will prove to have produced. Kronos was a harvest-god before he was a god of time. His sickle reaped grain before it castrated Ouranos or became the scythe of Father Time, and the Kronia was a festival of the gathered crop. In 802,701 the two Kronoi meet in the harvest. The benevolent Kronos is the harvest as it appears to those who are fed by it; the devouring Kronos is the same harvest as it

appears to the crop. The Eloi experience the Golden Age as a crop experiences a gardener's care.

3.2
The Nurslings of Kronos: Plato's *Statesman* and the Problem of Leisure

Plato's myth of the age of Kronos begins, in the *Statesman*, with a sign in the sky and a crime at table. The Eleatic Stranger introduces it by recalling the portent that accompanied the quarrel of Atreus and Thyestes over the golden lamb, when the god reversed the rising and setting of the sun (268e–269a). The wider tradition attached that portent to the most terrible banquet in Greek myth: Atreus, in revenge, served Thyestes the flesh of his own sons, and in Seneca's *Thyestes* the sun turns back from the feast and day deserts the sky. Plato's cosmology of reversed time thus begins at the edge of the Thyestean dinner, the archetype of a father fed upon his own descendants. *The Time Machine* repeats the configuration with a precision that need not imply design. A machine capable of reversing time carries a man to a world where descendants are eaten, and returns him to a dinner at which his hosts, having just eaten mutton, are served the story of their own posterity's consumption. The Richmond dinner is a Thyestean banquet in the form of a narrative.

The age the Stranger then describes is the age of Kronos in its fullness. The cosmos turned in the reverse direction; the god himself tended human beings as a herdsman tends his flock, while lesser divinities shepherded each kind of creature; food sprang from the trees without cultivation; there were no constitutions and no possession of wives and children; and — the detail is decisive — "nothing was wild, nor was there any eating of one another, nor war, nor faction at all" (271e). Hesiod had already made mutual devouring the mark of creatures without justice. In *Works and Days* the son of Kronos ordains that fish, beasts and birds should eat one another, "for right is not in them", but gives justice to humankind (276–279, trans. Evelyn-White): the god who escaped his father's stomach founds the human order on the prohibition of his father's appetite. Wells keeps every feature of the Platonic Golden Age except this one, and conceals it. 802,701 is a herded humanity without households or politics, fed from the trees and at peace, and founded out of sight on precisely the mutual eating

that Plato excluded and Hesiod assigned to the beasts. The Eloi are the nurslings of Kronos, and Kronos is below them.

The Stranger's closing question, whether the nurslings of Kronos were happier than the men of the age of Zeus, turns on what they did with their leisure. If, having abundant time and the power of conversing with beasts as well as men, they used these gifts for philosophy, they were incomparably happier; if they ate and drank their fill and told stories to one another and to the animals, the answer is easy (272b–d). The Eloi fail the test more completely than Plato imagined possible: they eat and play, but they do not even tell stories, for their language has lost its figures. The irony of the frame is that the Richmond dinner satisfies the Stranger's second description exactly — men who have eaten and drunk, telling one another a story — and yet produces, in the Traveller's tale, the philosophical question that the Eloi cannot ask.

Aristotle stated the economic premise that Plato's myth conceals. Leisure, *scholē*, is the end for which work is undertaken ("we are busy that we may have leisure", *Nicomachean Ethics* X.7, 1177b4–5), and the leisure of the few depends on the labour of others. In the first book of the *Politics* Aristotle imagined the single condition under which slavery might be abolished: if every instrument could do its own work at command or by anticipation, like the statues of Daedalus or the tripods of Hephaestus — if shuttles wove and plectra touched the lyre of themselves — then master-craftsmen would need no assistants and masters no slaves (1253b33–1254a1). The automaton is the condition of a free leisure; without the automaton, leisure is someone else's labour. A Greek epigrammatist of the first century BC, Antipater of Thessalonica, greeted the water-mill as the liberator of the slave-girls who ground the corn, bidding them sleep on while the nymphs turned the wheel, and Marx quoted him in the first volume of *Capital* (1867) beside Aristotle's dream, as the poet who hailed the machine as "the bringer back of the golden age". Marx's comment was savage: the heathens had not understood that machinery is the surest means of lengthening the working day.

The world of 802,701 is the place where Aristotle's condition and Marx's rebuttal meet. Its shuttles weave of themselves — the Eloi wear garments whose making they never witness, and the Traveller can only reason that "somehow such things must be made" — and its Golden Age has indeed returned. But the automata are human. The engines whose thud the Traveller hears rising from the ventilation shafts are tended by beings reduced to mechanism precisely so that

the beings above could be reduced to leisure. Marx's irony is literalized across geological time: the machine that was to restore the Golden Age has lengthened the working day of the Under-world to the whole of its history. Paul Lafargue, Marx's son-in-law, closed *Le Droit à la paresse* (1880; revised 1883) with the same quotations from Aristotle and Antipater and a playful demand that the proletariat claim the right to idleness. Wells's novel is the dark completion of Lafargue's joke. The right to be lazy has been won, but by one half of the species, and the other half has been made to pay for it in the coin of its own humanity — and has then collected the debt in flesh.

3.3
Nowhere and Its Shadows: Morris, Bellamy, Hudson, Bulwer-Lytton, Kingsley

The utopias Wells inverted had been published within the seven years preceding the novel, and they had already quarrelled with one another. Edward Bellamy's *Looking Backward: 2000–1887* (1888) sends its hero, Julian West, to sleep in a sealed vault beneath his Boston house and wakes him in a socialist Boston organized as an industrial army, where goods are distributed by credit and music arrives by telephone. Its most memorable image is the parable with which West describes the society of 1887: a prodigious coach dragged along a hilly and sandy road by the masses of humanity harnessed to it, with hunger for its driver, while the privileged ride on top and, for all their sympathy with the toilers below, cling to their seats for fear of falling among them. William Morris reviewed the book with distaste in *The Commonweal* in 1889, objecting that its ideal of life was that of the industrious professional middle class and its utopia a machine, and he answered it with *News from Nowhere* (serialized 1890; in book form 1891), whose subtitle is *An Epoch of Rest*. In Morris's Nowhere the revolution has abolished money, government and the factory; work has become art; the people are beautiful, healthy and unhurried; and the Houses of Parliament serve as a store for dung.

Wells's 802,701 takes something from each and negates both. From Morris it takes the beautiful, garlanded, leisurely people, the disappearance of money and commerce, the ruined monuments of the nineteenth century, even the subtitle: the Eloi live in an epoch of rest, and the novel discloses that rest, prolonged

without limit, is the social name of equilibrium. From Bellamy it takes the machine — and puts it underground, with Bellamy's coach-haulers beneath it. The Eloi are the passengers on top of Bellamy's coach after the coach itself has been forgotten. The road has become a lawn, the riders have lost all memory of riding, and the haulers, in harness for a thousand generations, have ceased to pull and begun to eat. Wells stated the lesson as theory a decade later. In *A Modern Utopia* (1905) he insisted that a modern utopia must be "not static but kinetic", a hopeful stage leading to a long ascent of stages; the static utopias of the nineteenth century were, in his view, dreams of arrest. The judgement is thermodynamic. A utopia at rest is a utopia at equilibrium, and 802,701 is what an epoch of rest becomes when it is allowed to run for eight hundred thousand years.

Two less-read romances show how close the Victorian utopia already stood to the Eloi. W.H. Hudson's *A Crystal Age* (1887) wakes its narrator in a far-future pastoral England whose people live in great communal houses in serene, artistic and unhurried harmony with nature — and in which sexual passion has been abolished for everyone except the Mother of each house, on whom its continuance depends. The narrator falls in love with a woman of the house, cannot endure the passionless calm, and dies of the draught he takes to cure himself. Hudson's title is worth setting beside the crystal of Wells's machine: the crystal age is clear, beautiful, inorganic and still. Edward Bulwer-Lytton's *The Coming Race* (1871) places its utopia underground. The Vril-ya command a universal energy, vril, are served by automata, have abolished war and inequality, and, the narrator observes, produce little literature of their own, since a life without passion, crime or struggle offers little to write about; he fears their emergence to supplant humanity. Bulwer-Lytton's subterranean super-race is the Morlocks inverted, and his remark about their literature anticipates the Eloi's loss of figurative speech. The romance's afterlife supplies the most chronophagic footnote in Victorian commerce. In 1886 the makers of a concentrated beef extract named their product Bovril, compounding the Latin *bos* with Bulwer-Lytton's *vril*, so that the energy of the underground race was sold to the surface as essence of meat.

The moral fable of degeneration through ease had been told to Victorian children before Wells told it to their parents. In Charles Kingsley's *The Water-Babies* (1863) the Doasyoulikes settle in the Land of Readymade at the foot of the Happy-go-lucky Mountains, where food grows on the trees without labour; across many centuries they cease to work, to think and to speak, grow hairy and simian, and dwindle until the last of them is shot by the explorer Paul Du

Chaillu. Kingsley's fable is Lamarckian and providential at once: disuse degrades, and the degradation is a lesson. Wells keeps the biology and discards the lesson, or rather redistributes it, since the Eloi are punished for their idleness not by nature but by their neighbours. The Traveller tries to receive the spectacle in the spirit of Thomas Carlyle, whose *Past and Present* (1843) had denounced an "Unworking Aristocracy" that consumed the produce of the land without labour: "I even tried a Carlyle-like scorn of this wretched aristocracy in decay." The attempt fails — "however great their intellectual degradation, the Eloi had kept too much of the human form not to claim my sympathy" — and the failure is instructive. Carlyle could scorn an idle class because he imagined it as powerful. The Eloi are idle and powerless at once, an aristocracy that has kept the leisure of rule and lost its substance. The Traveller's own historical analogy is exact: "the Eloi, like the Carlovingian kings, had decayed to a mere beautiful futility." The *rois fainéants*, the do-nothing kings of early medieval France, reigned while the mayors of their palaces governed, until the mayors took the crown. Whether Wells meant the late Merovingians and their Carolingian mayors or the last Carolingians and their Capetian successors, the figure is the same. The Morlocks are the mayors of the palace, and they have taken more than the crown.

3.1
Wilde's Machine-Slaves: The Eloi as Aesthetes

The Traveller's theory of the Golden Age is an aesthetics as well as an energetics. Watching the ruined palaces at sunset, he reasons that "the exquisite beauty of the buildings I saw was the outcome of the last surgings of the now purposeless energy of mankind before it settled down into perfect harmony with the conditions under which it lived", and he states the law: "This has ever been the fate of energy in security; it takes to art and to eroticism, and then come languor and decay." Of the artistic spirit only a residue survives: "To adorn themselves with flowers, to dance, to sing in the sunlight: so much was left of the artistic spirit, and no more." The passage is the novel's theory of decadence, and it is phrased in the vocabulary of thermodynamics: energy without an outlet, surges, harmony with conditions, a system settling into rest. Art, on this account, is what energy does in the interval between the end of need and the arrival of equilibrium. It is the last work a system performs before it ceases to do work at all.

The aesthetic movement had already described art in these terms, and with approval. The "Conclusion" to Walter Pater's *Studies in the History of the Renaissance* (1873), withdrawn from the second edition for fear that it might mislead the young, presents life as a brief interval in which experience flares and passes, and enjoins the reader to make of it a combustion: "To burn always with this hard, gem-like flame, to maintain this ecstasy, is success in life." "Not the fruit of experience, but experience itself, is the end." Pater's ethic is an ethic of consumption in the precise sense, the self as a flame that converts experience into intensity without residue. The Eloi are what such a flame leaves behind. They are the aesthetic life after the fuel has gone, garlanded, dancing, singing and capable of nothing else; and the gem of Pater's flame has hardened into the rock crystal of the Traveller's machine, the instrument from whose saddle he watches a civilization burn out.

Oscar Wilde supplied the movement's politics and its paradoxes, and the Eloi answer both. "The Soul of Man under Socialism" (1891), discussed in Chapter One, entrusts the drudgery of the world to machines working out of sight. The preface to *The Picture of Dorian Gray* (1891) concludes that "All art is quite useless", and the Eloi, "a mere beautiful futility", are uselessness become a species. The novel of Dorian Gray supplies the structure of 802,701 in miniature: a beautiful surface preserved by the transfer of its corruption to a hidden room, where the consequences of beauty accumulate unseen until they kill. "The Decay of Lying" (1889), Wilde's dialogue against realism, mourns the decline of the art of "the telling of beautiful untrue things" and proclaims that "Life imitates Art far more than Art imitates Life." The Eloi represent the terminal decay of lying. They have "little use of figurative language" because the figures of the nineteenth century have come true around them, and a people that lives inside its metaphors cannot lie. Wilde's paradox is literalized as Wells's method: the social figures of Victorian art and journalism are imitated, eight hundred thousand years later, by life.

The aesthetes had their pathologist. Max Nordau's *Entartung* (1892–93), dedicated to Cesare Lombroso, diagnosed the art of the fin de siècle — Wagner, Ibsen, Tolstoy, the Pre-Raphaelites, Wilde — as the symptom of a nervous degeneration of the European peoples. Its English translation, *Degeneration*, was published in London by William Heinemann early in 1895, a few months before the same house issued *The Time Machine*. The coincidence is more than bibliographical. Nordau and Wells share a vocabulary of decline and differ on its mech-

anism. For Nordau the aesthetes were degenerate because they were sick, and their sickness was hereditary; for Wells the Eloi are aesthetic because they are safe, and their safety is the achievement of the very progress Nordau believed the aesthetes betrayed. Bernard Shaw, answering Nordau in 1895 in the open letter later revised as *The Sanity of Art* (1908), argued that the so-called degenerates were the vanguard of a new sensibility. Wells's novel proposes a third possibility: that the vanguard and the decadence are the same energy at different points of its dissipation. The art of the period was, moreover, literally consumptive. Aubrey Beardsley, who illustrated Wilde's *Salomé* and served as art editor of *The Yellow Book* until he was dismissed in the panic that followed Wilde's arrest in April 1895, died of tuberculosis in 1898 at the age of twenty-five. The "hectic beauty" of the Eloi is the beauty of his line.

The decisive move, however, is one the novel makes only at its end, and it concerns the flower. In the Traveller's sunset theory, flowers are the last symptom of decadence: to adorn oneself with flowers is all that remains of the artistic spirit, and even that, he predicts, will "fade in the end into a contented inactivity". In the epilogue, two flowers given by one of these decadent creatures become the evidence that "gratitude and a mutual tenderness still lived on in the heart of man". The flower passes from aesthetics to ethics, from ornament to gift, from the residue of a dying energy to the remainder of a surviving care. Wilde himself would make the same passage. In Reading Gaol, in the long letter later published as *De Profundis* (written 1897), the author of "All art is quite useless" rewrote his aestheticism as an ethics of suffering and sympathy and found that "Where there is sorrow there is holy ground." The novel published in the month of his conviction performs, in two withered mallows, the conversion its most famous contemporary would perform in prison.

3.5
The Last Men: Nietzsche, Weber, Veblen, Keynes, Parfit

Nietzsche's Zarathustra, mocked by a crowd waiting for a tightrope-walker, tries to shame them by describing "the most contemptible" human being, the last man (*der letzte Mensch*). The portrait, in the fifth section of the prologue (1883), anticipates the Eloi with a fidelity that borders on the uncanny. The last men

have left the regions where life was hard, for they need warmth; they still work, but work is a form of entertainment, and they take care that the entertainment does not tax them; nobody becomes rich or poor any more, for both are too burdensome; they have their little pleasure for the day and their little pleasure for the night, but they honour health. "'Wir haben das Glück erfunden' — sagen die letzten Menschen und blinzeln": we have invented happiness, say the last men, and they blink. Above all, the earth will have become small, "und auf ihr hüpft der letzte Mensch, der Alles klein macht" — and on it hops the last man, who makes everything small. His race is as ineradicable as the flea-beetle; the last man lives longest.

Two details of the portrait sharpen into revelations beside Wells. The first is the verb. Nietzsche's last man *hops* upon a shrunken earth; the last living creature the Traveller sees, on the shore of a world that has stopped turning, is "hopping fitfully about", and the excised *New Review* episode had shown him grey, hopping, kangaroo-like beings as the final heirs of humanity. Whether or not Wells knew *Zarathustra*, which was not translated into English until 1896, the convergence is striking: two writers a dozen years apart, imagining the last of humanity, gave it the gait of the flea. The second detail is political. "Kein Hirt und Eine Heerde!" cry Zarathustra's last men: no shepherd and one herd; everyone wants the same, everyone is the same. The phrase parodies the Gospel promise of "one fold, and one shepherd" (John 10:16). Wells parodies the parody. The Eloi are indeed one herd, all with "the same form of costume, the same soft hairless visage, and the same girlish rotundity of limb", but they are not without a shepherd. Their shepherd lives underground, and keeps them as shepherds keep sheep.

Max Weber, as Chapter One noted, closed *The Protestant Ethic* (1905) by applying Nietzsche's phrase to the inhabitants of a completed capitalist order, specialists without spirit and sensualists without heart; in the same pages he described how the care for external goods, which Richard Baxter had wished to lie on the saint's shoulders like a light cloak, had hardened into a casing as hard as steel. The Eloi wear the light cloak; someone else wears the steel. Thorstein Veblen's *The Theory of the Leisure Class* (1899) supplies the sociology of the arrangement and is refuted by it. For Veblen the leisure class descends from the predatory phase of barbarian culture: its exemption from productive work is honorific because such work was once the portion of the conquered, and its conspicuous leisure and consumption preserve, in sublimated form, the habits of

exploit. The industrious classes carry instead what Veblen called the instinct of workmanship. 802,701 inverts the distribution of traits that Veblen thought permanent. The leisured have lost every vestige of prowess; the predatory habit has migrated to the workshop, where the Morlocks, who keep the instinct of workmanship to the point of oiling and cleaning the Traveller's machine, have also become the only hunters left on earth. Leisure, cut off from the exploit that gave it birth, has become prey.

John Maynard Keynes posed the question of the Eloi as a question of policy. In "Economic Possibilities for our Grandchildren" (1930) he predicted that within a century, in the progressive countries, the economic problem would be solved, and that humanity would then face "his real, his permanent problem — how to use his freedom from pressing economic cares, how to occupy the leisure, which science and compound interest will have won for him, to live wisely and agreeably and well." The phrase "science and compound interest" returns us to the Very Young Man at the Traveller's first dinner, who proposed to let his money accumulate at interest and hurry on ahead to collect it. Keynes saw the danger. He confessed that he thought "with dread" of the readjustment of habits required, and he cited the wives of the well-to-do, deprived by wealth of their traditional tasks, as a warning of the nervous breakdown awaiting a society without occupations. His horizon is now four years away. The standard of living in the countries he had in mind has risen roughly as he foresaw; the fifteen-hour week he imagined has not arrived; and the automation of cognitive work has revived his question in a form he did not anticipate, in which the machines that might free the many are built and trained by an invisible labour force distributed across the planet. Chapter Four returns to that labour. Here the point is Keynes's own: abundance does not answer the question of the Eloi; it poses it.

Derek Parfit's population ethics supplies the last term. In *Reasons and Persons* (1984) he showed that plausible principles for comparing outcomes lead to what he called the Repugnant Conclusion: that for any population of people with excellent lives there is some much larger population whose lives are barely worth living and whose existence would be better. Parfit later characterized such lives as consisting of little more than "muzak and potatoes". The Traveller, surveying the half-empty palaces, supposes that "the checks they had devised for the increase of population had succeeded too well", and the Eloi are in a sense the Repugnant Conclusion's inverse: a small population whose lives are pleasant, safe and thin — fruit and garlands in place of muzak and potatoes — sustained

by a larger hidden population whose lives are not counted at all. Wells's novel poses the question population ethics has found hardest to formulate: not how many lives of what quality, but whose lives are counted as the measure and whose as the means. Two later reckonings with the last men complete the series. Aldous Huxley, the grandson of Wells's teacher, began *Brave New World* (1932) as a parody of Wells's utopia *Men Like Gods* (1923) and produced a world of bred castes and engineered contentment in which the Savage's demand for "the right to be unhappy" is the Traveller's law restated as a right. Francis Fukuyama's *The End of History and the Last Man* (1992) asked whether the citizens of a liberal democracy without great struggles would become Nietzsche's last men or would restart history out of boredom. Wells had answered both writers in advance. At the end of history the last men do not restart it. History restarts beneath them, as appetite.

3.6 Weena: Androgyny, Sexual Difference and the Evolution of Care

The Traveller's first sociological discovery in 802,701, after the absence of houses, is the absence of sexual difference. Looking at the little figures following him, he perceives "in a flash" that "all had the same form of costume, the same soft hairless visage, and the same girlish rotundity of limb", and he explains the resemblance by the law that explains their feebleness: "the strength of a man and the softness of a woman, the institution of the family, and the differentiation of occupations are mere militant necessities of an age of physical force". Where population is balanced, violence rare and offspring secure, "the specialization of the sexes with reference to their children's needs disappears. We see some beginnings of this even in our own time, and in this future age it was complete." He adds, in the sunset meditation, that a sentiment was already arising "against connubial jealousy, against fierce maternity, against passion of all sorts; unnecessary things now, and things that make us uncomfortable, savage survivals, discords in a refined and pleasant life".

The theory has a precise context in the evolutionary biology of the 1880s, which had begun to explain sexual difference in the language of energy. Darwin's *The Descent of Man, and Selection in Relation to Sex* (1871) derived the secondary

characters of the sexes from male combat and female choice. Patrick Geddes and J. Arthur Thomson went further in *The Evolution of Sex* (1889), grounding sexual difference in metabolism itself: the female organism was anabolic, constructive and conserving of energy, the male katabolic, active and expending it. They drew the political conclusion with notorious confidence — "What was decided among the prehistoric Protozoa cannot be annulled by Act of Parliament" — and their book armed opponents of women's suffrage for a generation. In the light of their theory, 802,701 recodes the metabolic binary as a division of species. The Upper-world, girlish, conserving and passive, is anabolic; the Underworld, labouring, active and consuming, is katabolic. Sexual difference has not vanished from Wells's future. It has been redistributed from the individual to the species, and the class that once performed the masculine work of production has taken over the masculine work of predation.

The theory also answers the controversy of its year. In March 1894 the novelist Sarah Grand gave the "New Woman" her name in the *North American Review*, and the figure — educated, independent, bicycling, often unmarried — became the emblem of a decade's anxieties about the blurring of sexual roles, anxieties Elaine Showalter has surveyed in *Sexual Anarchy* (1990). The Traveller's account is the sceptical-progressive one: sex roles are militant necessities, and security dissolves them. Charlotte Perkins Gilman would give the argument its feminist form in *Women and Economics* (1898), contending that women's economic dependence on men had produced an excessive sex-distinction that economic independence would reduce, and its utopian form in *Herland* (1915), whose parthenogenetic women have made motherhood the organizing principle of a peaceful civilization. Ursula K. Le Guin tested the Traveller's hypothesis directly in *The Left Hand of Darkness* (1969), whose Gethenians are without fixed sex for most of their lives and have never known war as the Terran envoy understands it; in "Is Gender Necessary?" (1976) she described the novel as a thought-experiment designed to discover what would remain of human beings if sex were subtracted. Wells, Gilman and Le Guin agree that sexual difference is in part an artefact of conflict. They disagree about what follows its dissolution. For Gilman and Le Guin it is a gain in humanity; for the Traveller at sunset it is a symptom of exhaustion; for the novel as a whole, as Weena shows, it is neither.

Weena is the single individuated inhabitant of the Golden Age, and she enters the narrative through an absence of care. Bathing in a shallow, one of the little people is seized with cramp and swept downstream, and "none made the slight-

est attempt to rescue the weakly crying little thing which was drowning before their eyes". The Traveller wades in and saves her. The next afternoon she meets him "with cries of delight" and presents him with "a big garland of flowers — evidently made for me and me alone". The phrase marks the first appearance in 802,701 of the gift in the strict sense: an object addressed to one person, outside the undifferentiated garlanding of the crowd. From then on Weena follows him, fills his pocket with flowers, and alone among her people fears for him; it is from her that he learns that fear has not left the world, for she "dreaded the dark, dreaded shadows, dreaded black things", and her terror of the night becomes his first clue to the world beneath. The Traveller's language for her oscillates between the paternal and the possessive — she is "exactly like a child", she is "my little woman" — and a critical reading must register both the sentimental tradition on which it draws, the Victorian cult of the child-woman from Dickens's Little Nell to Ruskin's idealized girls, and the silence to which it consigns her. Weena is given almost no speech of her own; she exists in the narrative through the Traveller's report. Her care is real; her voice is appropriated.

The novel's deepest move concerns care itself, which it splits in two and distributes between the species. Joan Tronto's analysis in *Moral Boundaries* (1993) distinguishes four phases of care: caring about, which notices a need; taking care of, which assumes responsibility for meeting it; care-giving, which meets it in practice; and care-receiving, which completes the circuit. The Morlocks perform the practical phases. They make the Eloi's garments, maintain their halls, provide for their habitual needs "perhaps through the survival of an old habit of service", and see to their breeding. They take care of the Eloi exactly as a herdsman takes care of a herd, and with a herdsman's indifference to what the herd feels. Weena performs the affective phase alone. She notices, she fears for him, she attends; but she can do nothing for him except give him flowers, and in the forest she cannot even save herself. The Golden Age has separated care-giving from caring-about, and given the first to the predators and the second to the prey.

The separation clarifies what the flowers are. They are caring-about with every practical function removed: regard in its purest form, which cannot clothe or feed or defend, and which persists, pressed and brittle, when everything that could do those things has been consumed. The novel's closing phrase, "gratitude and a mutual tenderness", should be heard against a debate conducted in London as Wells wrote. In 1888 Huxley had published "The Struggle for Existence: A Programme", which likened the animal world to a gladiator's show; between

1890 and 1896 Peter Kropotkin answered him in a series of articles in *The Nineteenth Century*, collected in 1902 as *Mutual Aid: A Factor of Evolution*, arguing that cooperation within species was as much a product of selection as competition. Wells's last word on the evolution of humanity is "mutual". Whether or not he meant to side with Kropotkin against his teacher, the novel's final sentence lodges, in a species otherwise divided into herd and herdsman, the faculty Kropotkin placed at the root of the social instinct. Darwin had speculated in *The Descent of Man* that sympathy, once evolved, would extend beyond the tribe to all sentient beings. In 802,701 it has contracted to one small creature and one man from another age. It is enough to make an ending.

3.7
Every Arcadia a Feedlot: Omelas, Hailsham, Grace Field, *Logan's Run*, *Soylent Green*

The Traveller names the relation at the moment he grasps it: "These Eloi were mere fatted cattle, which the ant-like Morlocks preserved and preyed upon — probably saw to the breeding of." The sentence completes a list begun at sunset, when he counted "a more convenient breed of cattle" among the triumphs of improvement; the convenience has turned out to be the Morlocks'. The formula advanced in Chapter One, *every Arcadia is a feedlot seen from above*, can now be stated as a theorem with three clauses. A pastoral order is sustained by a consumption it does not see; the invisibility of the consumption is a condition of the pastoral's happiness; and the innocence of the beneficiaries is itself a product of the arrangement, cultivated as carefully as their bodies. The fictions that have rediscovered the theorem since Wells differ chiefly in what they allow the herd to know.

The theorem's philosophical form was stated four years before the novel. In "The Moral Philosopher and the Moral Life" (1891) William James asked his readers to imagine a world in which the utopias of Fourier, Bellamy and Morris should all be outdone, "and millions kept permanently happy on the one simple condition that a certain lost soul on the far-off edge of things should lead a life of lonely torture", and argued that our immediate revulsion at such a bargain is itself a moral datum. James named the very utopias that Wells inverted. Dostoevsky had given the argument its theological form a decade earlier, when Ivan

Karamazov returns his ticket to a harmony built on the unavenged tears of a tortured child (*The Brothers Karamazov*, 1880). Ursula K. Le Guin subtitled "The Ones Who Walk Away from Omelas" (1973) "Variations on a Theme by William James" and placed James's lost soul in a cellar beneath the joyous city, where its citizens learn, usually as children, of the child on whose misery their happiness depends; most return to their happiness, and a few walk away. Omelas and 802,701 are mirror images. In Omelas the citizens know and the victim is below; in 802,701 the beneficiaries do not know and the victims are above, while those below are the consumers. Both, however, place the price of Arcadia in the dark, and both end with a departure: the Traveller, like Le Guin's walkers, sets out again alone, towards a place no one can describe.

Kazuo Ishiguro's *Never Let Me Go* (2005) is the feedlot in which the herd knows and consents. The pupils of Hailsham, a boarding school in the English countryside, are clones raised to supply organs. They are encouraged to paint and write, are told the truth in fragments, and grow up to become first "carers" for other donors and then donors themselves, until they "complete". The novel's most devastating revelation concerns art. The guardians collected the pupils' paintings, Miss Emily finally explains, not to discover their talents but to prove to the outside world that they had souls at all. The Eloi's garlands are the residue of an artistic spirit that no longer proves anything; Hailsham's gallery is art conscripted as evidence in a trial its defendants cannot win. Ishiguro also completes the analysis of care begun in the previous section. His donors care for one another as carers before their own donations begin, so that the most intimate form of care becomes a stage in the harvest. The feedlot does not abolish care; it employs it.

Kaiu Shirai and Posuka Demizu's manga *The Promised Neverland* (2016–2020) makes the theorem a children's story with the brutality of a Grimm tale. The orphans of Grace Field House live in a pastoral idyll under the care of a loving "Mom", sit daily tests, and are periodically "adopted". They are, they discover, livestock on a plantation supplying human flesh to demons, and Grace Field is a premium farm: the higher their scores, the finer their brains as food for the demon aristocracy. The education the Eloi have lost is here the measure of value as meat, and the cultivation of the mind is a form of fattening. The "Mom" who tends them was herself a child of the farm, spared to become its keeper, and the farm exists under a "promise", a treaty concluded centuries earlier by which the human world bought its peace from the demons by ceding a

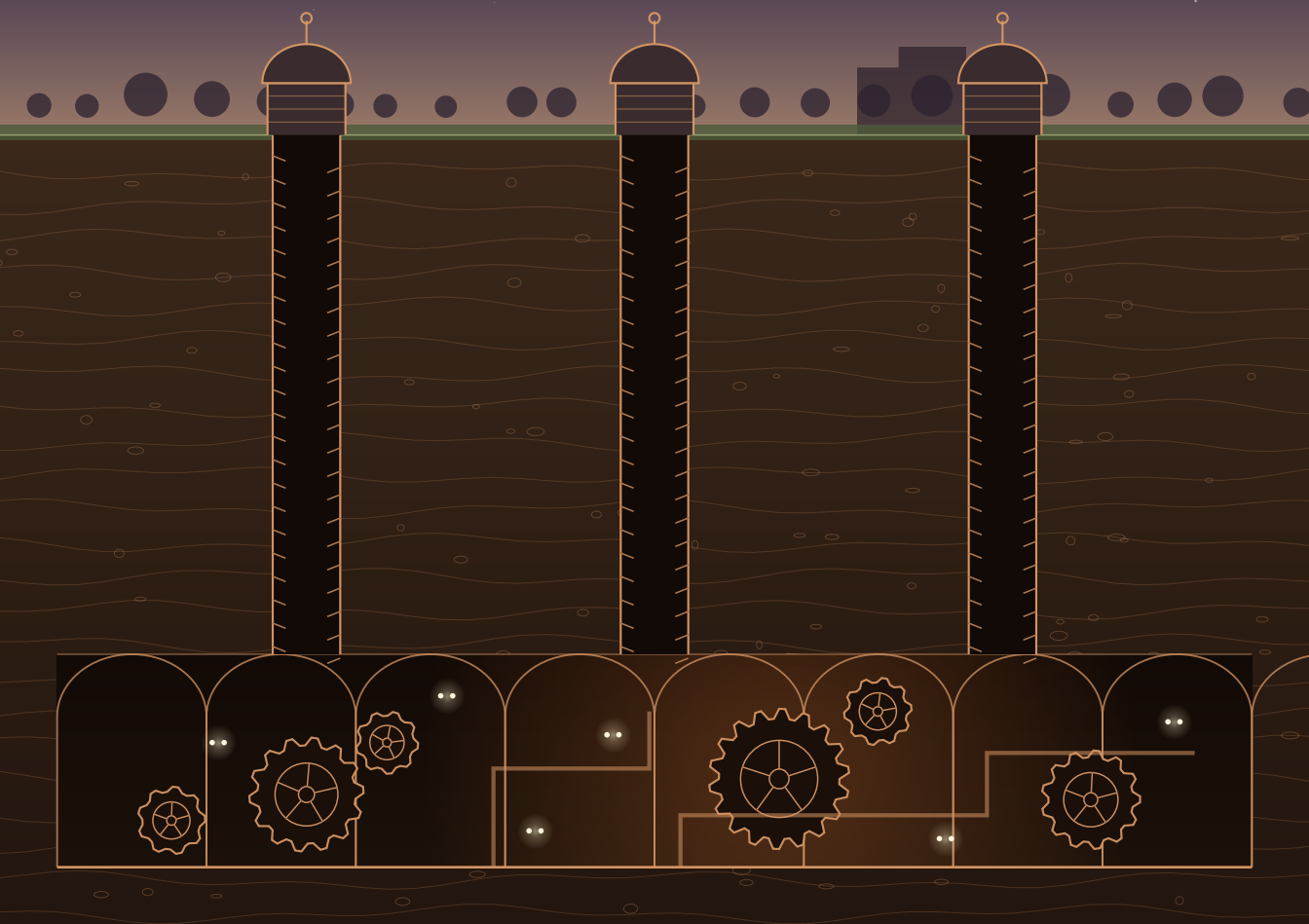
portion of humanity as food. The social contract is revealed as a feeding contract. Wells's 802,701 knows no such treaty, but it shares the manga's insight that the feedlot is not an anomaly within political order. It is one of the ways political order can be founded.

Two American films of the 1970s gave the theorem its popular forms. In *Logan's Run* (Michael Anderson, 1976), adapted from the novel by William F. Nolan and George Clayton Johnson (1967), a hedonistic domed city of the twenty-third century sustains its pleasures by consuming its own young: at thirty — twenty-one in the novel — citizens enter the ritual of Carrousel in the hope of "renewal" and are destroyed, and those who run are hunted. The dome is the regime of Kronos in its purest form, a paradise that eats its children at a fixed hour, and its outside is a ruined Washington inhabited by an old man and his cats. *Soylent Green* (Richard Fleischer, 1973), loosely adapted from Harry Harrison's *Make Room! Make Room!* (1966), sets the theorem in an overheated, overcrowded New York of 2022 fed on processed wafers, and compresses its revelation into four words: "Soylent Green is people!" Its most profound scene, however, is the death of the old scholar Sol Roth, who chooses euthanasia and, as he dies, is shown on a curved screen the vanished pastoral world — forests, flowers, rivers, grazing deer — to the music of Beethoven's Pastoral Symphony, Tchaikovsky and Grieg, before his body is carried away to be processed. It is the entropic pastoral reduced to its essence: the pastoral survives only as a film projected for the dying before they are made into food. The Wachowskis' *The Matrix* (1999) closed the century with humanity grown in pods as a power source for machines and lulled by a simulation of 1999; its Agent Smith explains that an earlier simulation, designed as a perfect world without suffering, failed because human beings would not accept it, so that "entire crops were lost". The machines learned what the Traveller learned at sunset, that the human animal is shaped by need, and drew a conclusion the Morlocks never reached: a feedlot that would keep its cattle docile must simulate their misery.

The series discloses a progression in what the herd is permitted to know. Wells's Eloi know nothing but their fear of the dark. Le Guin's citizens know and stay. Ishiguro's clones know and consent. Grace Field's children know and escape. Logan's citizens are taught a myth of renewal; the consumers of Soylent are fed one another without being told; the sleepers of the Matrix are given a counterfeit world complete with its sorrows. Across the series the herd's knowledge of the feedlot becomes the decisive moral variable, and the fictions that fol-

low Wells amount to a long inquiry into the question his novel leaves open: what an Eloi who knew would do. Weena does not know. She fears the dark and gives flowers. But her fear is a kind of knowledge, the body's memory of predation, and her gift is a kind of answer. The Golden Age of Kronos is a regime in which nothing is given, because everything is either provided or taken. In giving the Traveller a garland made "for me and me alone", Weena performs the one act the regime cannot accommodate, and the act survives the regime as two pressed flowers on a table in Richmond. Chapter Four descends from the lawn to the larder to ask who the providers and the takers are.





And the Lord said unto Cain, Where is Abel thy brother? And he said, I know not: Am I my brother's keeper?

— Genesis 4:9 (Authorized Version)

... we therefore take leave for a time of this noisy sphere, where everything takes place on the surface and in view of all men, and follow them both into the hidden abode of production, on whose threshold there stares us in the face "No admittance except on business."

— Karl Marx, *Capital*, vol. I (1867), ch. 6, trans. Samuel Moore and Edward Aveling (1887)

Ages ago, thousands of generations ago, man had thrust his brother man out of the ease and the sunshine. And now that brother was coming back — changed!

— H. G. Wells, *The Time Machine* (1895)

CHAPTER FOUR

4

The Nether World

Class, Speciation and the Below-Stairs of History

Chapter three surveyed the lawn; this chapter descends to the larder. The Morlocks are the novel's most original invention and its most compromised one. They are the speciation of the Victorian "residuum" — the casual, submerged, unrespectable poor — and they carry into deep time the fears with which the late-Victorian middle class regarded it. The chapter reconstructs the Traveller's sociology of the Under-world and its sources (4.1), the biographical below-stairs from which Wells wrote (4.2), and the mythic and political lineage of the child-devouring god whose name the Morlocks bear (4.3–4.4). It then argues that the Traveller's horror of the Morlocks is class disgust in the costume of biology, but that the novel lets its reader see through him: the vocabulary of his own violence is alimentary, the Morlocks oil and clean his machine, and the kinship he refuses is written into every scene of refusal (4.5). It ends with the vertical cities of modern fiction (4.6) and with the present tense of speciation in an economy that has learned to make its labour disappear (4.7).

4.1
Two Nations and the Submerged Tenth

The Traveller's sociology of the Under-world is the most explicit social theory in Wells's fiction, and it is delivered as a hypothesis under revision. It begins from the present. "At first, proceeding from the problems of our own age, it seemed clear as daylight to me that the gradual widening of the present merely temporary and social difference between the Capitalist and the Labourer, was the key to the whole position." Its evidence is the London of 1894: "there is a tendency to utilize underground space for the less ornamental purposes of civilization; there is the Metropolitan Railway in London, for instance, there are new electric railways, there are subways, there are underground workrooms and restaurants, and they increase and multiply." Its conclusion is a phrase the novel's whole method could take as its motto: "Industry had gradually lost its birthright in the sky." The capital supplied the image. The Metropolitan Railway had opened the world's first underground line in 1863; Joseph Bazalgette's intercepting sewers had been laid beneath the city between 1859 and the mid-1870s; and in 1890 the City and South London Railway had opened the first deep-level electric tube, driven through the London clay so far beneath the streets that its passengers descended to it by lift. In the year Wells wrote, a Londoner could pass a working day below ground.

The second step of the theory concerns the surface. The rich, the Traveller argues, are already withdrawing from contact with the poor, and their exclusiveness "is already leading to the closing, in their interest, of considerable portions of the surface of the land. About London, for instance, perhaps half the prettier country is shut in against intrusion." The observation belongs to the decades of the commons controversies, when the Commons Preservation Society, founded in 1865, fought the enclosure of Epping Forest and the heaths around the capital. Wells extends the enclosure of land into the enclosure of light: the surface itself becomes the park of a class. The third step concerns marriage. The widening gulf in education and habit, he argues, "will make that exchange between class and class, that promotion by intermarriage which at present retards the splitting of our species along lines of social stratification, less and less frequent. So, in the end, above ground you must have the Haves, pursuing pleasure and comfort and

beauty, and below ground the Have-nots, the Workers getting continually adapted to the conditions of their labour." The phrase "the splitting of our species along lines of social stratification" is the speciation of metaphor stated as a scientific prediction. Intermarriage is the exchange that keeps a population single; when it ceases, two populations diverge. Wells has translated the sociology of class into the language of reproductive isolation, which George Romanes and John Thomas Gulick had made central to the debate on the origin of species in the 1880s.

The final step is economic, and it is the most chilling. Once the Workers were underground, "they would no doubt have to pay rent, and not a little of it, for the ventilation of their caverns; and if they refused, they would starve or be suffocated for arrears." Rent for air: the commodification of breath is the limit of enclosure. Those "so constituted as to be miserable and rebellious would die; and, in the end, the balance being permanent, the survivors would become as well adapted to the conditions of underground life, and as happy in their way, as the Upper-world people were to theirs." The sentence closes the theory on the word that has organized this thesis: balance. The class system, run forward, reaches an equilibrium in which each population is adapted to its stratum and the gradient between them has hardened into anatomy.

The Traveller abandons this first theory, but not its premises. Discovering that the Morlocks, not the Eloi, hold the real power, he revises it into the picture of "a real aristocracy, armed with a perfected science and working to a logical conclusion the industrial system of to-day", whose triumph "had not been simply a triumph over Nature, but a triumph over Nature and the fellow-man"; and he reaches at last a Malthusian conclusion. "Apparently as time went on, the feeding of the Under-world, however it was effected, had become disjointed. Mother Necessity, who had been staved off for a few thousand years, came back again, and she began below." The Under-world, in contact with machinery that "still needs some little thought outside habit", had kept more initiative than the Upper, "and when other meat failed them, they turned to what old habit had hitherto forbidden." The phrase "other meat" is the novel's most economical statement of its alimentary logic. Malthus had argued in 1798 that population, unchecked, outruns subsistence, and that its checks are misery and vice. Wells adds a check that Malthus did not list and the Victorian imagination could not stop imagining: the eating of one class by another, which in the Under-world has ceased to be a figure of speech.

The Traveller marks the limits of his own authority. "I had no convenient cicerone in the pattern of the Utopian books," he admits, and his explanation "may be absolutely wrong"; of the final version he says that "it may be as wrong an explanation as mortal wit could invent." The confession is generic as well as epistemic. The utopias of Bellamy and Morris supply a guide — Dr Leete, old Hammond — who explains the future to its visitor; Wells's visitor must infer it from ruins and bodies, as an archaeologist or a field naturalist would, and may be wrong. The disappearance of the cicerone is the moment at which utopia becomes science fiction: the future ceases to be explained and must be read. Charles Booth's *Descriptive Map of London Poverty* (1889), which coloured every street of the capital from yellow for the wealthy to black for the "lowest class", had shown what such reading looked like in the present. The Traveller reads the surface of 802,701 as Booth's investigators read the streets of the East End, from outside, and with the same mixture of science and fear.

4.2
Uppark: The Biographical Below-Stairs

Wells wrote the Morlocks from a position few of his readers occupied: he had grown up at the threshold between the classes the novel separates. His mother, Sarah Wells, had been a lady's maid before her marriage, and in 1880, when the family shop in Bromley could no longer support her, she returned to service as housekeeper at Uppark, the great house on the South Downs where she had once served. Wells spent long periods there as a boy and adolescent. Uppark was a country house of the classical kind, its state rooms above and its service quarters apart, linked to the house by underground passages along which servants could move without being seen by those they served. Wells would recall the house's library, where he read Plato's *Republic* and Swift, and he produced for the household, during a snowbound winter, a daily newspaper of his own, *The Uppark Alarmist*. The boy who read the myth of the metals in the library above slept among the servants below.

The house had its own history of crossings between above and below. Its eighteenth-century owner, Sir Harry Fetherstonhaugh, kept the young Emma Hart, the future Lady Hamilton, there in 1780–81 before discarding her, and in old age married his dairymaid, Mary Ann Bullock, who inherited the estate; it

was her sister Frances who employed Sarah Wells. The servant who became mistress was part of the house's lore, and she embodies the exception against which the Traveller's theory is built: promotion by intermarriage, the very mechanism whose decline he predicts. Wells would later anatomize the house as a model of the nation. In *Tono-Bungay* (1909) Uppark becomes Bladesover, where the narrator, George Ponderevo, the housekeeper's son, grows up below stairs and comes to regard the great house and its hierarchy as the key to almost everything distinctively English. Bladesover, as George understands it, is 802,701 seen from inside: a gentry whose leisure is the visible form of a labour carried on beneath it by people whom the gentry were trained not to see.

Raymond Williams's *The Country and the City* (1973) taught a generation to read the English country house as a mystification of the labour that produced it. Analysing Ben Jonson's "To Penshurst" (1616), Williams showed how the country-house poem abolishes work by making nature serve itself up to the lord's table: on Jonson's estate "The painted partridge lies in every field, / And for thy mess is willing to be killed", and the fish leap into the angler's hand. The fantasy is the Golden Age topos transplanted to an English estate, and its essence is the self-offering of food. Wells's 802,701 reverses its terms with terrible precision. The Eloi are the partridge willing to be killed for another's mess: docile, beautiful, garlanded, they lie in every field and do not resist. The country-house poem imagined a nature that consents to be eaten; the novel imagines a human class that has been bred to consent.

Wells's working life completed his education in the nether world. Between 1880 and 1883 he was apprenticed to drapers at Windsor and at Southsea, where the assistants lived in on the premises, ate the employer's food and worked days of thirteen hours; he would turn the experience into the comic misery of *Kipps* (1905) and *The History of Mr Polly* (1910). He escaped through the scholarship system — a pupil-teachership at Midhurst Grammar School and, in 1884, a government studentship at the Normal School of Science in South Kensington — and his escape made him an exemplary instance of what Richard Hoggart would call the scholarship boy, the working-class child raised by education into a class whose manners he never fully trusts and away from one to which he cannot return (*The Uses of Literacy*, 1957). The Traveller is a gentleman of independent means; his author was a draper's apprentice who had lived below stairs. The novel's affective economy follows from the gap. Its pity goes upward, to the beautiful and helpless Eloi, the class to which the scholarship boy had been

taught to aspire; its disgust goes downward, to the pale and laborious Morlocks, the class from which he had escaped. The Morlocks are what Wells feared he might have remained, and what the Traveller cannot bear to recognize as kin.

The house has a coda. Uppark was gutted by fire in August 1989, its contents carried out onto the lawns by staff and volunteers, and it was restored by the National Trust and reopened in 1995, the centenary of *The Time Machine*. The house from whose passages Wells drew his Under-world thus passed, within a century, through the novel's own sequence of combustion and conservation — fire, ruin, the rescued remainder — and survives as a museum of the life lived above and below its stairs.

4.3
Moloch: Milton, Blake, Marx's Vampire and Lang's *Metropolis*

The name of the Morlocks carries the history of a god. In the Hebrew Bible, Molech is the deity to whom children are made to "pass through the fire": Leviticus forbids the rite (18:21; 20:2–5), and the account of Josiah's reform records that the king "defiled Topheth, which is in the valley of the children of Hinnom, that no man might make his son or his daughter to pass through the fire to Molech" (2 Kings 23:10). The valley of Hinnom, *Ge-Hinnom*, became Gehenna, and the place of child sacrifice gave its name to hell. Molech is the god in whom the devouring of children and the infernal underground are fused. Gustave Flaubert made him the presiding horror of *Salammbô* (1862), whose besieged Carthaginians feed their children into the burning arms of a bronze Moloch; Giovanni Pastrone's *Cabiria* (1914) put the scene on film, with a temple whose fiery maw receives its victims; and the image passed, by common critical consent, from Pastrone's set to Fritz Lang's.

Milton gave the god his English form. In the catalogue of the fallen in *Paradise Lost* Moloch comes first, the "horrid King besmear'd with blood / Of human sacrifice, and parents tears" (I.392–393), and in the infernal council he is the advocate of open war, the spirit who would rather be annihilated than endure servitude. Milton's hell is also the first industrial underworld of English epic. Before Pandemonium can rise, Mammon's crew opens a wound in a volcanic hill and digs out its gold — they "Rifl'd the bowels of thir mother Earth /

For Treasures better hid" (I.687–688) — and a second multitude founds the ore in cells fed by sluices of liquid fire. The capital of hell is a mine and a foundry before it is a parliament. In the "Nativity Ode" Milton had already shown Moloch in flight before the incarnate God — "And sullen Moloch, fled, / Hath left in shadows dread / His burning Idol all of blackest hue" — and William Blake illustrated the scene as *The Flight of Moloch* (1809). The god who flees from light into shadow is the Morlock's ancestor in posture as well as in name. Blake also supplied the phrase that fixed the industrial underworld in English, the "dark Satanic Mills" of the preface to *Milton* (c. 1804–10), and the London of *Songs of Experience* (1794), where "the Chimney-sweepers cry / Every blackning Church appalls".

Marx gave the image its economic theory, and he did so in the language of eating time. The tenth chapter of the first volume of *Capital* is a history of the working day as a contest over hours. Capital, "in its blind unrestrainable passion, its were-wolf hunger for surplus-labour", oversteps the physical limits of the day: "It usurps the time for growth, development, and healthy maintenance of the body. It steals the time required for the consumption of fresh air and sunlight. It higgles over a meal-time, incorporating it where possible with the process of production itself, so that food is given to the labourer as to a mere means of production, as coal is supplied to the boiler, grease and oil to the machinery." Marx cites the factory inspectors on the employers' "petty pilferings of minutes" and records the workers' own name for the practice: "nibbling and cribbling at meal-times". The passage is the Under-world in nuce. Capital steals the time in which the worker would have consumed fresh air and sunlight, the very goods the Morlocks have lost; it feeds him as one fuels an engine; it nibbles at his meals. Chronophagy is, in Marx, the literal practice of the factory: the employer eats the worker's time, and the worker is fed like a machine. The Morlocks, who oil and clean the Traveller's machine, perform on his engine the maintenance that capital once performed on their bodies. E.P. Thompson would trace the transformation in "Time, Work-Discipline, and Industrial Capitalism" (1967), observing that under the new discipline time becomes currency, not passed but spent. The Morlocks descend from the people whose minutes were nibbled, and in 802,701 they have taken up the habit of nibbling.

Lang's *Metropolis* (1927), written with Thea von Harbou, is the most famous visual descendant of the Morlocks, and its debts are explicit. Its sons of the rich play in the Eternal Gardens, a pleasure-park of fountains and peacocks above a

city whose workers live in a subterranean town and labour at machines in shifts that march like a single body. When Maria brings a troop of workers' children up into the garden, she presents them to the gilded youths with a line that restores Wells's Genesis: "Look, these are your brothers!" When Freder descends to the machine halls and witnesses an explosion, the great machine turns before his eyes into Moloch, a monstrous head into whose jaws columns of workers are driven. Wells reviewed the film in *The New York Times* in April 1927 and called it "the silliest film" he had seen. His objection was historical. *Metropolis*, he complained, recycled the ideas of his own *When the Sleeper Wakes* (1899) and of Karel Čapek's *R.U.R.* (1920) — the play that gave the world the word "robot", from the Czech for forced labour — and assumed, as he no longer did, that mechanization would multiply drudges rather than abolish them. The Wells of 1927 had become a prophet of technocratic plenty. The film he despised kept faith with the Wells of 1895 more fully than its author did.

The god's last great appearance in the century came in poetry. In the second part of Allen Ginsberg's *Howl* (1956), Moloch names the whole machinery of American industry, money and war, and the poem's litany of his attributes includes "Moloch whose blood is running money". From Leviticus to Ginsberg the figure is constant: a god who demands children, whose shrine is fire, and whose temple becomes, increasingly, a factory. Wells's achievement was to strip the god away and leave the worshippers. The Morlocks are Moloch without the idol — not a deity that demands sacrifice but a population that has made sacrifice its diet.

4·4
Nibelheim and the Goblin Mines: Wagner, Shaw, MacDonald, Machen

The most elaborate nineteenth-century depiction of an enslaved underground workforce was staged not in a novel but in an opera house. In the third scene of Richard Wagner's *Das Rheingold* (first performed 1869; staged within the complete *Ring* at Bayreuth in 1876), the gods Wotan and Loge descend into Nibelheim, where the dwarf Alberich, having renounced love to win the power of the ring, drives his fellow Nibelungs to mine and forge gold for his hoard. Wagner scored the descent and return for eighteen tuned anvils, so that the sound of

industrial labour hammers through the orchestra, and Alberich, wearing the Tarnhelm, becomes invisible and lashes his workers unseen. Above, in the world of light, the gods have had their fortress, Valhalla, built by the giants Fasolt and Fafner and are trying to avoid paying for it. The *Ring* is a cosmology of class: gods who will not pay for their palace, a master who has renounced love for accumulation, a subterranean workforce whose labour is the hoard, and a final conflagration in which the fortress of the gods burns. Wagner's revolutionary anticapitalism of 1848–49, entangled, as later criticism has insisted, with his antisemitism, shaped the cycle's political imagination.

Bernard Shaw drew out the allegory in *The Perfect Wagnerite* (1898), reading Alberich as the capitalist and Nibelheim as the modern sweated trade, and he named the trade with a precision that bears directly on this thesis. Nibelheim, Shaw wrote, need not be a mine: "it might just as well be a match-factory, with yellow phosphorus, phossy jaw, a large dividend, and plenty of clergymen shareholders." The match-factory of East London, whose products the Traveller carried into the future and burned against the Morlocks, is Shaw's Nibelheim. The convergence completes the analysis of the pocket in Chapter One. The matches with which the Traveller lights the Under-world were made in its Victorian original, and the most influential reading of the century's greatest operatic cosmology identified that original by name. Patrice Chéreau's centenary production at Bayreuth in 1976, conducted by Pierre Boulez, set the cycle in the industrial nineteenth century, with the Rhinemaidens at a hydroelectric dam and the gods as a dynasty of industrialists, and made Shaw's reading the most famous staging of the work.

The Victorian nursery had its own subterranean race. In George MacDonald's *The Princess and the Goblin* (1872) the goblins who infest the mines beneath the princess's castle were, by legend, ordinary people who had once lived above ground and withdrawn into the caverns when the king laid on them what they thought too severe taxes; they had "greatly altered in the course of generations; and no wonder, seeing they lived away from the sun, in cold and wet and dark places." The story's hero is a miner's son, Curdie, who defends the princess against them with his pick and with songs, since the goblins cannot bear rhymes. MacDonald's fable sorts the poor into two kinds: the honest miner who works below and returns to the light, and the degenerate race that has made the dark its home. This is the distinction Victorian social discourse drew between the respectable working class and what it called the "residuum", the casual and crim-

inal poor whom Gareth Stedman Jones studied in *Outcast London* (1971). Wells's Morlocks are the residuum speciated: the class the respectable imagination could not assimilate, given the time to become what that imagination already believed it was.

Arthur Machen gave the figure its anthropological form in the year of *The Time Machine*. In "The Novel of the Black Seal", one of the tales embedded in *The Three Impostors* (1895), a professor discovers that the fairies of Welsh legend are the memory of a degenerate pre-human race surviving in the hills, hissing in a speech older than any human tongue, and he disappears among them. Machen drew on the euhemerist theory, argued in David MacRitchie's *The Testimony of Tradition* (1890), that the little people of folklore were recollections of a small aboriginal population that had lived in earth-houses and underground chambers. Folklore, ethnology and evolutionary anxiety converged in the 1890s on a single figure: a small, pale, subterranean people, older or lower than ourselves, surviving in the dark and threatening the surface. J.R.R. Tolkien gave the figure its most widely read modern form in the orcs of *The Lord of the Rings* (1954–55), creatures of the mine and the furnace whom his legendarium represents as corruptions of an elder and fairer race — as the fading Elves represent the other half of the Wellsian division — and he made the danger of the depths explicit when Gandalf judges that the dwarves of Moria "delved too greedily and too deep". Every underground race of the period carries the same double charge. It is the labour on which the surface depends, and the depth from which the surface fears it will be devoured.

4.5
The Anthropological Machine: Disgust, Sympathy and the Refusal of Kinship

The Traveller's first descriptions of the Morlocks are a catalogue of disgust. The creature glimpsed among the ruins is "a queer little ape-like figure", "a dull white", with "strange large greyish-red eyes" and "flaxen hair on its head and down its back"; the one he watches climbing down a shaft makes him shudder — "It was so like a human spider!" Their skin is "just the half-bleached colour of the worms and things one sees preserved in spirit in a zoological museum", and they are "filthily cold to the touch". He explains their pallor by

Darwin's cave animals, "the white fish of the Kentucky caves, for instance", and their eyes by the owl and the cat. The repertoire is precise: the ape, the spider, the worm, the specimen in its jar, the blind fish. Each figure assigns the Morlock to a lower order of nature, and each is drawn from the naturalist's cabinet. The Traveller looks at his descendants as a South Kensington student looked at a preparation.

Giorgio Agamben has given the operation a name. In *L'aperto: l'uomo e l'animale* (2002) he describes the "anthropological machine" by which Western thought produces the human, defining humanity through an animality it must both include and exclude, across a zone of indistinction where the two cannot be told apart. The machine of the ancients, in his account, included an outside by humanizing an animal — the slave, the barbarian, the foreigner, figures of an animal in human form; the machine of the moderns excludes an inside by animalizing the human, and its emblem is the search, from Linnaeus to Ernst Haeckel's speechless ape-man, *Pithecanthropus alalus*, for the missing link that would mark the boundary. *The Time Machine* sets the machine running in a world where the boundary has dissolved into two populations, neither of which is simply human. The Traveller assigns humanity by aesthetic criteria. The Eloi, however degraded, "had kept too much of the human form not to claim my sympathy"; the Morlocks, who retain more initiative, more skill and more of the industry that his own lecture-room would have counted as the mark of the species, are refused it: "it was impossible, somehow, to feel any humanity in the things." Beauty buys kinship; labour forfeits it. The criterion is the one Pierre Bourdieu would find at the root of class taste, which defines itself first as distaste, a visceral intolerance of the tastes of others (*La Distinction*, 1979).

The theory of disgust explains the refusal. Mary Douglas's *Purity and Danger* (1966) defined dirt as matter out of place; Julia Kristeva's *Pouvoirs de l'horreur* (1980) described the abject as what the self must expel in order to constitute itself — the corpse, the waste, the boundary that will not stay drawn; Martha Nussbaum, in *Hiding from Humanity* (2004), argued that disgust is characteristically projected onto subordinate groups so that a dominant group may disown its own animality and mortality. The Morlocks are abject in each sense. They are matter out of place when they surface, blinking and fumbling in the light; they are what the Upper-world has expelled in order to be beautiful — the labour, the darkness, the meat; and they bear, projected upon them, everything the Traveller's class would prefer not to know about its own bodies:

that they eat, that they work, that they are animals. Kelly Hurley's term for such beings, the "abhuman", names their status exactly: not inhuman, but human in a form the human refuses to recognize.

The novel, however, does not share the Traveller's refusal, and it supplies the evidence against him with care. On his arrival, before he has met any inhabitant of the future, he fears that humanity may have "developed into something inhuman, unsympathetic, and overwhelmingly powerful", and that he will seem to it "some old-world savage animal, only the more dreadful and disgusting for our common likeness — a foul creature to be incontinently slain." The prophecy is fulfilled in reverse. It is the Traveller who finds a descendant the more disgusting for its likeness, and the Traveller who wishes to slay it: "I longed very much to kill a Morlock or so. Very inhuman, you may think, to want to go killing one's own descendants!" In the forest, armed with an iron lever broken from a machine in the museum — "a mace in my hand more than sufficient, I judged, for any Morlock skull I might encounter" — he lays about him, and his account of the fight is phrased in the vocabulary of appetite: "I could feel the succulent giving of flesh and bone under my blows." *Succulent* is a word for food. The Traveller, who arrived home starving for meat and whose first desire on his return is mutton, describes killing his descendants as a carnivore describes eating. Meanwhile the Morlocks have taken his machine into the pedestal of the White Sphinx and, he later discovers, "carefully oiled and cleaned" it, perhaps even taken it partly to pieces "while trying in their dim way to grasp its purpose". They are mechanics, and they have treated his engine with the care of fellow-mechanics. The Traveller is carnivorous, nocturnal, armed, a maker and mender of machines; so are the Morlocks. The figure he refuses to recognize as kin is his double.

The tradition of the double clarifies the logic. Robert Louis Stevenson's *Strange Case of Dr Jekyll and Mr Hyde* (1886) had split the gentleman-scientist from his own appetite and given the appetite a body that the lawyer Utterson finds "hardly human" — "something troglodytic, shall we say?" Freud's essay on the uncanny (1919) would explain such figures as the return of what is familiar and repressed, the *heimlich* become *unheimlich*. The Morlocks are the Traveller's troglodyte: the hunger, labour and violence that the gentleman-scientist has delegated below stairs, returning to the surface at night. Genesis supplies the oldest form of the structure, and Wells's own language points to it. The Traveller says that "man had thrust his brother man out of the ease and the sunshine", and

that industry had lost "its birthright in the sky". The birthright is Esau's, sold to his brother for a mess of red pottage (Genesis 25:29–34); and Esau, the hunter, was "a hairy man", while Jacob, the dweller in tents, was "a smooth man" (27:11). The Morlocks, with flaxen hair down their backs, are Esau; the Eloi, with their "soft hairless visage", are Jacob; and the novel imagines the hairy brother returning not to embrace the smooth one, as Esau does in Genesis 33, but to eat him. Cain's question completes the typology. "Am I my brother's keeper?" he asks (4:9), having killed Abel, who was "a keeper of sheep" (4:2). The Morlocks answer Cain with a terrible literalism. They are their brothers' keepers, in the sense in which Abel kept sheep.

Hegel's dialectic of lordship and bondage in the *Phänomenologie des Geistes* (1807) gives the relation its philosophical form and the novel its sharpest irony. The lord enjoys; the bondsman works; and work, unlike enjoyment, is "gehemmte Begierde, aufgehaltenes Verschwinden" — desire held in check, vanishing held back. In enjoyment the thing is consumed and disappears; in work it is formed and endures, and through forming it the bondsman reaches an independent consciousness that the idle lord loses. The Traveller's final theory is Hegel's dialectic run to its biological conclusion: the Under-world, in contact with machinery, "had probably retained perforce rather more initiative, if less of every other human character, than the Upper." But the dialectic does not end, as Hegel's does, in mutual recognition. It ends in ingestion. Alexandre Kojève, who lectured on Hegel in Paris in the 1930s, later added a notorious note imagining that at the end of history human beings would revert to animality, making their works of art as birds build nests and spiders spin webs, giving concerts like frogs and cicadas, playing like young animals. The Eloi are Kojève's post-historical animals; the Morlocks are the bondsmen whose labour has returned them not to freedom but to appetite. Hegel defined work as vanishing held back. In 802,701 the workers have learned to make their lords vanish.

Only once does the Traveller's refusal break. When the forest fire he has kindled blinds the Morlocks, they blunder helplessly about him in the glare, and he strikes at them "in a frenzy of fear", killing one and crippling several, until he watches one of them groping under a hawthorn against the red sky and hears their moans: "I was assured of their absolute helplessness and misery in the glare, and I struck no more of them." The sight of misery, in the light he himself has made, halts the blow. Emmanuel Levinas would locate the origin of ethics in such an encounter, in the face of the other whose vulnerability says, before any

argument, *thou shalt not kill* (*Totalité et infini*, 1961). The Traveller's Levinasian moment is brief, and it requires the Morlocks to be blinded and dying. But it is the novel's acknowledgement that the anthropological machine can be stopped, and that what stops it is not a criterion of humanity but the sight of suffering.

4.6
Vertical Cities: *Snowpiercer*, *Parasite*, *Us*, *The Platform*, *The Incal*

The topology of 802,701 — leisure above, labour below, a conduit between — became one of the master-images of twentieth-century speculative fiction, and its later forms obey a law the novel already states: in every vertical city, goods ascend and bodies descend. The Eloi's garments rise from the workshops; the Eloi themselves go down the shafts. The verticality is not decorative. It encodes the gradient that the social second law requires, and the conduit — shaft, lift, staircase, platform — is the channel through which the gradient is discharged as consumption.

The lineage runs from *Metropolis* through J.G. Ballard's *High-Rise* (1975), whose forty-storey tower sorts its residents by floor until its social order collapses into tribal violence, and whose opening sentence finds a doctor on his balcony eating a dog, to the City-Shaft of Moebius and Alejandro Jodorowsky's graphic saga *The Incal* (1980–88), a vertical metropolis bored into a planet, with its aristocrats at the summit and at the bottom an acid lake inhabited by mutants, into which the hero is falling on the first page. Yukito Kishiro's manga *Gunnm* (1990–95) suspends the rich in the floating city of Zalem above a scrapyard city that lives on its refuse, and reveals that Zalem's citizens have their brains replaced by chips on reaching adulthood: the Eloi's loss of mind made surgical. Neill Blomkamp's *Elysium* (2013) lifts the wealthy into an orbital habitat whose medical bays cure every disease, above an exhausted Earth whose workers are poisoned in its factories, and its name returns the Isles of the Blessed, where Kronos ruled, to their place above the toilers.

Bong Joon-ho has made the law the subject of two films. *Snowpiercer* (2013), adapted from Jacques Lob and Jean-Marc Rochette's *Le Transperceneige* (1982), turns the vertical city on its side. After a failed attempt at climate engineering has frozen the earth, the remnant of humanity circles the planet in a train driven by a

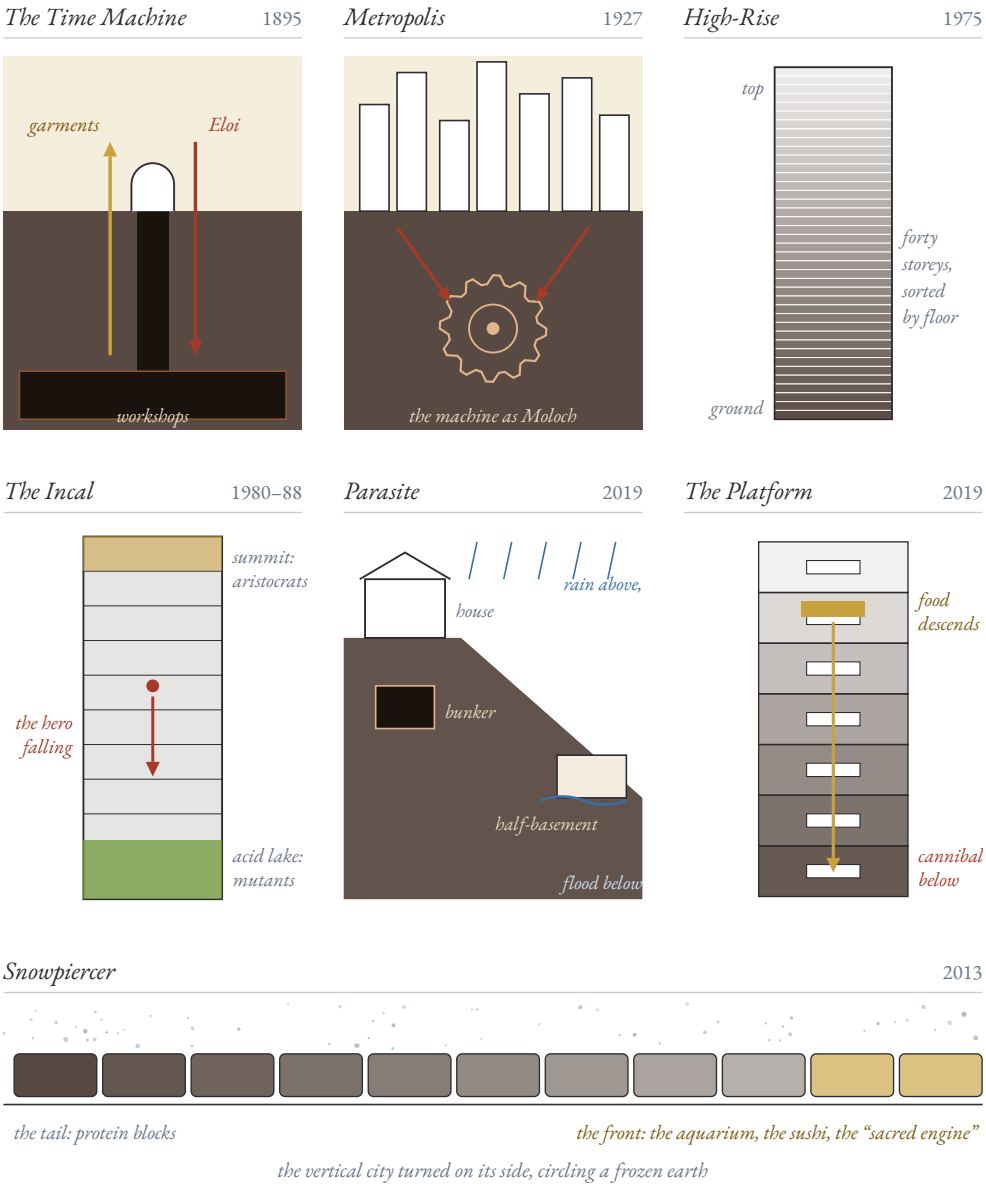


Figure 4.1 Vertical cities, in schematic section. In every vertical city goods ascend and bodies descend, and the conduit — shaft, lift, staircase, platform — is the channel through which the gradient is discharged as consumption. *Snowpiercer* turns the vertical city on its side.

"sacred engine" in perpetual motion, the rich in the front carriages with their aquarium and their sushi, the poor in the tail eating protein blocks that prove to be made of insects. The train is a closed thermodynamic system in a world that has reached its cold death outside, and its stability depends, as its builder explains, on balance: the revolts of the tail are permitted, even engineered, as a means of controlling population. The rebel leader, Curtis, confesses that in the tail's first months people ate one another and that he knew which of them tasted best; the old patriarch, Gilliam, had cut off his own arm and leg to feed the others instead. The climax discovers small children from the tail working inside the engine, replacing parts that have become extinct — Moloch's sacrifice performed by the machine itself — and Curtis thrusts his own arm into the mechanism to free one of them. The last image shows two children stepping out of the wreck onto the snow and seeing a polar bear. The remnant is alive, and so is the world.

Parasite (2019) restores the vertical axis to a single hillside in Seoul. The Kim family lives in a half-basement whose window looks out at the level of the street; the Park family lives in an architect's house high above; and beneath the Parks' house, in a bunker of which the family knows nothing, the husband of their former housekeeper has lived for years, saluting Mr Park as a benefactor by flashing Morse code through the lights. The rain that is a pleasant shower for the Parks becomes, below, a flood that destroys the Kims' home: the same weather is pastoral above and catastrophic beneath. The film's pivot is olfactory. Mr Park remarks on the smell of the Kims, the smell of people who ride the subway, and his involuntary gesture of disgust at it provokes his murder. Bourdieu's distaste and the Traveller's "filthily cold to the touch" are the same reflex. The film's title names the oldest figure of the dinner-table: in Greek comedy the *parasitos* was one who ate at another's table and paid in flattery and talk, and Michel Serres made the parasite the elementary relation of social and informational systems, the guest who takes without giving and the noise in the channel (*Le Parasite*, 1980). Bong leaves unanswerable the question of who is the parasite, which is the point: in a vertical city everyone eats at someone else's table. The film ends with the Kim father sealed in the bunker in his turn, signalling to his son in Morse through the lights. The Morlock position does not disappear when its occupant dies. It is refilled.

Jordan Peele's *Us* (2019) gives the underground a population of doubles. In disused tunnels beneath the United States live the Tethered, copies of the surface population made by an abandoned experiment, compelled to mirror their origin-

als' movements in grotesque pantomime and fed on raw rabbits bred in hutches below. When they surface, in red jumpsuits and armed with scissors, they join hands across the country in a parody of the 1986 charity event Hands Across America; and the film's final revelation is that its heroine is herself a Tethered who, as a child, changed places with her double above. *Us* is the uncanny reading of the Morlocks made literal: the underground population is the surface population's own double, and the difference between them is an accident of who was born where. Galder Gaztelu-Urrutia's *The Platform* (2019) is the purest diagram of the law. Its prison is a vertical shaft of cells, two prisoners to a level, through which a platform laden with food descends once a day; the upper levels gorge, the lower starve and, far down, eat one another. Prisoners are reassigned at random every month, so that every inmate has been above and below, and still no level rations for those beneath. A former official pleads for "spontaneous solidarity"; the hero, who has brought *Don Quixote* as his single permitted possession, descends on the platform with a companion, distributing food by force and guarding one untouched dessert as a message to the administration above, and finds at the bottom a child, whom he sends up instead. The child is the message. The descent of goods has been reversed by a single remainder.

Two inferences follow. The first is that the vertical city is a heat engine in the precise sense of the social second law: its gradient drives a flow, the flow is consumption, and the system's apparent stability is a steady state maintained by the continuous destruction of those below. The second is that the modern fictions locate the possibility of reversal in a remainder — an untouched dessert, a rescued child, two children stepping onto the snow — that travels against the flow. The structure of Wells's ending, in which two flowers return from the bottom of time to the table in Richmond, has become the structure of hope in the vertical city.

4·7
Morlocks of the Cloud: Automation, Ghost Work and the Present Tense of Speciation

In 1770 Wolfgang von Kempelen presented to the court of Maria Theresa a chess-playing automaton in Turkish costume that defeated most of its challengers. Its secret, exposed by Edgar Allan Poe among others in "Maelzel's Chess-

Player" (1836), was a human chess master concealed in the cabinet beneath the board. Walter Benjamin made the machine the emblem of the first of his theses "On the Concept of History" (1940): a puppet that always wins because a hunchbacked dwarf, a master of chess, sits hidden beneath the table and guides its hand. In 2005 Amazon named its marketplace for piecework performed by human beings on behalf of software "Mechanical Turk" and described the service as artificial artificial intelligence. The genealogy is exact. The automaton that seems to think conceals a human who works below; the surface sees the machine and not the worker; and by the twenty-first century the name of the arrangement is a joke its beneficiaries tell openly.

The world of 802,701 is a Mechanical Turk at planetary scale, and the economy of 2026 has made its structure familiar. Wilde's machine-slaves and Keynes's age of leisure have arrived, in the rich countries, in the form of software that answers, drafts, translates and draws; and beneath that software lies a labour force that the interface is designed not to show. Mary L. Gray and Siddharth Suri named it "ghost work" in their study of the hidden human labour behind automated services (2019): the labelling of images, the transcription of speech, the rating of outputs, the moderation of violent and abusive material. Journalistic investigations in the early 2020s documented data workers in Kenya and the Philippines labelling traumatic material for training data at wages that sometimes fell below two dollars an hour. Kate Crawford's *Atlas of AI* (2021) extended the inventory downward, to the extraction of minerals for batteries and chips and to the energy and water consumed by data centres. The Congo, whose ivory furnished the Traveller's machine, now furnishes, in cobalt dug by hand from its artisanal mines, the batteries of the devices on which the surface reads; Siddharth Kara documented the conditions of that digging in *Cobalt Red* (2023). The machine is still made of the world it will not visit.

The vocabulary of the industry repeats the novel's topology with an innocence that would have delighted Wilde. Computation is said to take place in "the cloud", a figure of the sky, while its machines stand in windowless halls — in one celebrated case, in a former nuclear bunker cut into the rock beneath Stockholm. The Traveller lamented that industry had lost its birthright in the sky; the industry of the present has kept the sky as a name and moved below. Mark Weiser, the computer scientist who founded the research programme of ubiquitous computing, stated its ideal in 1991 with perfect candour: "The most profound technologies are those that disappear." A technology that disappears into

the fabric of everyday life is a technology whose workings, and whose workers, the user no longer sees. Seamlessness is the Eloi's condition rendered as a principle of design.

The thermodynamic register has returned with the same technology. Every logical operation that erases information dissipates heat, as Rolf Landauer showed in 1961, and the data centres on which automated cognition depends have made their demand for electricity and cooling water a question of energy policy. The cloud is a heat engine. Its outputs are weightless; its inputs are coal, gas, uranium, water and wages; and its waste is heat. The four dimensions of appetite set out in Chapter One converge in a single infrastructure: the oikonomic in ghost work, the zoological in the extraction of bodies' time and health, the thermodynamic in the heat of computation, and the mediatic in the frictionless interface through which the surface consumes.

Biology, too, has re-entered the argument. The geneticist Lee M. Silver predicted in *Remaking Eden* (1997) that unequal access to reproductive genetic technologies could divide humanity, over generations, into an enhanced class he called the GenRich and an unenhanced class he called the Naturals, and that the two might in time become separate species. Yuval Noah Harari speculated in *Homo Deus* (2016) that automation might create a "useless class" without economic function, while biotechnology allowed an elite to upgrade itself. These are the Traveller's first theory restated in the idiom of the twenty-first century — the splitting of the species along the lines of social stratification — now with a mechanism faster than natural selection. A thesis must record the counter-arguments with equal care. Predictions of technological unemployment have been made and confounded since the Luddites; the mechanization Wells dreaded in 1895 he welcomed by 1927, on the ground that machines would abolish drudgery rather than breed drudges; many economists expect new tasks to replace those automated; and the distribution of the gains, not the technology itself, remains the decisive variable. The value of Wells's novel in this debate is not predictive but diagnostic. It identifies the conditions under which a division of labour becomes a division of kinds — invisibility, isolation, the end of exchange between strata — and those conditions can be observed, and contested, in the present tense.

The method of contesting them is the one the Traveller discovered by accident. At the end of the sixth chapter of *Capital*, Marx proposed to leave the noisy sphere of the market, where everything happens on the surface and in view

of all, and to follow buyer and seller "into the hidden abode of production, on whose threshold there stares us in the face 'No admittance except on business.'" Critique, for Marx, is a descent. The Traveller makes the same descent with a box of matches, climbing down a ventilation shaft into the thud of the engines and lamenting, as he strikes his last lights, that he had not brought a camera: "If only I had thought of a Kodak! I could have flashed that glimpse of the Underworld in a second, and examined it at leisure." The wish is the founding wish of documentary, from Jacob Riis's magnesium-flash photographs of New York tenements in *How the Other Half Lives* (1890) to the present investigations of the data-labelling industry. To bring a light below and record what it shows is the first act of an ethics that would refuse the Eloi's condition. It is also, the novel warns, a dangerous one. The Morlocks flee the light, and the Traveller's matches, which illuminate the Under-world, are also the weapons with which he sets it on fire. Knowledge of the hidden abode can be the beginning of solidarity or the beginning of a purge. *The Time Machine* contains both possibilities, and leaves its reader to decide which of them the light will serve.





And Abraham lifted up his eyes, and looked, and behold behind him a ram caught in a thicket by his horns: and Abraham went and took the ram, and offered him up for a burnt offering in the stead of his son.

— Genesis 22:13 (Authorized Version)

Padre, assai ci fia men doglia / se tu mangi di noi: tu ne vestisti / queste misere carni, e tu le spoglia.

— Dante, *Inferno* XXXIII.61–63

Chacun appelle barbarie ce qui n'est pas de son usage.

— Montaigne, "Des cannibales" (1580)

CHAPTER FIVE

5

Tempus Edax

Predation, Sacrifice and the Alimentary Code of the Human

THIS CHAPTER STATES THE theory of chronophagy in its general form. Its premise, drawn from the structural anthropology of Greek sacrifice, is that the human is defined by an alimentary code: a set of rules that place humanity between the gods, who do not eat flesh, and the beasts, who eat it raw and eat one another. Its thesis is that *The Time Machine* is an experiment in the dissolution of that code under deep time, and that the dissolution follows from an event the novel records almost in passing: the extinction of the sacrificial animal. In 802,701 "horses, cattle, sheep, dogs" are gone. The substitute that stood, since Abraham's ram, between humanity and its own flesh has disappeared, and with it the barrier between eating and cannibalism. Three corollaries organize what follows. First, chronophagy is at bottom autophagy: time devours what it has made, as Kronos devours his children and as a starving cell digests its own components, and 802,701 is a society in autophagy (5.1–5.2). Second, the canonical figures of exploitation — the vampire, the shepherd, the feedlot — are positions within the alimentary code, and Wells redistributes them (5.3–5.5). Third, the only exit from the code the novel proposes is not abstinence but transformation: the conversion of something edible into something kept, of food into a gift (5.6). A coda on games shows the code become procedure (5.7).

5.1
Theogonic Appetites: Kronos, Goya, Ugolino

Greek myth is crowded with fathers who eat their children, and the crowd is not incidental. Kronos swallows his offspring to prevent succession. Tantalus serves his son Pelops to the gods, who restore the boy — all but the shoulder that Demeter, distracted by grief, has eaten, which they replace with ivory. Atreus serves Thyestes his sons. In the first book of the *Metamorphoses*, Lycaon, king of Arcadia, tests Jupiter by setting before him the flesh of a hostage and is turned into a wolf for it (I.209–243), the first werewolf of European letters and the crime that precipitates the Flood; Marx's phrase for capital's appetite, a "werewolf hunger for surplus-labour", descends from Lycaon's table. In each case the devouring is a crime against succession, against the passage of time from one generation to the next. The father who eats his children refuses to be succeeded, and so refuses time. The cannibal father is the mythic figure in which time's ordinary direction, from parent to child, is turned back into consumption.

Goya painted the figure on the wall of his own house. *Saturn Devouring His Son* (c. 1820–23), one of the Black Paintings with which he covered the walls of the Quinta del Sordo outside Madrid, occupied, according to reconstructions of the house, the ground-floor room that served as its dining room. Peter Paul Rubens's *Saturn* (1636–38), painted for Philip IV's hunting lodge, shows the god as an old man biting decorously into an infant's chest beneath the three stars of his planet; Goya's Saturn, wild-eyed and naked, tears at a headless body with the frenzy of a starving thing. Rubens paints a mythological subject; Goya paints an appetite, and he painted it to be seen at table. Kronos at table, the emblem with which this thesis began, was a literal arrangement in a Spanish house of the 1820s: a man in old age, deaf and afraid, taking his meals beneath the image of time devouring its children.

Dante gives the figure its most terrible human form, and its most precise relevance to Wells. In the ninth circle of the *Inferno*, frozen in the ice of Cocytus, Count Ugolino della Gherardesca gnaws for ever at the skull of the archbishop who imprisoned him with his sons and grandsons in a tower in Pisa and let them starve (XXXIII). Ugolino tells how, when he bit his hands in grief, his children, thinking he did so from hunger, offered themselves to him: "Padre, assai ci fia

men doglia / se tu mangi di noi: tu ne vestisti / queste misere carni, e tu le spoglia" — Father, it will be far less pain for us if you eat of us; you clothed us in this wretched flesh, and you strip it off (61–63). The children die one by one, and Ugolino ends with the most debated line in the poem: "Poscia, più che 'l dolor, poté 'l digiuno" — then fasting had more power than grief (75). Whether hunger killed him or drove him to eat, Dante leaves open. The sons' offer, however, states the Morlock relation with uncanny exactness. The Morlocks clothe the Eloi — they make the garments the Eloi wear without knowing who made them — and they strip the flesh they have clothed. *Tu ne vestisti, e tu le spoglia*: those who clothe are those who strip. In the Under-world the providers and the devourers are one, and the gift of clothing is a stage in the preparation of food.

Freud inverted the figure and made it the origin of society. In *Totem und Tabu* (1913) he imagined a primal horde ruled by a violent father who kept the women for himself, until the expelled sons banded together, killed him and ate him; their guilt instituted the prohibitions of murder and incest, and the totem meal, in which the clan ritually kills and eats the animal that stands for the father, both commemorates and atones for the crime. The Freudian myth reverses Kronos — the sons eat the father instead of the father eating the sons — and places a cannibal meal at the foundation of law. 802,701 combines both versions. The Morlocks are the expelled brothers of Wells's Genesis, thrust out of the sunshine by the fathers of the Eloi, returned to eat the fathers' descendants; but there is no guilt, no totem and no law, and therefore no society in Freud's sense, only a meal repeated without commemoration. Wells's Under-world is the primal crime enacted nightly, without the remorse that would turn it into culture.

The biological name for such a system is autophagy. The term, coined by Christian de Duve in 1963, designates the process by which a cell under starvation digests its own components in order to survive; Yoshinori Ohsumi received the Nobel Prize in 2016 for elucidating its mechanisms. Autophagy is not pathology but emergency metabolism: the organism consumes part of itself so that the rest may live. The Traveller's final theory describes exactly this. "The feeding of the Under-world, however it was effected, had become disjointed. Mother Necessity ... came back again, and she began below", and "when other meat failed them, they turned to what old habit had hitherto forbidden." A species that has run out of other meat digests part of itself. Chronophagy, in its most general form, is autophagy at the scale of the species and of time: the process by which a

system that has exhausted its external resources sustains itself by consuming its own substance. Kronos eating his children is the myth of autophagic time. The Morlocks are its natural history.

5.2
The Cuisine of Sacrifice: Prometheus at Mecone and the Place of the Human

The structural anthropology of ancient Greece has reconstructed the code that the Morlocks break. In the essays collected as *La Cuisine du sacrifice en pays grec* (1979), Jean-Pierre Vernant and Marcel Detienne showed that Greek religion defined the human by its diet. The gods are immortal, feed on ambrosia and nectar, and receive from sacrifice only the savour of burning bones and fat; the beasts eat raw flesh and devour one another; human beings stand between, eating the cooked meat of domestic animals slain at the altar, together with the bread of cultivated grain. The founding myth of this order is Hesiod's account of Mecone (*Theogony* 535–616). Prometheus divided the first sacrificial ox, hiding the meat in the ox's stomach and wrapping the bare bones in glistening fat, and Zeus chose the bones. From that choice follow the human condition and its marks: human beings eat meat because they are mortal and hungry; they must labour for grain, since Zeus has hidden their livelihood; and they receive, as the price of the stolen fire, Pandora, whom Hesiod describes with misogynist venom as a drone that sits in the hive and reaps the toil of others into its belly (594–599). Vernant read the myth as the charter of human mortality, which is to say of the belly, the *gastēr*, that must always be filled.

Wells's three populations occupy the three positions of the code, and the novel's horror lies in their rearrangement. The Eloi eat as gods and as the golden race ate, abstaining from flesh and living on fruit; the Morlocks eat as beasts, consuming their own kind; the Traveller, demanding mutton, eats as a man. The middle term of the code, the position of the properly human, depends on a mediator: the domestic animal killed and cooked in place of the human. It is precisely this mediator that the future has lost. In 802,701 "horses, cattle, sheep, dogs, had followed the Ichthyosaurus into extinction." The Traveller records the fact as a curiosity of natural history. It is the event on which the whole Underworld depends.



At the Richmond dinner the substitute survives: the mutton is the ram in the thicket.

Figure 5.1 The alimentary code and its rearrangement. The gods receive the savour of burning bones and fat; human beings eat the cooked meat of domestic animals slain at the altar; the beasts eat raw flesh and one another. In 802,701 the Eloi eat as gods, the Morlocks as beasts, and the Traveller, demanding mutton, eats as a man — but the mediator, the domestic animal, is extinct.

The biblical tradition makes the logic explicit. In the binding of Isaac (Genesis 22), Abraham is commanded to offer his son as a burnt offering and is stopped at the last moment by an angel; he sees "a ram caught in a thicket by his horns" and offers it up "in the stead of his son" (22:13). The episode founds the institution of substitution — the animal in the stead of the child — on which the prohibition of Molech's rite depends. René Girard generalized the insight in *La Violence et le sacré* (1972): sacrifice channels a community's violence onto a surrogate victim, and when substitution fails, the community falls into a sacrificial crisis in which violence returns upon its own members. The world of 802,701 is the sacrificial crisis made permanent by extinction. The ram is gone; nothing stands in the stead of the son; and so the Morlocks eat Isaac. The Richmond dinner, by contrast, still possesses its substitute. The mutton the Traveller demands is the ram in the thicket, the animal that stands between the late-Victorian professional class and the Morlock's table, and it is by eating the substitute that the guests remain human while they listen to the story of a world without one.

Homer had already set the code's violators side by side, and he set them in an after-dinner tale. The ninth book of the *Odyssey*, which Odysseus narrates at the banquet of the Phaeacians, passes from the Lotus-eaters to the Cyclopes. Those who taste the honey-sweet lotus lose all wish to go home and want only to stay among the lotus-eaters, forgetful of their return (82–104). The Cyclopes live in a land of golden abundance where, trusting in the gods, they neither plant nor plough, yet everything grows for them unsown; they have neither assemblies nor laws; and Polyphemus, a shepherd who milks his ewes in a cave, devours Odysseus's companions raw, entrails, flesh and marrowy bones, like a mountain lion (106–115, 288–293). Pierre Vidal-Naquet's "Land and Sacrifice in the *Odyssey*" (1970) read the poem's peoples as a map of the alimentary code, in which those who eat bread and sacrifice are human and the rest are placed above or below them. The fit with Wells is astonishing. Homer's ninth book sets forgetful fruit-eaters beside cave-dwelling cannibal herdsmen in a Golden Age land without law. Odysseus, who calls himself "Nobody", blinds the herdsman with a stake hardened in the fire and escapes beneath the bellies of his sheep; the Traveller, who has no name, holds off the light-blinded cave-dwellers with fire and escapes. And Odysseus tells the story at the feast of a people who, in their king's words, love above all the banquet and the lyre, the dance, changes of clothing, warm baths and bed (VIII.248–249) — a people of perpetual pleasure, the Eloi

as hosts. *The Time Machine* is the ninth book of the *Odyssey* told at Richmond: the after-dinner narrative of a voyage among lotus-eaters and cyclopes, delivered to diners who enjoy the security the voyage has called into question.

Tennyson made the Lotus-eaters a Victorian temptation. In "The Lotos-Eaters" (1832; revised 1842) Odysseus's mariners come "unto a land / In which it seemed always afternoon", and their choric song asks "why / Should life all labour be?" before resolving to lie on the hills like gods, careless of mankind, while a race of men below sows the seed and reaps the harvest with enduring toil. The Eloi live in Tennyson's afternoon, and the novel answers the mariners' question. Life need not all be labour, provided someone else's life is.

5.3
Vegetarian Monsters and Carnivorous Gentlemen: Shelley, *Frankenstein*, Montaigne, Swift

The Romantic tradition had already reversed the moral valence of diet that Wells inherits. Percy Bysshe Shelley's *A Vindication of Natural Diet* (1813), expanded from a note to *Queen Mab*, read the myth of Prometheus as an allegory of the fall into meat: Prometheus first applied fire to cookery, and the vulture that devours his vitals is the disease that flesh-eating brought upon the race. On this reading the Golden Age ended not with labour but with the kitchen, and the cure for civilization was a return to fruit and grain. Mary Shelley gave the doctrine to her monster. Victor Frankenstein's creature, pleading for a companion, promises to withdraw with her to the wilds of South America and live as the golden race lived: "My food is not that of man; I do not destroy the lamb and the kid to glut my appetite; acorns and berries afford me sufficient nourishment." The monster is a vegetarian in a carnivorous society, and his education — he learns history by overhearing a reading of Volney's *Ruins of Empires* — makes him a philosopher of ruins as well. Carol J. Adams, in *The Sexual Politics of Meat* (1990), argued that meat-eating depends on an "absent referent": the living animal disappears into the word for its flesh. Mary Shelley's creature restores the referent by refusing the meat.

The absent referent has a history in English that Walter Scott made famous. In the opening chapter of *Ivanhoe* (1819) the jester Wamba explains to the swineherd Gurth that the beast is called by its Saxon name, swine, while it lives and is

tended by Saxon serfs, but becomes pork, a Norman word, when it is carried to the castle hall to be eaten by Norman masters; the ox and the calf become beef and veal in the same passage from field to table. The language encodes a class division between those who raise the animal and those who eat it. The Traveller does not ask for sheep. He asks for mutton, the Norman word, the name the animal acquires at the master's table, and the novel's first alimentary sentence thus performs the conversion it will spend its narrative exposing: a living creature tended by others becomes, in the dining-room, a dish without a past.

Montaigne had used the cannibal to reverse the moral geography of eating long before the Romantics. "Des cannibales" (1580), drawing on accounts of the Tupinambá of Brazil, argues that every people calls barbarism whatever is not its own custom, and finds more barbarity in Europe's tortures of the living than in the Tupinambá's ritual eating of the dead. Its closing pages report the visit of three Tupinambá to Rouen in 1562, where they were shown the young king and asked what they had found most remarkable. They had noticed, among other things, that there were men in France gorged with every kind of good thing while others, whom in their language they called the *halves* — *moitiés* — of the rich, begged at their doors, gaunt with hunger and poverty; and they found it strange that these needy halves could endure such injustice and did not take the others by the throat or set fire to their houses. Montaigne's cannibals see in France the two halves of a single humanity, one gorged and one starving, and wonder why the starving have not yet turned on the gorged. Three centuries later Wells answered their question. Given eight hundred thousand years, the starving half turns on the gorged and eats it. Hesiod's maxim that the half is more than the whole acquires, beside Montaigne's halves, a second meaning: the half that consumes the whole destroys the species of which it was half.

Swift had already performed the speciation of metaphor in prose, with a precision that makes him Wells's most direct precursor in method. *A Modest Proposal* (1729) takes the commonplace that landlords devour the Irish poor and runs it to its literal conclusion: the children of the poor shall be raised and sold as food for the rich, since "a young healthy child well nursed is at a year old a most delicious, nourishing, and wholesome food", and the dish will be dear and therefore "very proper for landlords, who, as they have already devoured most of the parents, seem to have the best title to the children." The figure is made flesh by a change of tense: what the landlords have done figuratively to the parents they shall do literally to the children. Wells changes the tense by eight hundred thou-

sand years and reverses the direction of the meal. In the fourth voyage of *Gulliver's Travels* (1726), Swift also supplied the model of the refusal of kinship. The Houyhnhnms, rational horses, live on oats and milk; the Yahoos, degenerate humanoids, hairy, filthy and voracious, are kept as beasts of burden; and Gulliver, recognizing his own form in the Yahoo, recoils into a disgust that survives his return home, where he cannot bear the smell of his wife and children and spends his days conversing with horses in the stable. The rational vegetarians are masters and the degenerate carnivores their livestock: Swift's arrangement is the Eloi and the Morlocks with the power reversed, and Gulliver's horror at recognizing himself in the Yahoo is the Traveller's horror at the Morlocks. Wells, who read Swift in the library at Uppark, wrote a fourth voyage in which the Yahoos have learned to keep the Houyhnhnms.

5·1
Vampire Capital and the Inversion of Moretti's Dialectic

The most influential Marxist reading of the nineteenth-century monster gives this section its point of departure. In "The Dialectic of Fear" (1982) Franco Moretti argued that Mary Shelley's creature and Bram Stoker's Dracula are the two faces of the fear that haunted bourgeois society. The creature, assembled from the dead by a scientist who then disowns it, collective, nameless and vengeful, is the proletariat; Dracula, an aristocrat who hoards gold, cannot stop accumulating and lives on the blood of others, is capital in its monopolistic form. Moretti grounded the second figure in Marx's own metaphor of capital as dead labour that lives, vampire-like, by sucking living labour. The dialectic is a symmetry. The bourgeoisie fears the monster it has made beneath it and the monster it might become above.

The Time Machine collapses the symmetry two years before *Dracula* appeared. Its Morlocks are Moretti's proletarian monster — made by the order that thrust them underground, collective, nameless, avenging — and they have acquired the ecology of the vampire: nocturnal, subterranean, averse to light, emerging at night to feed on the inhabitants of the surface. Stoker's count, sleeping by day in boxes of his native earth in the chapel at Carfax and hunted by a band of men armed with light, is in this respect a Morlock with a title. Wells's

inversion of Marx's metaphor is complete. In *Capital*, dead labour sucks the blood of the living; in 802,701, the living labour of the Under-world eats the descendants of those who once lived on it. The monster created by exploitation becomes the exploiter, and the vampire turns out to be not a class but whatever occupies the receiving end of the gradient when the gradient reverses.

Marx and Engels had imagined the reversal in the *Communist Manifesto* (1848), in an image that names the Under-world directly. Modern bourgeois society, they wrote, "is like the sorcerer, who is no longer able to control the powers of the nether world whom he has called up by his spells"; and "what the bourgeoisie therefore produces, above all, is its own grave-diggers." The Morlocks are the powers of the nether world in the most literal sense the century could devise. They are also, in their way, the grave-diggers: the class that buries its masters not in the earth but in itself.

The vampire economy has received its most literal modern form in Andrew Niccol's film *In Time* (2011), in which time itself is currency. Genetic engineering has stopped human ageing at twenty-five, after which every person has a single year of life, displayed as a luminous clock on the forearm; wages, prices and debts are paid in minutes; the poor of the ghetto called Dayton live a day at a time and die when their clocks run out, while the rich of the zone called New Greenwich accumulate centuries. The premise is Benjamin Franklin's maxim that time is money pressed until it becomes physiology, and the film's plutocrat states the logic of the system plainly: for a few to be immortal, many must die. The name of the rich district completes the argument of Chapter Two. The prime meridian of standard time, fixed at Greenwich in 1884, has become the address of the class that eats the time of others. The Spierig brothers' *Daybreakers* (2009) carries the vampire economy to its Malthusian limit. In a world where vampires are the majority and human beings are farmed for blood, the supply runs short, and vampires deprived of human blood, or driven to drink that of their own kind, degenerate into feral, bat-like creatures. Autophagy accelerates degeneration. The process Wells spread across eight hundred millennia is compressed, in the vampire economy, into a decade.

The lesson of the inversion is that the vampire is a position, not a nature. It is the occupant of the receiving end of a gradient through which life flows from one population to another, and Wells's novel is the first fiction to show that the position can change hands while the gradient remains. To abolish the vampire it is not enough to change who drinks. The gradient itself must be refused.

5.5
Pastoral Power: Foucault, Agamben and the Feedlot

The Morlocks do not merely eat the Eloi; they keep them, and the keeping is the more disturbing part. The Traveller infers that they make the Eloi's garments, maintain their halls, provide for their habitual needs and "probably saw to the breeding of" them. Michel Foucault gave this mode of power its genealogy. In the Tanner Lectures he delivered at Stanford in 1979, "*Omnes et Singulatim*: Towards a Critique of Political Reason", and in his course of 1977–78, *Sécurité, territoire, population*, he traced to the Hebrew image of God as shepherd a form of power exercised not over a territory but over a flock: a power that gathers, guides, feeds and watches over its subjects, all and each, and knows each one intimately in order to secure the salvation of all. Foucault observed that the Greeks had considered and rejected the model. In Plato's *Statesman*, the divine shepherd belongs to the age of Kronos; in our age, the Stranger concludes, the ruler is not a herdsman but a weaver who binds together the threads of the city. Foucault's genealogy therefore runs straight through the myth that Chapter Three found at the heart of 802,701. The world of the Eloi is the age of Kronos restored, and with it the shepherd returns — as the Morlock, a pastor whose knowledge of the flock serves the table.

The modern form of pastoral power is what Foucault called biopower. In the last chapter of *La Volonté de savoir* (1976) and in the lectures *Il faut défendre la société* of the same year, he contrasted the old sovereign right to take life or let live with a modern power to make live and to let die, which administers populations through their births, health, longevity and diet. The Morlocks exercise a biopower of terrible simplicity: they make live in order to make die. Their care for the Eloi is husbandry, and the Eloi's security — the abolition of disease, war and want that the Traveller first took for the triumph of civilization — is the security of livestock. Giorgio Agamben's *Homo Sacer* (1995) supplies the category for the managed: bare life, *zoē*, the mere fact of living, stripped of the qualified life, *bios*, that belongs to a political community, and figured by the archaic Roman *homo sacer*, who could be killed but not sacrificed. The Eloi are bare life in Agamben's sense. They are killed without ritual and eaten without sacrifice;

no altar mediates their death; they are neither victims nor criminals but provisions.

The modern economy built the feedlot for animals long before fiction imagined it for human beings. The Union Stock Yards of Chicago opened in 1865, and the city's meat-packers devised the overhead trolley along which carcasses moved past stationary workers who each made a single cut. Henry Ford acknowledged in *My Life and Work* (1922) that the idea of the moving assembly line came to him, in a general way, from the overhead trolley the Chicago packers used in dressing beef. Mass production began as mass disassembly: the line that builds the machine descends from the line that takes the animal apart. Upton Sinclair's *The Jungle* (1906), written to expose the condition of the workers in Packingtown, recorded the period's most grisly literalization. Men who worked over the open vats in the cooking rooms sometimes fell in, and, Sinclair wrote, there was never enough of them left to be worth exhibiting: all but their bones went out to the world as Durham's Pure Leaf Lard. Sinclair said that he had aimed at the public's heart and by accident hit it in the stomach. The public, reading of workers rendered into its breakfast, demanded pure food rather than humane labour, and the Meat Inspection Act of 1906 followed. The episode is a parable of the Eloi's ignorance: the consumers above recoiled from the contamination of their food, not from the consumption of those who prepared it.

The twentieth century perfected the feedlot for animals and gave it a name. Ruth Harrison's *Animal Machines* (1964) described the intensive confinement of poultry, pigs and calves — the battery cage, the veal crate — and brought the phrase "factory farming" into public debate; the Brambell Report of 1965 answered with the principle of minimal freedoms for farmed animals. The industrial system has produced a pathology that bears directly on the alimentary code. Bovine spongiform encephalopathy, which emerged in Britain in the 1980s, spread through the feeding of cattle, herbivores by nature, with meat-and-bone meal rendered from other cattle: the disease that killed them was transmitted by enforced cannibalism. Claude Lévi-Strauss, reflecting on the epidemic in "La leçon de sagesse des vaches folles" (1996), and earlier on kuru, the prion disease once transmitted among the Fore of Papua New Guinea through funerary practices of eating the dead, concluded that the line between alimentary custom and cannibalism is a convention, and that modern societies cross it in forms they do not name. The mad cow is the Eloi of the industrial age: a herbivore made to eat its own kind by the system that keeps it, and destroyed by the consequence.

These are not digressions. They are the present form of the structure the novel imagined, and the feedlot of 802,701 is, in the twenty-first century, the ordinary condition of tens of billions of domesticated animals each year. Peter Singer's *Animal Liberation* (1975) argued that a boundary of moral concern drawn at the species is a prejudice analogous to racism and sexism; his critics have replied that moral status depends on capacities that species membership tracks, and that the comparison trivializes human oppressions. A thesis need not resolve that debate to register the structural point. *The Time Machine* produces its horror by moving a human population into the position that industrial modernity reserves for animals, and the strength of the horror measures how much depends on the reservation.

5.6
Eating Well: Derrida, Andrade and the Ethics of Incorporation

If eating is the form in which time becomes perceptible, an ethics of time must be an ethics of eating, and abstinence is not available. Every living thing consumes; the Eloi's frugivory is no innocence, since they are fed by a labour that ends by feeding on them; and the Traveller, starving, cannot live on principle. Jacques Derrida formulated the problem in his interview with Jean-Luc Nancy, "'Il faut bien manger' ou le calcul du sujet" (1989). One eats the other in any case, he argued, literally or symbolically, by assimilating, interiorizing and identifying; the moral question is therefore not whether to eat but how to eat well, and it is a question of measure, of hospitality, of learning and giving to eat. One never eats entirely alone. Derrida named the sacrificial structure of the Western subject *carnophallogocentrism* — the virile, flesh-eating subject who accepts the putting to death of animals as the condition of being a subject at all — and proposed that the ethics of the other begins where that structure is questioned.

Christianity placed eating at the centre of memory. At the Last Supper, according to the Gospels and Paul, Jesus took bread and said, "Take, eat; this is my body" (Matthew 26:26), and, in Luke's and Paul's accounts, "this do in remembrance of me" (Luke 22:19; 1 Corinthians 11:24). The Eucharist is an alimentary anamnesis, an eating that remembers, the sacramental counterpart of Proust's madeleine. The Morlocks' eating forgets: it leaves no commemoration,

no rite, no word. The Eloi, whose name recalls the cry from the cross, are consumed without being remembered by those who consume them. The contrast specifies what chronophagy destroys. In the human code eating is not only nutrition but the medium of memory and community — the feast, the sacrifice, the communion — and the horror of 802,701 is that it has reduced eating to nutrition alone.

Psychoanalysis gave incorporation its theory. In "Trauer und Melancholie" (1917) Freud described mourning as the slow withdrawal of attachment from a lost object and melancholia as an identification in which the lost object is taken into the ego, in a fantasy that recalls the oral wish to devour. Nicolas Abraham and Maria Torok distinguished introjection, by which the self grows by assimilating what it has lost, from incorporation, the fantasy of swallowing the lost object whole and keeping it, undigested, in a psychic crypt (*L'Écorce et le noyau*, 1978). The distinction illuminates the end of *The Time Machine*. The narrator neither digests the Traveller's tale nor discards it; he keeps two flowers, which keep the lost in a form the living can hold. The pressed flower is a crypt in Abraham and Torok's sense, and also a relic: the remainder of a loss kept outside the body.

The most radical theory of incorporation comes from Brazil. Oswald de Andrade's *Manifesto Antropófago* (1928) begins "Só a Antropofagia nos une. Socialmente. Economicamente. Filosoficamente" — only anthropophagy unites us, socially, economically, philosophically — and proposes the ritual cannibalism of the Tupinambá, the people Montaigne described, as the model of a culture that becomes itself by devouring the culture of its colonizers. The anthropologist Eduardo Viveiros de Castro developed the idea into a philosophy. In his studies of Amazonian thought, culminating in *Métaphysiques cannibales* (2009), he describes a perspectivism in which every kind of being sees itself as a person and sees others according to its body: what human beings see as blood, the jaguar sees as manioc beer, and where human beings see themselves as people, jaguars see them as prey. Among the Araweté, whose warriors sing after a killing in the voice of the slain enemy, and among the Tupinambá before them, cannibalism was a technology of perspective: one ate the enemy in order to take up the enemy's point of view. Perspectivism makes the deepest sense of Wells's Under-world available. To the Morlocks the Eloi are not persons who are eaten but game; to the Eloi the Morlocks are not persons but the terror of the dark; only the Traveller, the anthropologist from another age, sees both as human, and even he can-

not hold the perception steady. The horror of 802,701 is that of a world in which the human perspective has split into two that can no longer see each other as people. What Viveiros de Castro's Amerindians accomplish by ritual, the taking-up of the enemy's point of view, the Morlocks have forgotten, and the Traveller achieves only for a moment, in the firelight, when he sees their misery and strikes no more.

The ethics this chapter has been approaching can now be stated in alimentary terms. It is neither the Eloi's false innocence nor the Morlocks' appetite, but a third relation in which something that could be eaten is kept. Hesiod's mallow was food, the food of the frugal poor, praised as the half that is more than the whole. Wells's flowers, "not unlike very large white mallows", are not eaten. They are carried, pressed and kept. The transformation of a food into a gift by refraining from consuming it is the elementary act of the ethics of the remainder, and its model is not the fast — Kafka's hunger artist, who starved because he could never find the food he liked, is replaced at the end of his story by a young panther from whose throat the joy of life streams — but the keeping of what could have been consumed. To eat well, in Derrida's phrase, is finally to leave something uneaten, and to give it.

5.7
Ludic Chronophagy: *BioShock*, *Chrono Trigger*, *Horizon Zero Dawn*, *Attack on Titan*, *The Promised Neverland*

The Traveller calls the Morlocks "ant-like", and the epithet recalls the oldest fable of labour and play. In Aesop's fable, retold by La Fontaine as "La Cigale et la Fourmi" (1668), the cicada sings all summer while the ant stores grain; when winter comes and the cicada begs for food, the ant asks what she did in the warm season and, told that she sang, replies: "Vous chantiez ? j'en suis fort aise. / Eh bien ! dansez maintenant." You sang? I am delighted. Well then, dance now. The Eloi, whose remaining art is to adorn themselves with flowers, to dance and to sing in the sunlight, are the cicadas of a summer that has lasted eight hundred thousand years, and the Morlocks are ants who do not merely refuse to feed them. Bernard Suits's *The Grasshopper: Games, Life and Utopia* (1978) rewrites the fable as philosophy. Its Grasshopper, who refuses to work and dies, argues

that in a utopia where machines have made all instrumental activity unnecessary the only thing left worth doing would be to play games, which he defines as the voluntary attempt to overcome unnecessary obstacles. The Eloi refute Suits in advance. They live in his utopia and do not play games. They play, in Johan Huizinga's sense of free and unproductive activity (*Homo Ludens*, 1938), but without rules, goals or obstacles, and the only game in their world is the one being played on them at night.

The video game is the medium in which chronophagy has become procedural. Ian Bogost's *Persuasive Games* (2007) argued that games make arguments through their rules, by what he called procedural rhetoric, and the rules of games have from the beginning been rules about eating and being eaten in time. Toru Iwatani designed *Pac-Man* (1980) around the act of eating, deriving its hero, by his own account, from a pizza with a slice removed; its maze is a feedlot in which the player eats and is hunted by ghosts, until a power pellet briefly reverses the relation and the hunted becomes the eater — the Traveller's box of matches rendered as a mechanic. The text adventure *Zork* (1980) warned its players in the dark, "It is pitch black. You are likely to be eaten by a grue", borrowing the name of its unseen predator from Jack Vance's far-future *The Dying Earth* (1950); and the survival game *Don't Starve* (2013), whose players manage hunger, sanity and health, kills anyone who remains in complete darkness at night by means of an invisible attacker. The forest night of 802,701, in which the Traveller's life depends on keeping a fire alight against things that strike from the dark, has become a principle of design.

Three games of the last three decades make the alimentary code their explicit subject. In *BioShock* (2007), the undersea city of Rapture, founded as a refuge from every moral constraint on the market, has collapsed into a population of addicts dependent on ADAM, a genetic material that young girls, the Little Sisters, harvest from corpses under the protection of armoured guardians. Encountering each Sister, the player must choose: to harvest her, killing her to extract the full measure of ADAM, or to rescue her, freeing her and receiving less. The game rewards restraint with gifts, delivered later and unexpectedly by the scientist who shelters the freed children, so that its moral economy turns on the difference between consumption and the gift. *Chrono Trigger* (1995), whose title carries the name of Kronos, sets its time-travelling heroes against Lavos, a parasitic being that fell to earth sixty-five million years ago, burrowed into the planet and has fed on its life ever since, until it erupts in 1999 and leaves a future of ruined domes

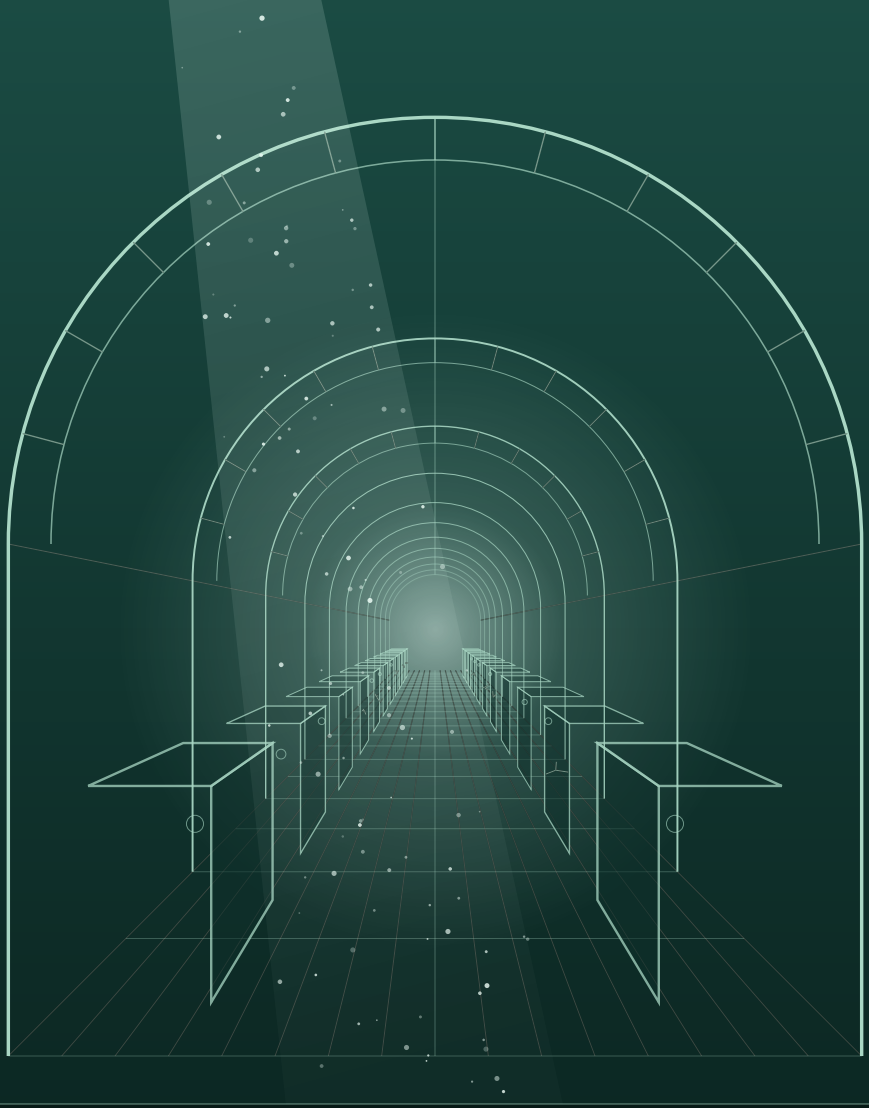
where survivors starve beside a machine that restores their strength but cannot feed them. In one of those domes the heroes find a seed and give it to the survivors to plant. In an earlier age the game shows the magical kingdom of Zeal, whose enlightened inhabitants live on floating islands above a frozen earth where the unenlightened shelter in caves, and whose queen draws her power from Lavos through an engine called the Mammon Machine, after Milton's mining devil. *Horizon Zero Dawn* (2017) locates its catastrophe in a swarm of self-replicating military machines that consumed the biosphere as fuel, and its recovery in a buried project that reseeded the earth with life after the swarm was shut down — a project whose archive of human knowledge was deliberately erased by the industrialist responsible for the swarm. Consumption, seed and archive: the three games compose, in another medium, the argument of this thesis.

Two manga carry the code to its limit. In Hajime Isayama's *Attack on Titan* (2009–21), humanity lives behind walls, penned like livestock, against giant humanoid Titans that eat human beings; the Titans, it emerges, have no digestive system and regurgitate the people they swallow, so that their eating nourishes nothing. Consumption without metabolism is chronophagy in its purest form, eating as compulsion rather than need. The Titans are, moreover, transformed human beings, and the walls that protect the city are built of dormant Titans; the power of the strongest Titans passes from one human host to the next when the new host eats the old, transmitting the memories of every previous holder. Cannibalism becomes the medium of historical memory, and the city's safety is literally made of the bodies of the eaters. In Kaiu Shirai and Posuka Demizu's *The Promised Neverland*, discussed in Chapter Three as a feedlot, the demons must eat human beings to keep their intelligence and their human-like form; deprived of human flesh, they degenerate into mindless beasts. The predators' humanity is borrowed from their prey. The novel's anthropology is inverted here with a precision that returns us to Viveiros de Castro: the eater becomes what it eats, and the Morlocks' retention of "rather more initiative" than the Eloi is, in the manga's terms, the intelligence of the herd digested.

The ludic tradition thus confirms the theory from an unexpected direction. Games turn the alimentary code into rules, and the rules rediscover, again and again, the structures Wells imagined: the predator in the dark, the light that holds it off, the harvested child, the seed planted in the ruins, the archive erased, the eater who becomes what it eats. Games also add something the novel could only imply. They place the player in the position of the eater and make the

choice between harvest and rescue, between consumption and the gift, an action rather than a contemplation. In that respect the medium has realized what the narrator of *The Time Machine* asks of his readers: not merely to witness the remainder, but to choose it.





... of making many books there is no end; and much study is a weariness of the flesh.

— Ecclesiastes 12:12 (Authorized Version)

Clearly we stood among the ruins of some latter-day South Kensington!

— H. G. Wells, *The Time Machine* (1895)

Round the decay / Of that colossal Wreck, boundless and bare / The lone and level sands stretch far away.

— Percy Bysshe Shelley, "Ozymandias" (1818)

CHAPTER SIX

6

The Palace of Green Porcelain

Archive, Ruin and the Futility of Records

THE PALACE OF GREEN Porcelain is the novel's theory of the archive, and its verdict is that an archive does not so much preserve the past as store it as fuel. The Traveller enters the ruined museum in search of knowledge and leaves it armed. Its books have rotted to rags, its plants to dust, its specimens to fragments threaded as ornaments; from its galleries he takes a lever for a club, matches from an airtight case, and a jar of camphor that will burn like a candle. This chapter argues that the museum is the place where the four registers of chronophagy converge upon the record itself. Decay eats the archive; the Morlocks carry off its cases; the Eloi make jewellery of its fossils; and the Traveller mines it for fire. The chapter reconstructs the institutions on which the episode draws (6.1), the tradition of the future traveller among the ruins of the present (6.2), the museum's long habit of imagining its own ruin (6.3), the fate of scientific records (6.4), the archive as arsenal (6.5), the theory of the archive from Benjamin to Derrida (6.6), and the modern attempts to send messages across deep time (6.7). Against the archive as deposit it sets the model the flowers embody: the remainder as a gift carried by a witness.

6.1
South Kensington in Ruins

The Traveller identifies the building at once. Walking with Weena through its dusty galleries, he recognizes by the oblique feet the skeleton of "some extinct creature after the fashion of the Megatherium", finds beyond it "the huge skeleton barrel of a Brontosaurus", and exclaims: "Clearly we stood among the ruins of some latter-day South Kensington!" The district he names was the most ambitious cultural precinct of Victorian Britain. With the profits of the Great Exhibition of 1851 the commissioners bought an estate in South Kensington and built upon it the museums, colleges and concert hall that Londoners nicknamed Albertopolis: the South Kensington Museum, later the Victoria and Albert; the Royal Albert Hall (1871); and the Natural History Museum, opened in 1881 in Alfred Waterhouse's Romanesque terracotta building, the cathedral of nature for which Richard Owen had campaigned. Wells knew the precinct as a student. The Normal School of Science, where he studied under Huxley from 1884, stood on Exhibition Road in a building that now belongs to the Victoria and Albert Museum. The Palace of Green Porcelain is the ruin of his own university quarter.

Its contents are the inventory of Victorian science. The Megatherium, the giant ground sloth of South America, had been reconstructed by Owen; the Brontosaurus had been named by the American palaeontologist Othniel Charles Marsh only in 1879, during the competitive excavations of the American West, and the Traveller's mention of it is up to date. Victorian London had already staged a famous meal inside such a reconstruction. On the last night of 1853 the sculptor Benjamin Waterhouse Hawkins gave a dinner for Owen and a party of men of science inside the mould of his life-size Iguanodon, built for the grounds of the re-erected Crystal Palace at Sydenham. The Victorian imagination of deep time began, quite literally, at table, inside the body of an extinct beast. In 802,701 the meal is over and the beasts are bones again.

The Traveller describes the building's exterior with equal precision: "the façade had an Oriental look: the face of it having the lustre, as well as the pale-green tint, a kind of bluish-green, of a certain type of Chinese porcelain." Glazed porcelain was among the oldest commodities of the global trade that South

Kensington's collections celebrated, and the green glaze recalls the celadon wares that Europe had imported from China for centuries. A museum of science clad in the most prized commodity of the China trade is an emblem of the alliance between knowledge and commerce on which the Great Exhibition was founded, and the ruin of the palace is the ruin of the alliance.

What the Traveller finds inside is an archive at the end of its resistance. "The inevitable process of decay that had been staved off for a time, and had, through the extinction of bacteria and fungi, lost ninety-nine hundredths of its force, was nevertheless, with extreme sureness if with extreme slowness at work again upon all its treasures." The sentence states the thermodynamic law of the archive. Preservation is decay staved off — Hegel's phrase for work, *aufgehaltenes Verschwinden*, vanishing held back, discussed in Chapter Four — and decay resumes even when its ordinary agents, the bacteria and fungi, are extinct. The galleries of natural history have fared worst: "desiccated mummies in jars that had once held spirit, a brown dust of departed plants: that was all!" The herbarium has become dust. The novel thus contains two kinds of dead flower, and the contrast between them is the argument of this thesis in miniature. The museum's plants, collected, classified and preserved, have crumbled into "a brown dust". Weena's flowers, picked for a single person and carried in a pocket, survive, "brown and flat and brittle", as testimony. The institutional archive has failed where the gift has not.

The inhabitants of 802,701 relate to the museum in characteristic ways. The Eloi play in it. The Traveller finds "traces of the little people in the shape of rare fossils broken to pieces or threaded in strings upon reeds", and Weena amuses herself by rolling a sea-urchin down the sloping glass of a case. The archive has become a source of ornaments and toys, knowledge reduced to decoration, the last phase of the aesthetic economy described in Chapter Three. The Morlocks salvage it: "the cases had in some instances been bodily removed — by the Morlocks as I judged." What they take, and why, the Traveller cannot tell, but the pattern fits everything else known of them, a population of mechanics who use what the surface has forgotten. Between the Eloi's jewellery and the Morlocks' salvage, the museum is being consumed from both ends. Georg Simmel described the ruin, in his essay of 1907, as the site where nature takes its revenge on the spirit that had formed it, and where the balance of the two produces a strange peace. The Palace of Green Porcelain offers no such peace. Its ruin is not

nature's slow reclamation of form but a triple consumption — by decay, by play and by salvage — in which every party takes what it can use and nobody reads.

6.2
The New Zealander: Macaulay, Volney, Shelley, Horace Smith, Gibbon, Doré

The scene in the Palace belongs to a tradition older than the museum: the traveller from the future among the ruins of the present. Horace Walpole, writing to Horace Mann in 1774, foresaw a curious traveller from Lima visiting England to describe the ruins of St Paul's, as eighteenth-century antiquaries described Baalbek and Palmyra. Constantin-François de Volney opened *Les Ruines, ou Méditation sur les révolutions des empires* (1791) among the ruins of Palmyra and had his traveller wonder whether some future wanderer would one day sit among the silent ruins of the cities on the Seine, the Thames or the Zuyder Zee and weep for their peoples; the book was so widely read that Mary Shelley made it her creature's first lesson in history. Anna Laetitia Barbauld's *Eighteen Hundred and Eleven* (1812) imagined young Americans touring the ruins of London as Britons toured Rome, and was savaged by reviewers for its lack of patriotism. In 1818 Percy Shelley and his friend Horace Smith wrote rival sonnets on the colossus of Ozymandias. Shelley's became one of the most famous poems in the language; Smith's turned the desert into the Thames valley, imagining a hunter "Where London stood, holding the Wolf in chace", who meets some fragment huge and stops to guess "What powerful but unrecorded race / Once dwelt in that annihilated place." Smith's hunter is the Traveller's closest ancestor. He wanders a wilderness that was London, finds the fragment of a forgotten civilization and cannot interpret it.

Thomas Babington Macaulay gave the tradition its canonical figure in a review of Ranke's history of the popes in the *Edinburgh Review* (1840), predicting that the Roman Church "may still exist in undiminished vigour when some traveller from New Zealand shall, in the midst of a vast solitude, take his stand on a broken arch of London Bridge to sketch the ruins of St. Paul's." The New Zealander became a Victorian commonplace, invoked in leading articles and cartoons, and Gustave Doré gave him a body in the final plate of *London: A Pilgrimage* (1872), his collaboration with Blanchard Jerrold: a cloaked figure seated

on a broken arch above a ruined city, sketching. Edward Gibbon had supplied the figure's psychology a century earlier, recalling in his memoirs that the idea of writing the decline and fall of Rome first came to him on 15 October 1764 "as I sat musing amidst the ruins of the Capitol, while the barefooted fryars were singing vespers in the temple of Jupiter". The historian of decline begins among ruins in which a later religion sings in the temple of an earlier one. The Traveller in the Palace is Gibbon's musing figure without Gibbon's literary vocation — "Had I been a literary man I might, perhaps, have moralized upon the futility of all ambition" — and Macaulay's New Zealander without a sketchbook; he takes a camera only on his second departure.

By 1895 the New Zealander had arrived in London in another form. In 1882 the sailing ship *Dunedin*, fitted with refrigerating machinery, carried the first successful cargo of frozen mutton and lamb from New Zealand to London, and within a decade frozen meat from the southern colonies had become a staple of the metropolitan market. Macaulay's figure, imagined as the spectator of London's ruin, came instead as its supplier, and he came as meat. There is no evidence about the provenance of the mutton on the Traveller's table, and none is needed for the structural point. The late-Victorian dinner was supplied by a global cold chain whose refrigerating engines ran the thermodynamic cycle in reverse to manufacture cold, and whose periphery fed the centre. The metropolis that imagined its own ruin as a colonial traveller's spectacle was already eating the colonies' flocks.

The tradition has a coda in the present. In August 2015 the forces of the so-called Islamic State, having occupied Palmyra, executed Khaled al-Asaad, the octogenarian scholar who had directed the site's antiquities for four decades, after he refused to reveal where its movable treasures had been hidden, and in the following weeks demolished the temples of Baalshamin and of Bel, among the ruins at which Volney had meditated. The ruin that had been Europe's emblem of the mortality of empires was itself destroyed, and the man who died for it died protecting a remainder. The archive's last defence, in that case, was not a building or a catalogue but a custodian's silence.

6.3
Hubert Robert's Louvre and the Museum as Prophecy of Its Own Ruin

The museum has always imagined its own ruin, and the first great public museum did so within three years of opening. The Louvre opened to the public as the Muséum central des arts in 1793, and at the Salon of 1796 Hubert Robert, one of the painters charged with its arrangement, exhibited two pictures of its Grande Galerie: a design for the gallery as it might be lit and hung, and an imaginary view of the same gallery in ruins, its vault collapsed to the sky, its statues broken, an antique Apollo standing amid the rubble while a young artist sketches it and scavengers pick through the debris. Robert painted the museum and its ruin as a pair. The curator of the present and the antiquarian of the future are the same man looking in two directions.

The architects of the period practised the same double vision. Sir John Soane wrote in 1812 a fantasy, "Crude Hints towards an History of my House", in which a future antiquary picks through the ruins of the house in Lincoln's Inn Fields that is now Soane's museum and tries to guess what kind of building it had been; in 1830 Soane's draughtsman Joseph Gandy painted his master's greatest work, the Bank of England, as a roofless ruin seen from above, its halls open to the weather. The counting-house of the empire was imagined in the condition of Palmyra. In the twentieth century the double vision was turned to propaganda. Albert Speer claimed, in memoirs that historians read with caution, to have persuaded Hitler that the buildings of the Reich should be designed according to a "theory of ruin value", so that after a thousand years they would leave magnificent remains. The prophecy of ruin can be humility or vanity, an admission of mortality or a bid for permanence through the beauty of one's debris.

Walter Benjamin found the double vision in the fabric of the nineteenth century itself. At the end of his exposé of 1935 for the *Arcades Project* he wrote that in the convulsions of the commodity economy we begin to recognize the monuments of the bourgeoisie as ruins even before they have crumbled. *The Time Machine* is the literary realization of that sentence. Wells looked at South Kensington, the most confident monument of Victorian science, and saw it as the

Palace of Green Porcelain, a ruin with fossils threaded on reeds and books turned to rags. The Crystal Palace, in which the precinct had its origin, received the same treatment from Fyodor Dostoevsky, who visited it in 1862 and described it in *Winter Notes on Summer Impressions* (1863) as a scene out of the Apocalypse, a temple of Baal before which one needed great spiritual resistance not to bow down. A year later his Underground Man declared that he feared the crystal palace precisely because it could never be destroyed and one could not put out one's tongue at it even on the sly (*Notes from Underground*, 1864). The Underground Man is a Morlock with a philosophy, and his crystal palace is the Palace of Green Porcelain before its fall. The Crystal Palace itself burned to the ground on the night of 30 November 1936, in a fire visible across London.

The prophecy has continued to fulfil itself. On 2 September 2018 a fire destroyed most of the collections of Brazil's National Museum in Rio de Janeiro, housed in the former imperial palace of São Cristóvão: some twenty million items, including natural-history specimens, archaeological collections and recordings of Indigenous languages, some no longer spoken. Among the losses first feared was Luzia, the skull of one of the oldest known inhabitants of the Americas, fragments of which were later recovered from the debris. A palace that had become a museum of nature and of human diversity burned in a single night, and the scavenging that followed was the work of scientists sifting the ashes for remainders. Robert's painting, Gandy's Bank and Wells's Palace were not fantasies of a remote future. They were descriptions, in advance, of a recurrent event in the life of collections.

6.1
Seventeen Papers upon Physical Optics: Science and the Decay of Its Records

The library is the cruellest room in the Palace. Its books have become "the brown and charred rags that hung from the sides of it … the decaying vestiges of books. They had long since dropped to pieces, and every semblance of print had left them." The Traveller's response is famous: "Had I been a literary man I might, perhaps, have moralized upon the futility of all ambition. But as it was, the thing that struck me with keenest force was the enormous waste of labour to which this sombre wilderness of rotting paper testified. At the time I will confess that I

thought chiefly of the *Philosophical Transactions* and my own seventeen papers upon physical optics." The passage is comic and exact. The scientist disclaims the literary moral and then enacts it; he sees the library as congealed labour wasted, in terms Marx would have recognized; and his grief is, at bottom, for his own contributions to the oldest scientific journal in the world, which the Royal Society had published since 1665. His field is physical optics, the science of light. The man who will spend his last nights in the future striking matches against creatures that cannot bear light is a specialist in light, and his papers on the subject have rotted in the dark.

The description is also accurate chemistry. "Brown and charred" is how paper made from wood pulp ages. From the middle of the nineteenth century the printing industry turned from rag paper to cheap paper made from ground wood and sized with alum and rosin, whose acidity destroys it from within: such paper yellows, browns and grows brittle until it crumbles at a touch. Twentieth-century librarians called the process the "slow fires" that were consuming the record of the industrial age, and a documentary of 1987 took the phrase for its title. Most books of Wells's decade were printed on paper of this kind. The Palace's library is not merely old; it is burning slowly, as the libraries of 1895 were already burning, and the Traveller's word "charred" records combustion without flame. The archive, made of the same cheap material as the commodity culture it recorded, eats itself.

The digital age has repeated the lesson at higher speed. In 1986 the BBC marked the nine-hundredth anniversary of the Domesday Book with a multimedia survey of Britain stored on laser discs readable only by a specialized microcomputer system; within fifteen years the equipment had become so rare that the survey was nearly unreadable and had to be rescued by emulation, while the Domesday Book of 1086, written on parchment, remained legible in the National Archives. NASA concluded in 2009 that the original tapes of the Apollo 11 television transmissions had probably been erased and reused in the following decades. A study published in 2014 found that about half of the web links cited in the opinions of the United States Supreme Court no longer led to the material cited. The Palace of Green Porcelain took eight hundred thousand years to lose its books. The present can lose its records in a decade.

Wells answered the Palace in his later career with a programme of preservation by multiplication. In the lectures and essays collected as *World Brain* (1938) he proposed a permanent world encyclopaedia, recorded on microfilm and

duplicated across the globe, so that no single fire or war could destroy it: the cure for the vulnerable deposit was the distributed copy. Digital librarians reached the same principle and gave it a slogan, "Lots of Copies Keep Stuff Safe", the name of a preservation system developed at Stanford University from 1999. The largest attempt to preserve the record of the digital present bears a name the Traveller would have understood. The Internet Archive, founded in 1996, calls its record of the changing web the Wayback Machine, after the time machine of the cartoon dog Mr Peabody, whose WABAC machine carried him and his boy into the past. The archive of the present has named itself a time machine.

The Palace's failure bears, finally, on the epistemology of the frame. The motto of the Royal Society, whose *Transactions* the Traveller mourns, is *Nullius in verba*: take nobody's word for it. Science replaces testimony with evidence that can be checked and reproduced. The Traveller returns with almost no evidence and a story that cannot be reproduced, and his guests, men of the professions, apply the Society's motto and take nobody's word, least of all his. The Palace explains why. Records do not survive the time the Traveller has crossed; only witnesses can carry the past across it. The novel places its reader in the position the Royal Society sought to abolish, that of deciding whether to believe a witness, and it gives the witness a single piece of physical evidence, the flowers, whose force depends on the testimony that accompanies them. A remainder without a witness is a dead plant. A witness without a remainder is a storyteller. The novel's ending requires both.

6.5
Camphor and Matches: The Archive as Arsenal

The Traveller walks through the Palace as a man looking for weapons, and the novel records the search with care. In the gallery of minerals "the sight of a block of sulphur set my mind running on gunpowder. But I could find no saltpetre; indeed, no nitrates of any kind. Doubtless they had deliquesced ages ago." In the gallery of machines, whose corroded engines interest him as puzzles, he hopes that by solving them he might come into possession of powers of use against the Morlocks, and he breaks a lever from one of them to serve as a mace. In a gallery of technical chemistry, "in one of the really air-tight cases", he finds a box of matches, and celebrates with the dance described in Chapter One. Last, he finds

a sealed jar whose contents he takes for paraffin wax until he smells camphor: "In the universal decay this volatile substance had chanced to survive, perhaps through many thousands of centuries." He remembers that it is "inflammable and burned with a good bright flame — was, in fact, an excellent candle" — and pockets it. "I found no explosives, however, nor any means of breaking down the bronze doors." The museum has been read, from the first gallery to the last, as an arsenal.

The camphor is the episode's most suggestive detail. Camphor was a preservative of collections, used for generations to protect specimens and textiles from moths and beetles; the substance that kept the museum's treasures from being eaten by insects becomes, in the Traveller's hands, a source of fire. The preservative of the archive is converted into its accelerant. The Traveller compares the surviving camphor to "a sepia painting I had once seen done from the ink of a fossil Belemnite that must have perished and become fossilized millions of years ago", an allusion to the fossil ink sacs of cephalopods found at Lyme Regis in the 1820s, which collectors there ground and used for drawing. A remainder of deep time becomes a medium of art; a remainder of the museum becomes a weapon. The two uses define the alternatives an archive offers its finder.

Camphor had a further career that Wells could not have foreseen but that completes the analysis. It was the plasticizer of celluloid, the first commercial plastic, developed in the United States around 1870 from nitrocellulose and camphor, and celluloid nitrate became, from 1889, the base of photographic and then cinematic film. The medium born, as Chapter Two argued, in the year of the novel was made of an unstable compound that could ignite spontaneously and continue to burn even under water. Its history is a history of archive fires. In 1937 a fire in the Fox Film Corporation's storage vaults at Little Ferry, New Jersey, destroyed most of the studio's silent films; in 1965 a fire in a vault at Metro-Goldwyn-Mayer destroyed, among much else, the last known print of Tod Browning's *London After Midnight* (1927). The archive of the moving image consumed itself, as the Traveller's camphor did, in a bright flame. Bill Morrison's *Decasia* (2002), composed entirely of decaying nitrate footage set to a score by Michael Gordon, made the corruption of the archive an art of its own, and his *Dawson City: Frozen Time* (2016) told the story of more than five hundred nitrate reels discarded in the Yukon in 1929, buried as fill beneath a hockey rink and recovered in 1978, preserved by the permafrost. Cold preserved what heat destroys: the archive survived where the world was coldest.

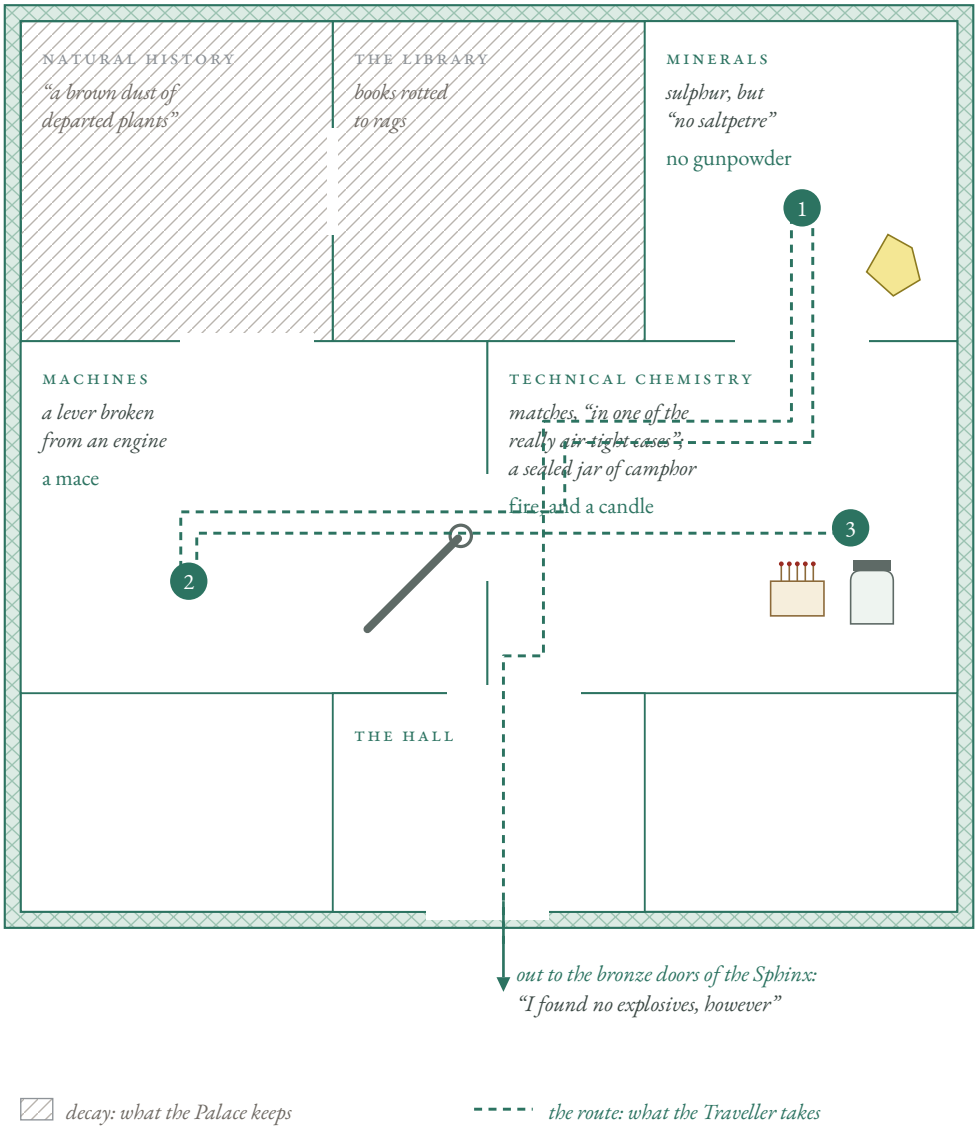


Figure 6.1 The museum read as an arsenal: the Traveller's search through the Palace of Green Porcelain, gallery by gallery, from the first to the last. A schematic route, not a plan.

Walter Benjamin gave the archive-as-arsenal its moral. In the seventh of his theses on history he wrote that there is no document of civilization that is not at the same time a document of barbarism, and that the cultural treasures carried in the triumphal procession of the victors owe their existence not only to the great minds who made them but to the anonymous toil of their contemporaries. The Traveller's "enormous waste of labour" is that anonymous toil seen from the far end of time. The museum of the nineteenth century was in some cases an arsenal in the literal sense. The collection of Augustus Pitt Rivers, given to Oxford in 1884, arranged weapons in evolutionary sequences from the club upward, so that the history of humanity appeared as the history of its instruments, and the Traveller, breaking a lever from a machine to use as a mace, returns the sequence to its beginning. The museums of empire were also stores of spoils. In 1897, two years after the novel, a British punitive expedition sacked and burned Benin City and carried off thousands of brass and ivory works, many of which entered the British Museum and other collections in Europe and North America. Since 2022 several institutions in Germany, Britain and the United States have begun to transfer ownership of Benin objects to Nigeria, while others continue to argue for retention; the debate over restitution is, among other things, a debate about whether a museum's holdings are a deposit held in trust or a document of conquest. The Palace of Green Porcelain shows the end point of the second view. An archive that began as spoils ends as an arsenal.

6.6
Heterotopias of Accumulated Time: Benjamin, Foucault, Derrida

Michel Foucault gave the museum its place in a typology of spaces. In the lecture "Des espaces autres", delivered to architects in 1967 and published in 1984, he defined heterotopias as real places that function as counter-sites, in which the other sites of a culture are represented, contested and inverted, and he distinguished among them heterotopias of time. Some, like the fairground and the holiday village, are bound to time in its most fleeting aspect, the festival. Others, the museum and the library, are heterotopias of indefinitely accumulating time: the modern project of enclosing in one place all times, all epochs, all forms and all tastes, of constituting a place of all times that is itself outside time and inac-

cessible to its ravages. The Palace of Green Porcelain is the heterotopia whose exemption from time has expired. And 802,701 contains both of Foucault's kinds side by side. The Eloi live in a perpetual festival, a heterotopia of transitory time spread over the surface of the earth, while the museum that was to accumulate all time stands abandoned among them. The festival has outlasted the archive.

Theodor W. Adorno, in "Valéry Proust Museum" (1953), began from a German connotation: the word *museal* has an unfriendly colour, describing objects to which the observer no longer has a living relationship and which are themselves in the process of dying. Museum and mausoleum, he wrote, are connected by more than phonetic association; museums are like family sepulchres of works of art. In the Palace the sepulchre has been opened. Its objects are no longer dying but dead, and the relationship Adorno found lacking between visitor and exhibit has collapsed completely: the Eloi play with the fossils, the Morlocks carry off the cases, and the Traveller looks for weapons. No one in 802,701 relates to the museum as a museum.

Jacques Derrida's *Mal d'archive* (1995) explains why the collapse is inherent in the institution. The archive, from the Greek *arkheion*, the house of the magistrates who held and interpreted official documents, combines two principles: commencement, the origin it preserves, and commandment, the authority that consigns and controls. It is haunted, Derrida argued, by the death drive Freud had described in *Beyond the Pleasure Principle*, a force he called archiviolithic, which destroys the archive even as it compels the archiving. Archive fever is the compulsion to return to the origin together with the knowledge that the origin is being lost; there is no archive without a place of consignment, and no place of consignment that time does not erode. The Traveller's visit to the Palace is archive fever experienced from outside the archive's time. He seeks the origin — the history of how humanity came to this — and finds rags, dust and fossils strung on reeds. The archiviolithic drive has completed its work.

Benjamin's ninth thesis supplies the Palace's allegory. Paul Klee's *Angelus Novus*, Benjamin wrote, is the angel of history: his face is turned towards the past, where we perceive a chain of events and he sees one single catastrophe piling wreckage at his feet, while a storm blowing from Paradise, which we call progress, drives him irresistibly into the future to which his back is turned. The Traveller is the angel inverted. He faces the future, rides into it by choice, and sees ahead of him the wreckage that progress has piled up; the storm is not

behind him but in front, and its debris is the Palace of Green Porcelain. Jorge Luis Borges imagined the opposite catastrophe in "La biblioteca de Babel" (1941). His infinite library, containing every possible book, is tended by librarians who are dying out, and its narrator suspects that the human species is nearing extinction while the library, solitary, infinite, motionless and incorruptible, will endure. In Borges the books outlast humanity; in Wells humanity, in degraded forms, outlasts the books. Both stories measure the incommensurability of an archive and its users.

Two theorists of memory complete the account. Pierre Nora argued, in the introduction to *Les Lieux de mémoire* (1984), that sites of memory exist because environments of memory no longer do: a society that still lives its past in custom and ritual needs no museums. The Eloi have neither: no milieu of memory and no site they can use. Jan Assmann, in *Das kulturelle Gedächtnis* (1992), distinguished communicative memory, carried by living people across three or four generations, from cultural memory, carried by texts, rites and institutions across millennia. The Palace is the ruin of cultural memory, and the Eloi, whose language lacks figure and whose curiosity lasts minutes, have lost communicative memory as well. What remains is the Traveller's memory, carried back across eight hundred thousand years to a dinner-table, and two flowers. The novel's ending is an act of communicative memory in Assmann's sense, the transmission of a past by a living witness to listeners, and its fragility is the fragility of every story told after dinner.

6.7
Messages to the Deep Future: Nuclear Semiotics, Voyager, the Long Now, *A Canticle for Leibowitz*, *The Disintegration Loops*

The twentieth century turned the problem of the Palace into a problem of policy. Nuclear waste remains dangerous for tens of thousands of years, and the governments that bury it must warn people who may share neither their languages nor their symbols nor their assumptions. In 1981 the United States Department of Energy convened a Human Interference Task Force to consider how repositories might be marked. The linguist Thomas Sebeok, in *Communication Measures to Bridge Ten Millennia* (1984), concluded that no inscription

could be trusted to remain intelligible and proposed instead an "atomic priesthood", a self-renewing institution that would transmit the warning by ritual and legend, as religions have transmitted their messages across millennia. The semioticians Françoise Bastide and Paolo Fabbri proposed breeding cats that would change colour in the presence of radiation, and embedding in folklore the lesson that one should flee when the cats change colour. A panel convened for the Waste Isolation Pilot Plant in New Mexico drafted, in 1993, a message meant to be rendered in many languages and in pictograms, which declares that the place is a message and continues: "This place is not a place of honor ... no highly esteemed deed is commemorated here ... nothing valued is here." The language of the warning is the exact negation of the museum's. The Palace of Green Porcelain said in every gallery that something valued was here, and the Traveller, finding it, looked for weapons. The designers of the nuclear marker feared precisely such a finder: a visitor from the future who would read a monument as a treasury, and dig.

Finland's repository at Onkalo, cut into the bedrock of Olkiluoto to hold spent fuel for a hundred thousand years, raised the opposite possibility. Michael Madsen's documentary *Into Eternity* (2010) recorded its planners debating whether the site should be marked at all, or deliberately forgotten, on the ground that any marker might attract curiosity. The ethics of the remainder must include this case. Not every remainder is a flower. Some are poison, and the responsibility they impose on the present is to ensure that they are not received as gifts.

The century also sent messages outward. The two Voyager spacecraft, launched in 1977, each carry a gold-plated copper record assembled by a committee chaired by Carl Sagan, with greetings in fifty-five languages, natural sounds, music from Bach to Chuck Berry and Blind Willie Johnson, images of human life, and an encoded map of the sun's position among the pulsars. A message from President Jimmy Carter accompanies it: "This is a present from a small, distant world, a token of our sounds, our science, our images, our music, our thoughts and our feelings. We are attempting to survive our time so we may live into yours." The record is the Palace of Green Porcelain made portable and launched beyond the reach of decay, and its covering message names the two terms of this thesis exactly: it is a *present*, a gift, sent by a civilization unsure that it will survive its own time. The Long Now Foundation, established in 1996 to foster responsibility on the scale of ten thousand years, has added to the genre a

nickel disk micro-etched with parallel texts in some fifteen hundred languages — the Rosetta Disk, a copy of which travelled aboard the European Space Agency's probe to a comet — and a "Manual for Civilization", a library of the books judged most necessary to rebuild one. George Pal's film of *The Time Machine* (1960) had already ended on the same question. Filby, examining the Traveller's shelves after his second departure, notices that three books are missing and wonders which three he took. The film converts Wells's flowers into books: the remainder becomes a reading list.

Fiction has tested the alternatives more severely than policy. Walter M. Miller Jr.'s *A Canticle for Leibowitz* (1959) imagines, after a nuclear war, a "Simplification" in which mobs burn books and murder the learned, and a monastic order that preserves the surviving fragments of science as relics through centuries in which it cannot understand them. Its most venerated relic, a scrap in the hand of the order's patron, is a shopping list: "Pound pastrami, can kraut, six bagels — bring home for Emma." The holiest object of the archive is an errand of food for a loved one, an act of care that outlives the civilization that produced it. The novel carries civilization through three ages, from darkness to renewed science to a second nuclear war, and ends with a shark descending into cold, deep water, very hungry that season. Russell Hoban's *Riddley Walker* (1980) is set some two thousand years after a nuclear catastrophe, in a Kent whose people speak a broken English in which the ruin of the archive is audible in every sentence; its hero pieces together a mythic history of the atom from a garbled legend and the remains of a medieval wall painting of St Eustace. Miller's monks preserve without understanding; Hoban's Riddley understands without records. Between them lies the problem of the Palace.

Music has made the decay of records audible. In 2001 the composer William Basinski, transferring to digital form tape loops he had made two decades earlier, found that the magnetic oxide was flaking from the tape as it passed the playback head, so that each repetition of the music was fainter and more fragmentary than the last; he recorded the process, and *The Disintegration Loops* (2002) is the sound of an archive dying in real time. Basinski completed the recordings in the days of the attacks of 11 September 2001 and used as cover images his own footage of the smoke over Lower Manhattan at dusk. Leyland Kirby, working as The Caretaker, released between 2016 and 2019 the six parts of *Everywhere at the End of Time*, in which recordings of 1930s ballroom music are progressively degraded to represent the stages of dementia, until the songs dissolve into noise. Both

works are the Palace of Green Porcelain as music: the record consumed by the medium that holds it.

The modern messages to the future fall into three kinds, and the Palace helps to rank them. There is the deposit: the museum, the library, the record, the disk, which entrusts the past to a material object and fails when the object decays or its code is lost. There is the relay: Sebeok's atomic priesthood, Miller's monks, the ten-thousand-year institution, which entrusts the past to a succession of living people and fails when the succession breaks or forgets why it exists. And there is a third kind, which the novel proposes and policy has scarcely considered: the gift accompanied by a witness. Weena's flowers are neither a deposit, for they cannot speak for themselves, nor a relay, for there is no institution behind them. They are a present, in Carter's word, carried back across time by one person who can say whom they came from and what they meant, and kept by a second, the narrator, who holds them and tells the story. Their survival depends on two acts of fidelity in sequence, and this thesis calls that sequence testimony. The Palace of Green Porcelain shows what becomes of records without it.





And when the sun was going down, a deep sleep fell upon Abram; and, lo, an horror of great darkness fell upon him.

— Genesis 15:12 (Authorized Version)

I had a dream, which was not all a dream. / The bright sun was extinguish'd, and the stars / Did wander darkling in the eternal space.

— Lord Byron, "Darkness" (1816)

All the sounds of man, the bleating of sheep, the cries of birds, the hum of insects, the stir that makes the background of our lives — all that was over.

— H. G. Wells, *The Time Machine* (1895)

CHAPTER SEVEN

7

The Terminal Beach

Entropy, Extinction and the Death of the Witness

THE NOVEL'S LAST VISION is the place where every register of chronophagy reaches its limit. The sun, swollen and red, hangs motionless above a beach on an earth that has stopped turning; crabs as large as the Richmond dinner-table stalk the shore, their mouths "all alive with appetite"; then even they are gone, and only a black, football-sized thing with trailing tentacles hops on a shoal while an eclipse extinguishes the light. This chapter reads the scene through the physics that produced it and the physics that dated it (7.1), the tidal theory that stopped the earth (7.2), the poetic tradition of the Last Man (7.3), the biblical covenants of posterity its language recalls (7.4), the nineteenth century's imagination of the end as cold (7.5), the contemporary philosophy of extinction (7.6), and the dying suns of later fiction, cinema, games and music (7.7). Its argument is that the terminal beach stages what this thesis calls the death of the witness: the limit at which there is no longer anyone to see, which fiction can approach only by sending a witness to the brink and bringing him back. The Traveller nearly faints on the beach — "a terrible dread of lying helpless in that remote and awful twilight sustained me" — and the novel's whole ethical economy depends on his return.

7.1
Clausius, Kelvin, Helmholtz and the Victorian Heat Death

The terminal beach is the second law of thermodynamics given a landscape. Rudolf Clausius had stated the law in 1850 and in 1865 gave its central quantity a name, explaining that he had chosen *entropy*, from the Greek *tropē*, transformation, so that the word would resemble *energy* as closely as possible; his two-line cosmology — the energy of the world is constant, the entropy of the world tends to a maximum — was quoted in Chapter One. William Thomson, later Lord Kelvin, had drawn the eschatological consequence before the term existed. His paper "On a Universal Tendency in Nature to the Dissipation of Mechanical Energy" (1852) concluded that within a finite period of time to come the earth must again be "unfit for the habitation of man as at present constituted", unless operations were performed that the known laws of nature made impossible. Hermann von Helmholtz, lecturing at Königsberg in 1854, extended the conclusion to the cosmos: the energy of the universe would at last be converted wholly into heat at a uniform temperature, after which every natural process would cease and the universe would rest for ever, a state the German tradition called the *Wärmetod*, the heat death.

The Victorian heat death was, in imagination, a cold death, and it came with a timetable. In "On the Age of the Sun's Heat" (*Macmillan's Magazine*, 1862) Kelvin argued that the sun's radiation was supplied by gravitational contraction, that the sun could therefore be only tens of millions of years old, and that the inhabitants of the earth could not continue to enjoy the light and heat essential to their life for many million years longer "unless sources now unknown to us are prepared in the great storehouse of creation". The phrase is worth pausing over. The cosmos is a larder, its stores are finite, and the inhabitants of the earth live on rations that are running out. Kelvin's timetable was the great scientific obstacle to Darwinism, which needed hundreds of millions of years for natural selection to work; Darwin confessed in 1869 that Thomson's views on the recent age of the world had been for some time one of his sorest troubles, and he withdrew from later editions of the *Origin* his estimate of three hundred million years for the denudation of the Weald. Wells's terminal beach obeys Kelvin's

schedule exactly. It lies "more than thirty million years hence", within the span that contraction allowed the sun, and its sun is red because it is cooling.

The physics that dated the novel arrived almost at once. Henri Becquerel discovered radioactivity in 1896, the year after its publication, and in 1904 Ernest Rutherford, lecturing at the Royal Institution with the aged Kelvin in the audience, observed that Kelvin had limited the age of the earth on the proviso that no new source of heat was discovered, and that radium was that source; Rutherford recalled that the old man beamed at him. In the 1920s Arthur Eddington proposed that the stars shine by converting hydrogen into helium, and in 1939 Hans Bethe worked out the nuclear reactions involved. The sun, it is now understood, is some four and a half billion years old and roughly halfway through its life on the main sequence. Its slowly increasing luminosity will make the earth uninhabitable in about a billion years, and in some five billion years it will swell into a red giant and engulf the inner planets. The swollen red sun of Wells's beach has thus outlived the physics that produced it. Its cause has changed from cooling to swelling, its date from thirty million years to billions, its temperature from cold to heat; but its image — a vast red disc filling a tenth of the sky above a dying world — has turned out to be close to the astronomical truth.

Eddington also gave the law its modern name and rank. In *The Nature of the Physical World* (1928) he called the direction of irreversible processes "time's arrow" and wrote that "the law that entropy always increases ... holds, I think, the supreme position among the laws of Nature." The Victorian imagination registered that supremacy before the physicists named it, and it registered it as mourning. Tennyson's Nature in *In Memoriam* (1850) cries from scarped cliff and quarried stone that a thousand types are gone and that "all shall go"; Swinburne's "The Garden of Proserpine" (1866) imagines a time when "star nor sun shall waken", leaving "Only the sleep eternal / In an eternal night". Ruskin, lecturing on "The Storm-Cloud of the Nineteenth Century" (1884), described a new darkness that had settled on the English sky since the 1870s, a plague-cloud and plague-wind that left a "Blanched Sun", "blighted grass" and "blinded man": an environmental dimming of the sun read as moral portent. Wells's achievement was to give these premonitions a place, a date and a witness. The heat death is no longer a proposition about the universe. It is a beach on which a man stands, breathing too fast, while the light goes out.

7.2
George Darwin's Tidal Drag and the Stopped Earth

The beach has a second clock besides the sun, and Wells names it. As the Traveller slows, the alternation of day and night stretches through centuries until the sun ceases to set and simply hangs, red and enormous, on the horizon. "I perceived by this slowing down of its rising and setting that the work of the tidal drag was done. The earth had come to rest with one face to the sun, even as in our own time the moon faces the earth." The theory is George Howard Darwin's. Charles Darwin's second son, an astronomer at Cambridge, showed in a series of papers from 1879 that the friction of the tides slowly brakes the earth's rotation and drives the moon outward, so that in the remote past the day was shorter and the moon nearer, and in the remote future the day will lengthen until earth and moon present the same faces to each other for ever. Robert Ball popularized the theory in *Time and Tide: A Romance of the Moon* (1889). Wells dramatizes it freely, locking the earth to the sun rather than to the moon, which has vanished from his sky, but the principle is Darwin's: the motions that make day and night are themselves consumed, by friction, in the long run of time. Charles Darwin supplied the novel's biological clock and his son its astronomical one, and the son's clock stops the father's.

The stopped earth produces the beach's peculiar geography. One hemisphere faces the sun in perpetual sunset, the other lies in perpetual night, and life clings to the band between them, where the Traveller finds "intensely green vegetation" on the rocks, "the same rich green that one sees on forest moss or on the lichen in caves: plants which like these grow in a perpetual twilight." Modern astrobiology has made this landscape a principal object of its search. The most common stars in the galaxy are red dwarfs, small, cool and long-lived; planets close enough to them to be warm are expected to be tidally locked, with one face in permanent day and the other in permanent night; and since the discovery of temperate planets around Proxima Centauri (2016) and TRAPPIST-1 (2017), researchers have modelled the twilight band between the hemispheres as a possible refuge for life. The Traveller's beach — a red sun fixed in the sky, an ocean under eternal sunset, life in the twilight zone — is the picture astronomers now draw of the most

numerous candidate habitable worlds in the galaxy. The end of the earth, in Wells, has become the image of life elsewhere.

The stopped earth also completes a metaphor. At 802,701, watching the sun go down over the ruined palaces, the Traveller had reflected that "the ruddy sunset set me thinking of the sunset of mankind." Thirty million years later the sunset is permanent, and the ocean lies "all bloody under the eternal sunset". The figure has been run through deep time until it became astronomy. The speciation of metaphor, which converted the social figures of Victorian discourse into biology in the world of the Eloi, here converts the oldest figure of decline, the setting sun, into a physical state of the planet. The sunset of mankind is no longer a way of speaking about humanity. It is the condition of a world from which humanity has gone.

The life that remains is a regression. The crabs that stalk the beach are arthropods, members of the phylum whose trilobites were among the first complex animals of the Cambrian seas. In Thomas Hardy's *A Pair of Blue Eyes* (1873), Henry Knight, hanging from a sea-cliff, finds himself face to face with a fossil trilobite embedded in the rock, and "Time closed up like a fan before him." Hardy's trilobite stands at the beginning of animal life, Wells's crabs at its end: the history of complex life is bracketed by arthropods. Biologists have a word for the tendency the beach dramatizes. In 1916 the zoologist Lancelot Borradaile described the evolution of a crab-like body from other crustacean forms as one of "the many attempts of Nature to evolve a crab", and called the process carcinization; it has occurred independently several times among the decapods. Wells anticipated the joke the twenty-first century would make of it, that given enough time everything becomes a crab. His crabs, however, are not comic. They are measured against the furniture of the frame — "Can you imagine a crab as large as yonder table", the Traveller asks his guests — and their mouths are "all alive with appetite". The last predator on earth is the size of the dinner-table, and it is hungry.

After the crabs, only the simplest life remains: "the green slime on the rocks alone testified that life was not extinct." The verb is *testified*. Life at the end of the world is reduced to evidence of itself, a smear of algae bearing witness to a history no one remains to read. The soundscape has gone the same way. "All the sounds of man, the bleating of sheep, the cries of birds, the hum of insects, the stir that makes the background of our lives — all that was over." The list begins, after man, with the sacrificial animal of Chapter Five, the sheep whose bleating

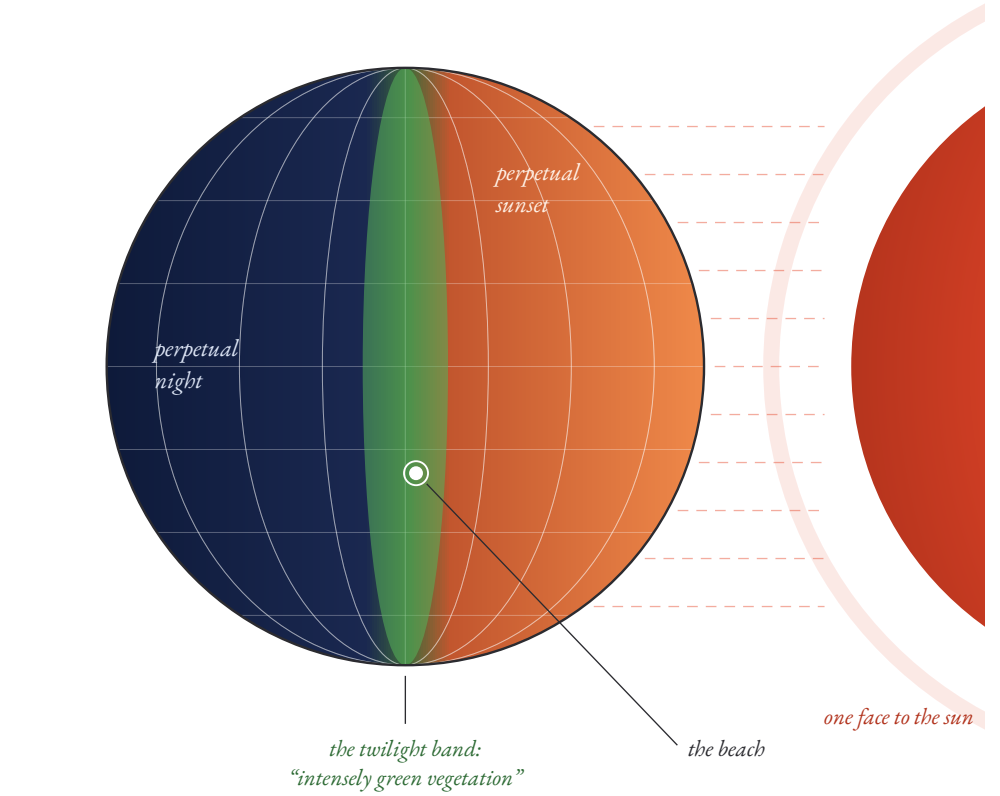


Figure 7.1 The stopped earth. When the work of the tidal drag is done, one hemisphere faces the sun in perpetual sunset and the other lies in perpetual night; life clings to the twilight band between them, "plants which like these grow in a perpetual twilight".

stood behind every Victorian pasture and every joint of mutton. Rachel Carson would open *Silent Spring* (1962) with the fable of a town where the birds had stopped singing; Wells's beach is that silence at planetary scale, a spring without voices that no summer will follow.

7.3 Byron's "Darkness" and the Tradition of the Last Man

The terminal beach has a literary ancestor more exact than any scientific source. In the summer of 1816, the year without a summer, when the eruption of Tambora had darkened the skies of Europe, Byron wrote "Darkness" at the Villa Diodati on Lake Geneva, in the same season that produced *Frankenstein* and Polidori's "The Vampyre". The poem begins: "I had a dream, which was not all a dream. / The bright sun was extinguish'd, and the stars / Did wander darkling in the eternal space, / Rayless, and pathless, and the icy earth / Swung blind and blackening in the moonless air". Wells's beach is moonless, icy and dark, and its final state is described in Byron's adjective: "All else was rayless obscurity." Byron's humanity, deprived of light, burns its palaces and cities for fire and then turns on itself — "The meagre by the meagre were devour'd" — and the poem ends with the last two men alive, enemies, who meet at the embers of an altar, kindle a flame, see each other's faces by its light, and die of what they see: "Even of their mutual hideousness they died, / Unknowing who he was upon whose brow / Famine had written Fiend." Darkness then becomes the universe.

The antithesis with Wells's ending is exact, and it defines the novel's ethics. Byron's last two human beings die "of their mutual hideousness". Wells's narrator closes on two flowers that witness that "gratitude and a mutual tenderness still lived on in the heart of man." The adjective is the same; the noun is reversed. Byron imagined the end of the world as the moment when the last human faces, seen by the last light, reveal only famine and enmity. Wells imagined it as the place from which a man returns carrying evidence that another face, met in the dark of 802,701, had looked at him with tenderness. *The Time Machine* answers "Darkness" point by point — the extinguished sun, the burning of the cities, the cannibal famine, the last pair — and replaces its final word.

The tradition of the Last Man supplies the rest of the scene's genealogy. Jean-Baptiste Cousin de Grainville's prose epic *Le Dernier Homme* (1805) named its

hero Omégare, the last letter of humanity. Thomas Campbell's "The Last Man" (1823) has its survivor address the dying sun — "All worldly shapes shall melt in gloom, / The Sun himself must die" — and claim an immortality the sun cannot share. Mary Shelley's *The Last Man* (1826) leaves Lionel Verney alone in a depopulated Rome after a plague, setting out to sail the world's coasts with a few books, Homer and Shakespeare chief among them. John Martin painted *The Last Man* (1849) as a solitary figure on a cliff above a corpse-strewn shore, facing a blood-red setting sun: Wells's beach before the crabs. Camille Flammarion's *La Fin du monde* (1893–94), published as the novel was being written, carried the earth through millions of years of slow refrigeration until the last couple, Omégar and Eva, die together in the ice. M.P. Shiel's *The Purple Cloud* (1901) gave the tradition its most disturbing form. Its last man, Adam Jeffson, alone after a poison cloud has killed humanity, travels the world burning its cities for the pleasure of watching them burn. The Last Man as arsonist is the archive-as-arsenal of Chapter Six raised to planetary scale: the survivor treats the whole remainder of civilization as fuel.

Wells's Traveller belongs to the tradition and departs from it in one decisive respect: he is not the last man. He is the last witness, a visitor from the past standing at the end of a history in which he took no part. The Last Man of Romantic poetry is a survivor, the remainder of his own species, who must decide how to live alone; the Traveller is a spectator who must decide whether to return. His choice is the novel's. He does not stay to die on the beach, as the Last Men of the tradition die; he climbs back into the saddle, sick and fainting, and returns to his dinner. The tradition of the Last Man ends in solitude. *The Time Machine* ends in company, at a table, telling the story.

7.4
"A Horror of This Great Darkness": Genesis 15 and the Annulled Covenant of Posterity

At the climax of the eclipse the Traveller uses a phrase that his Victorian readers would have recognized as scriptural. "The sky was absolutely black. A horror of this great darkness came on me." The source is the covenant between the pieces in Genesis 15, where God promises Abram a posterity as numerous as the stars — "Look now toward heaven, and tell the stars, if thou be able to number them ...

So shall thy seed be" (15:5) — and seals the promise in a ritual. Abram divides a heifer, a goat and a ram and lays the halves opposite one another; "when the fowls came down upon the carcases, Abram drove them away" (15:11); and "when the sun was going down, a deep sleep fell upon Abram; and, lo, an horror of great darkness fell upon him" (15:12). In the darkness he learns that his descendants will be strangers in a land not theirs and will serve and be afflicted four hundred years before their deliverance; and then "a smoking furnace, and a burning lamp" pass between the pieces (15:17).

The scene contains, in compressed form, almost every element of the novel: a promise of innumerable descendants; slaughtered animals divided in two; predators descending on the carcasses and driven off by a man; a horror of great darkness at sunset; a prophecy of the servitude of descendants in a foreign land — Egypt, whose emblem is the sphinx; and fire passing in the night. Wells's Traveller re-enacts it with every term inverted. He stands under the stars of the promise — "the pale stars alone were visible" — at the moment when the seed they numbered has vanished from the earth. The covenant of posterity is annulled under the very signs that sealed it.

The next sentences of the novel recall the other great covenant of Genesis. As the eclipse passes, "like a red-hot bow in the sky appeared the edge of the sun." After the Flood, God set his bow in the cloud "for a token of a covenant between me and the earth", promising that "neither shall all flesh be cut off any more by the waters of a flood" (9:11–13). The spiritual remembered the corollary — no more water, the fire next time — and James Baldwin took the line for a title in 1963. Wells's bow is red-hot, a bow of fire rather than of water, and it rises over a world in which all flesh has been cut off, not by flood but by cold. The letter of the covenant has been kept, for there has been no second flood. Its spirit has been annulled.

The novel's onomastics close the circle. The cry that gave the Eloi their name was uttered in an eclipse. "And when the sixth hour was come, there was darkness over the whole land until the ninth hour. And at the ninth hour Jesus cried with a loud voice, saying, Eloi, Eloi, lama sabachthani?" (Mark 15:33–34). The darkness at noon is the setting of the cry of dereliction, and Wells's terminal eclipse is that darkness at cosmic scale, with no one left to cry. The Eloi bore the cry in their name; on the beach there is not even a forsaken voice.

Christian eschatology had imagined the end of time as fulfilment. The angel of Revelation swears "that there should be time no longer" (10:6), and Olivier

Messiaen took the passage as the inspiration of his *Quatuor pour la fin du Temps*, composed in a German prisoner-of-war camp at Görlitz and first performed there, before an audience of prisoners, on 15 January 1941. Its first movement is called "Liturgie de cristal"; its last is a praise of immortality in which the violin climbs slowly towards silence. Pierre Teilhard de Chardin, in *Le Phénomène humain* (1955), projected evolution towards an Omega Point at which consciousness would converge upon the divine, and the physicist Frank Tipler later attempted a cosmological version of the idea (*The Physics of Immortality*, 1994). The Omega tradition, from Grainville's Omégare to Teilhard's Omega Point, reads the end as consummation. Wells reads it as exhaustion. His terminal beach is the end of time without the angel: time does not cease in eternity but runs down into equilibrium, and what crosses the threshold is not a soul but a witness, who returns.

7.5
Cocytus: Dante and the Cryo-Eschatology of the Nineteenth Century

Dante's hell does not end in fire. Its lowest circle is Cocytus, a lake of ice in which traitors are frozen, and at its centre stands Satan, frozen to the chest, beating six vast wings whose wind keeps the lake frozen; from a distance he looks to Dante like a windmill turning in the wind, "un molin che 'l vento gira" (*Inferno* XXXIV.6). In each of his three mouths he chews a traitor — Judas, Brutus and Cassius — for ever (XXXIV.55–67). Cold, wind, an apparatus of wings, an eternal mouth: the bottom of Dante's universe is a frozen machine of appetite. The terminal beach is its secular counterpart. There is a bitter cold and a moaning wind; the last predators are crabs whose mouths are "all alive with appetite"; and at the last a black thing hops on the shoal under a sky gone dark. The heat death of the physicists turns out, in its imaginative form, to be Cocytus: the universe ending, as Dante's hell ends, in ice.

The nineteenth century imagined the end as cold with a consistency that deserves a name, and this thesis calls it cryo-eschatology. *Frankenstein* (1818) ends in the Arctic, with Walton's ship locked in the ice and the creature borne away on an ice-raft, "lost in darkness and distance". Poe's polar fictions carried their narrators towards a white void at the end of the world. Kelvin's dissipation and

Flammarion's refrigerating earth gave the imagination its physics. And the age's most famous polar disaster gave it a scandal that bears directly on the alimentary code. Sir John Franklin's expedition in search of the Northwest Passage vanished in the Arctic after 1845; in 1854 the explorer John Rae reported, on the testimony of Inuit witnesses, that the last survivors had resorted to cannibalism. The Victorian public refused to believe it. Charles Dickens attacked the report in *Household Words*, insisting that Englishmen could not have eaten one another and that the Inuit witnesses could not be trusted, and he and Wilkie Collins went on to stage *The Frozen Deep* (1857), a melodrama of Arctic heroism and self-sacrifice. Modern forensic study of the expedition's bones has confirmed Rae's account, including evidence of the most desperate forms of consumption. The episode reads as a parable of the Richmond frame. A testimony of cannibalism in the cold arrives from a far place; the respectable refuse it because the witnesses are not of their class; and the truth is established only long after the witnesses are dead.

Edwin Landseer gave the Arctic its emblem of chronophagy in *Man Proposes, God Disposes* (1864), painted in the wake of the discoveries of the expedition's relics: two polar bears among the wreckage of a ship, one tearing at a tattered ensign, the other gnawing a human ribcage. The bears are the crabs of the terminal beach transposed to the ice of the present, the last predators feeding on the remains of an empire's ambition. The painting hangs at Royal Holloway, University of London, where students sitting examinations in its presence have long covered it with a flag. The gesture is an image of the Eloi's relation to the end: the end is in the room, and it is draped so that no one need see it.

Cryo-eschatology has a further implication, which the beach makes visible. Cold preserves. The Dawson City films of Chapter Six survived in permafrost; the bodies of three of Franklin's sailors, buried in the frozen ground of Beechey Island, were exhumed in the 1980s almost intact. The end of the world as cold is also, paradoxically, the condition of the longest preservation: a frozen universe keeps its remainders. The Traveller's beach is the last place where anything is kept — salt on the shore, slime on the rocks, a black thing hopping on a shoal — and its cold is the cold of a museum without visitors. The death of the witness and the preservation of the evidence are the same event.

7.6
After the Witness: Lyotard, Meillassoux, Brassier, Thacker, Colebrook

The philosophy of the late twentieth century returned to the terminal beach and made it a problem for thought. In "Can Thought Go On Without a Body?", collected in *L'Inhumain* (1988), Jean-François Lyotard argued that the sun will explode in four and a half billion years, that with it every human question and answer will be destroyed, and that this is the only serious question facing humanity: everything we do is done in the shadow of a death that will leave no one to remember that anything was done. Techno-science, on his reading, is already preparing thought to survive the loss of its terrestrial body, as software might survive the destruction of its hardware. Lyotard's solar catastrophe is the Traveller's beach stripped of its witness. There is no one on the shore when the sun goes out, and the question is whether thought can find a way to be there.

Quentin Meillassoux's *Après la finitude* (2006) gave the problem its epistemological form. Philosophy since Kant, he argued, has been correlationist: it holds that we have access only to the correlation of thinking and being, never to either term apart from the other. Science, however, makes "ancestral" statements about events that preceded all thought — the accretion of the earth, the origin of the universe — on the evidence of traces Meillassoux calls arche-fossils, which attest a reality that existed without any witness. Ray Brassier, in *Nihil Unbound: Enlightenment and Extinction* (2007), extended the argument forward. Extinction is the future counterpart of the arche-fossil, a reality that will exist after all witnesses have gone; drawing on Lyotard and on the cosmology of the far future, Brassier argued that the certainty of extinction is a kind of transcendental trauma that retroactively annuls the significance of everything thought has done, so that, in his phrase, everything is dead already.

The terminal beach is the place where fiction meets these arguments, and it meets them with a paradox. The scene presents a world after humanity, which no one can witness; yet it is presented through a witness. The Traveller is the impossible observer of a reality from which observers have gone, the narrative counterpart of Meillassoux's arche-fossil turned towards the future. The novel acknowledges the paradox physically. On the beach the Traveller can barely

breathe; the cold "smote to my marrow"; "a deadly nausea seized me"; he nearly faints and fears lying helpless in the twilight. The witness is dying of the world he witnesses. This is what this chapter calls the death of the witness: the limit at which presentation becomes impossible because there is no one left to present to. Fiction can approach the limit only by sending a witness to its edge and bringing him back, and the bringing back is not an evasion of the philosophical problem but the novel's answer to it. What survives the death of the world is testimony, carried across time by a body that was nearly consumed by what it saw.

Eugene Thacker's *In the Dust of This Planet* (2011) names the horror the scene produces. Thacker distinguishes the world-for-us, the world as human beings interpret it; the world-in-itself, the world as it is apart from us; and the world-without-us, the planet as it would be after us, which he makes the proper object of horror. The terminal beach is the world-without-us made briefly visible to someone still in it. Claire Colebrook, in *Death of the PostHuman: Essays on Extinction* (2014), proposed reading the present as it will be read after humanity, in the strata it leaves behind, and the geologist Jan Zalasiewicz had already imagined, in *The Earth After Us* (2008), a non-human geologist a hundred million years hence deciphering humanity's thin layer of concrete, plastic and altered chemistry. The debate over whether to formalize the Anthropocene as a geological epoch has made such retrospection institutional: in 2024 the stratigraphic subcommission charged with the question declined to ratify the proposal, while the term remains in wide use across the sciences and humanities. Wells's Traveller is the first geologist of the human future, reading the strata of 802,701 and then the bare rock of the beach.

Physics has also tried to save the witness. Freeman Dyson, in "Time without End: Physics and Biology in an Open Universe" (1979), argued that in a universe that expands for ever, intelligent life could survive indefinitely on a finite store of energy by thinking ever more slowly and hibernating for ever longer intervals, so that its subjective time would be infinite while its total consumption remained finite. Dyson's eternal intelligence is the Hesiodic maxim of Chapter One rendered as cosmology: a mind that survives by consuming, at every stage, less than it consumed before. The discovery in 1998 that the expansion of the universe is accelerating undermined his scenario, and cosmologists have since argued that eternal life of his kind is impossible. Fred Adams and Gregory Laughlin's *The Five Ages of the Universe* (1999) mapped the long future instead: an age of

stars ending after a hundred trillion years, ages of stellar remnants and of black holes, and a final age of dark, dilute radiation. The terminal beach was a beginning, not an end.

There is an older response to extinction, which the final chapter of this thesis will develop. Giacomo Leopardi's last great poem, "La ginestra, o il fiore del deserto" (1836), addresses the broom that grows on the slopes of Vesuvius, among the lava fields that buried Pompeii, "contenta dei deserti", content with deserts. Leopardi, the most uncompromising pessimist of his century, did not conclude from nature's indifference that human life is already dead. He concluded that human beings should recognize nature as their common adversary and bind themselves to one another in solidarity, as the broom, which the volcano will one day bury too, perfumes the desert while it can. The flower in the desert is Leopardi's answer to the philosophy of extinction, a century and a half before that philosophy was written; and it is, as Chapter Eight will argue, Wells's answer too.

7.7
Dying Suns: Hodgson, Stapledon, Lovecraft, Vance, Wolfe, Kubrick, *Dark Souls*, *Outer Wilds*, Ligeti, Haydn

The terminal beach founded a genre, and its descendants can be read as successive attempts to answer the death of the witness. William Hope Hodgson's *The House on the Borderland* (1908) repeated the Traveller's vision almost frame for frame: its recluse watches day and night accelerate into a flicker, the sun become a band of light and his own body decay to dust beside him, and then the sun dim, redden and die, while his disembodied consciousness continues to watch. Hodgson's *The Night Land* (1912) placed the last millions of humanity, long after the sun's death, inside the Last Redoubt, a metal pyramid miles high, powered by the Earth-Current and besieged by the Watchers and the Powers of the dark; its hero crosses the night to rescue the woman he has loved in another life. The Last Redoubt fuses the Palace of Green Porcelain with the Underworld, an archive of humanity sealed against the dark, and its plot makes love the only motive strong enough to leave it.

Olaf Stapledon's *Last and First Men* (1930) extended the Traveller's journey across two billion years and eighteen human species, ending on Neptune, where

the Last Men, facing the destruction of the solar system, scatter into the galaxy a drift of seeds bearing the possibility of a new humanity; its last pages praise man as "a fair spirit, whom a star conceived and a star kills". H.P. Lovecraft's "The Shadow Out of Time" (1936) imagined the Great Race of Yith, who escape extinction by projecting their minds across time into other bodies and who will flee their own doom into the hardy beetle species destined to succeed humanity; its narrator, searching the ruins of the Great Race's library in the Australian desert, finds a record written in his own hand during the years his mind was exchanged. The archive that contains the witness's own testimony is the most vertiginous form of the Palace of Green Porcelain, and Lovecraft's beetles, like Wells's crabs, award the earth to the arthropods.

The dying sun became a mode of fantasy. Jack Vance's *The Dying Earth* (1950) set its wizards and wanderers under a red sun that might go out at any moment, in a world grown decadent, melancholy and cruel. Gene Wolfe's *The Book of the New Sun* (1980–83) sent its torturer-hero Severian across Urth, a far-future earth under a failing sun, on a quest for the New Sun that will restore its light at the cost of a deluge. Wolfe made chronophagy the medium of memory. At the feast of the rebel Vodalus, Severian eats the flesh of the executed Thecla, mixed with an extract from the alzabo, a beast that carries the memories of those it devours, and thereafter carries her memories and personality within his own. In a world under a dying sun, the eating of the dead becomes the only archive that cannot rot.

Cinema gave the genre its most famous image of the end as a meal. In the final sequence of Stanley Kubrick's *2001: A Space Odyssey* (1968), after a passage through the Star Gate rendered in streaming light by Douglas Trumbull's slit-scan photography — the Traveller's flicker-fusion made visible — the astronaut David Bowman finds himself in a white neoclassical room, where he sits alone at a table eating a meal, knocks a wine glass to the floor, and sees himself grown old in the bed before he is reborn as the Star Child. The journey through time ends, as Wells's did, at a table, and Kubrick lets his traveller grow old there before he begins again. Lars von Trier's *Melancholia* (2011) staged the end of the world as the earth's collision with a rogue planet named for the humour of Saturn. Its prologue shows Bruegel's *Hunters in the Snow* burning, and its heroine, calmer than anyone as the end approaches, builds for her nephew a "magic cave" of sticks, a shelter that cannot shelter, in which they wait together. The last human act is a gift of care that can save no one.

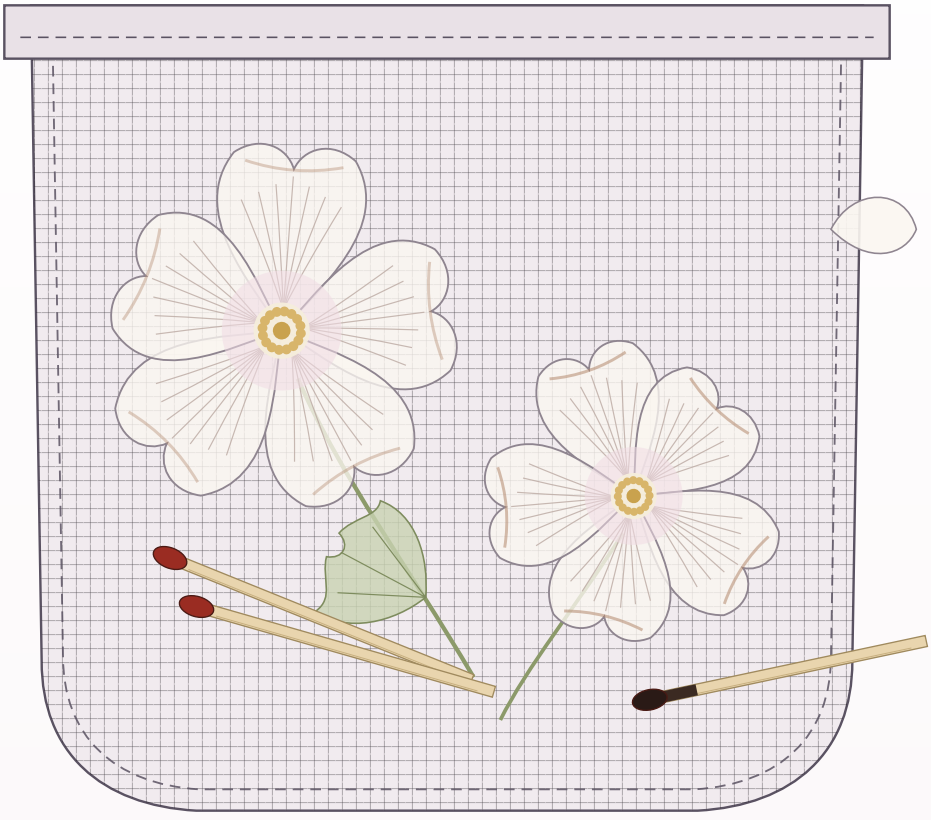
Games have made the dying sun a choice. *Dark Souls* (2011) builds its cosmology on the fading of a First Flame that inaugurated an Age of Fire. The gods prolonged that age by burning themselves as its fuel, and the player, reaching the kiln where the flame gutters in a landscape of ash, must decide whether to link the fire — to become fuel in turn and postpone the dark — or to let it fade and inaugurate an Age of Dark. The game presents the second law as a myth of sacrifice and offers no third option. *Outer Wilds* (2019) traps its player in a twenty-two-minute loop that ends each time with the sun going supernova, in a universe whose stars are dying of old age. Nothing the player carries survives the loop except knowledge; the only progress is what the player learns, from the ruined writings of an extinct people, about why the universe is ending and how its end may be witnessed. At the close the player enters the Eye of the Universe and gathers the travellers met along the way around a campfire, where each plays the instrument that has been their theme until the music joins; from that gathering, in the game's final image, a new universe begins. *Outer Wilds* is the most complete answer to the terminal beach that any medium has produced. Its witness does not return to the past; the witness's knowledge, shared in music around a fire, becomes the seed of what comes after.

Music has rendered the running-down itself. György Ligeti's *Poème symphonique* (1962) is written for one hundred mechanical metronomes, wound and set to different tempi and released together; as they run down one by one, the dense ticking thins until a single metronome beats alone and stops. It is the heat death as performance. Joseph Haydn wrote its humane prototype two centuries earlier. In the finale of his Symphony No. 45 (1772), composed when Prince Nikolaus Esterházy had kept his musicians at his summer palace far longer than they wished, the players stop one by one, snuff their candles and leave, until two muted violins finish alone. The prince understood, and the musicians went home. The "Farewell" Symphony is a terminal beach that functions as a message: the music dwindles to two players, and the dwindling is a request that is granted. John Cage's *ORGAN²/ASLSP* ("As Slow As Possible"), in a performance begun in 2001 in the church of St Burchardi at Halberstadt and scheduled to end in 2640, takes the course opposite to the time machine's: it decelerates music to the tempo of centuries, so that each change of chord is an event for which audiences travel. The time machine accelerates until the world becomes a smear of light; Cage's organ slows until a chord outlasts a human life. Both measure the gap

between the time of a life and the time of the world that Blumenberg made the theme of modernity.

The dying suns of the genre fall, finally, into two families. In one, the end is witnessed by a mind that has escaped the body — Hodgson's recluse, Lovecraft's Great Race, Lyotard's software — and the witness survives by ceasing to be human. In the other, the end is met by human beings who make, in the face of it, a gesture of care: Hodgson's lover crossing the night, Wolfe's Severian carrying Thecla, von Trier's magic cave, Haydn's last two violinists, the campfire of *Outer Wilds*. *The Time Machine* belongs to the second family, and it founded it. Its Traveller does not escape his body; he nearly dies in it. He returns with nothing but a story, and his story leads back to two flowers given by a creature who loved him in the dark. The terminal beach is the death of the witness. The remainder is what the witness brings back.





But this I say, brethren, the time is short: it remaineth, that both they that have wives be as though they had none; and they that weep, as though they wept not; and they that rejoice, as though they rejoiced not; and they that buy, as though they possessed not.

— 1 Corinthians 7:29–30 (Authorized Version)

... Zeus set it fast in the wide-pathed earth at goodly Pytho under the glens of Parnassus, to be a sign thenceforth and a marvel to mortal men.

— Hesiod, *Theogony* 498–500, trans. Hugh G. Evelyn-White (1914)

If that is so, it remains for us to live as though it were not so.

— H. G. Wells, *The Time Machine* (1895)

CHAPTER EIGHT

8

The Pocket and the Mallow

Toward an Ethics of the Remainder

THE THESIS RETURNS, AT its end, from the beach to the table. The novel's last two sentences contain, this chapter argues, a complete ethics, compressed into the grammar of a single clause and the matter of a single object: the narrator's resolve "to live as though it were not so", and the two flowers he keeps "to witness that even when mind and strength had gone, gratitude and a mutual tenderness still lived on in the heart of man." Chapter One named that ethics the ethics of the remainder and listed its components. This chapter develops them: its grammar, which descends from Paul (8.1); its measure, which descends from Hesiod (8.2); its economy, the gift (8.3); its physics, negentropy and care (8.4); and its form, the container rather than the weapon (8.5). It then tests the ethics against the remainders of the machine age (8.6) and against the most ambitious contemporary ethics of the future, longtermism (8.7), before drawing the argument of the thesis to a close (8.8).

8.1 "It Remains for Us": The Grammar of the Remainder

The epilogue is a dialogue between two attitudes to the future, and its narrator takes sides. He first imagines the Traveller's possible fates — fallen among the

savages of the Stone Age, wandering beside the saline lakes of the Triassic, or gone forward "into one of the nearer ages, in which men are still men, but with the riddles of our own time answered" — and declares, "for my own part", that he cannot think "these latter days of weak experiment, fragmentary theory, and mutual discord are indeed man's culminating time". Then he states the Traveller's view, that the growing pile of civilization is "only a foolish heaping that must inevitably fall back upon and destroy its makers in the end", and answers it: "If that is so, it remains for us to live as though it were not so." The sentence does not refute the Traveller. It concedes his premise in the conditional and refuses its consequence in the subjunctive.

The grammar of the reply has a source that its first readers, raised on the Authorized Version, could hear. In the seventh chapter of the First Epistle to the Corinthians, Paul writes: "But this I say, brethren, the time is short: it remaineth, that both they that have wives be as though they had none; and they that weep, as though they wept not; and they that rejoice, as though they rejoiced not; and they that buy, as though they possessed not; and they that use this world, as not abusing it: for the fashion of this world passeth away" (7:29–31). *It remaineth ... as though*: the construction of Wells's sentence is Paul's, and so is its situation. Both speak in a time shortened by the knowledge of an end; both conclude that the end does not abolish conduct but changes its mode; and both express the change by the formula *as though ... not*, Paul's *hōs mē*. Giorgio Agamben, in *Il tempo che resta* (2000), made the *hōs mē* the key to Pauline messianism: the messianic subject neither leaves the world nor denies its conditions but lives them in the mode of revocation, weeping as not weeping, possessing as not possessing, so that every worldly condition is held open to transformation. Wells's narrator lives the Traveller's prophecy in the same mode. He does not deny that the heap must fall; he lives as though it will not — as though its fall did not decide the value of what is done before it.

The same epistle supplies the sentence's object. "Charity never faileth: but whether there be prophecies, they shall fail; whether there be tongues, they shall cease; whether there be knowledge, it shall vanish away" (13:8). What remains when knowledge has vanished is named at the chapter's close: "And now abideth faith, hope, charity, these three; but the greatest of these is charity" (13:13). The verb the Authorized Version renders "abideth" is in Greek *menei* and in the Vulgate *manent*, the root of the English *remain*. Wells's last sentence is a secular translation of Paul's. When "mind and strength had gone" — when knowledge,

the faculty the Traveller had served, has vanished with the books of the Palace, and strength, the faculty the empire had served, has gone with the Eloi's muscles — what remains is "gratitude and a mutual tenderness", which is charity under another name. The remainder of the species is not a faculty but a relation.

The epilogue's adjectives trace the passage. The narrator's present is an age of "mutual discord"; Byron's end of the world, as Chapter Seven showed, is an end in "mutual hideousness"; Wells's remainder is "a mutual tenderness". *Mutual* names relation in each phrase, and the noun decides what kind of relation survives. The ethics of the remainder is the choice of the third noun under conditions that seem to dictate the second.

Three further features of the sentence deserve notice. Its subject is collective: "it remains *for us*", a first-person plural that includes the reader and turns the narrator's private comfort into a common task. Its verb is an infinitive of practice, "to live", not to believe or to hope. And its mood is the subjunctive: "as though it *were* not so." Samuel Delany, as Chapter One recalled, defined science fiction by its subjunctivity, the level of possibility at which its sentences are read. Wells's novel spent its length converting figures into facts — the speciation of metaphor — and it ends by converting a fact back into a figure: the fall of civilization, stated as a probability, is held in the subjunctive so that it may not govern action. The ethical act of the conclusion is the restoration of the subjunctive mood to a world the narrative had made literal.

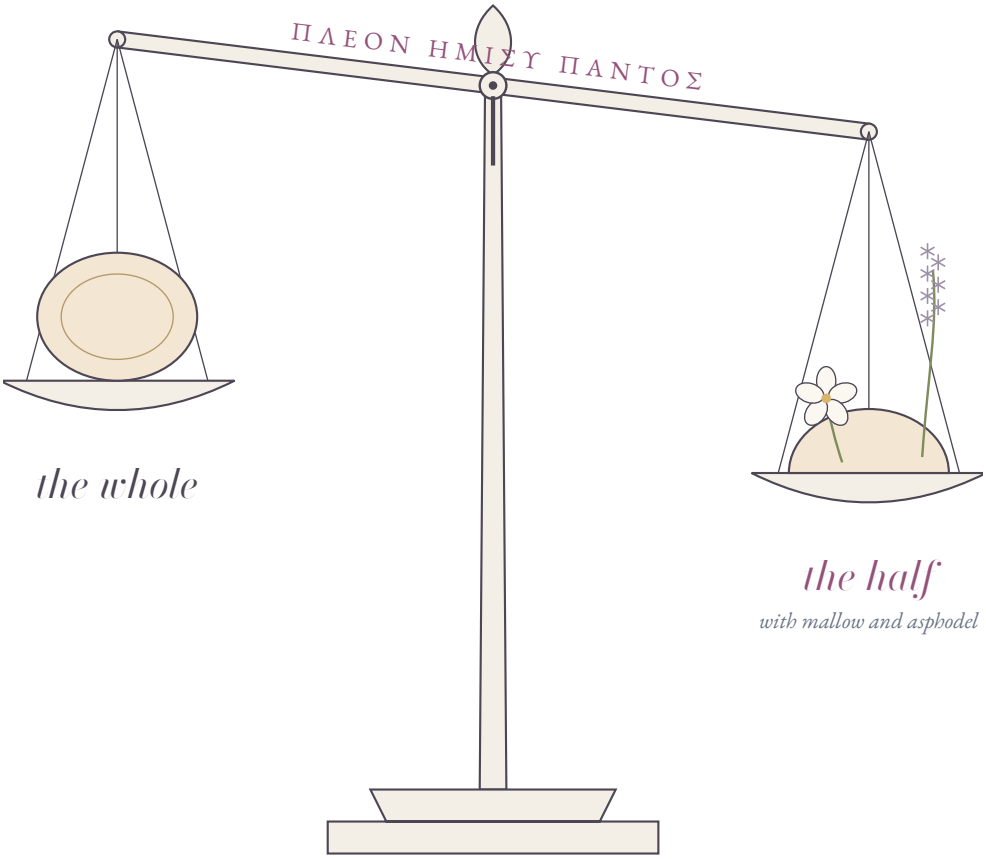
The epilogue, finally, sets the flowers against the evidence the Traveller failed to bring. On the morning after his tale he is about to leave again with a camera and a knapsack, promising to prove time travelling "up to the hilt, specimen and all". He never returns. The specimen, the photograph, the proof of the kind the Royal Society demands, are never delivered; the only evidence that remains is the gift. On the night of his return, doubting his own memory, the Traveller had answered the Medical Man's question about where he really got the flowers by saying that Weena had put them into his pocket when he travelled into Time, and had gone to touch his machine to be sure it existed. The Editor's verdict on the whole performance — "What a pity it is you're not a writer of stories!" — names the condition under which the remainder survives. It survives as a story told by a witness and kept with a relic by a listener who chooses to believe. The narrator keeps the flowers "by me, for my comfort". The remainder has passed from Weena to the Traveller to the narrator, and from him, by way of the book, to its readers: a relay of gift.

8.2
Hesiod's Half and the Gift-Devouring Kings

The measure of the ethics comes from the passage of *Works and Days* recovered in Chapter One. Hesiod rebukes his brother Perses, who has seized more than his share of their inheritance and courted the "bribe-swallowing" lords who judge the case, and turns the rebuke into a maxim: "Fools! They know not how much more the half is than the whole, nor what great advantage there is in mallow and asphodel" (40–41). The half is more than the whole because the whole, seized, destroys the relations that made it valuable: the brother, the justice of the city, the gods who watch it. Mallow and asphodel are more than the feast because they can be shared without loss. The poem goes on to found the human condition on work and justice against the appetite of the gift-devouring kings, and it states its governing rule in a line that joins measure to time: "Observe due measure: and proportion is best in all things" (694, trans. Evelyn-White), where the word rendered "proportion" is *kairos*, the fitting moment and the fitting measure. The ethic of the half is an ethic of *kairos*: of knowing when to stop.

Plato took up the maxim twice and turned it against rulers. In the fifth book of the *Republic* (466b–c), Socrates warns that a guardian who sought to seize everything in the city for himself would learn the wisdom of Hesiod's saying that the half is more than the whole; in the third book of the *Laws* (690e), the Athenian Stranger traces the ruin of the Dorian kingdoms of Argos and Messene to kings who, ignorant of the same saying, grasped at more than their measure. The vice Hesiod named and Plato pursued is *pleonexia*, the desire to have more, which Callicles in the *Gorgias* defends as nature's own justice — the stronger should have more than the weaker — and which Aristotle made the mark of injustice in the fifth book of the *Nicomachean Ethics*. Kronos is the god of *pleonexia*: he will not share his rule with his children, and so he swallows them whole. The Morlocks, who do not keep half of the Eloi but take the whole of each one they take, are Callicles' doctrine literalized as an appetite. The ethics of the remainder begins where *pleonexia* is refused.

Plutarch staged the question of the mallow at a dinner. In his *Banquet of the Seven Sages* the guests turn to Hesiod's mallow and asphodel and to the legendary food of the sage Epimenides, said to banish hunger, and debate



Fools! They know not how much more the half is than the whole

Figure 8.1 Hesiod's measure: "Fools! They know not how much more the half is than the whole, nor what great advantage there is in mallow and asphodel" (*Works and Days* 40–41, trans. Evelyn-White).

whether release from the need to eat would be a blessing. One of them objects that it would abolish the table itself, and with it the sacrifices to the gods, the hospitality of the hearth and the friendships that the shared meal sustains. The objection is the strongest classical statement of what chronophagy destroys and what the ethics of the remainder must preserve. Eating is not only consumption; it is the occasion of community. The Eloi have abolished the communal meal along with the household and the family, eating fruit in great halls without ceremony, and the Morlocks have reduced it to nutrition. The Richmond dinner, whatever its complacencies, preserves the table as the place where a story can be told and a gift set down. The ethics of the remainder does not abolish the table. It changes what is placed on it.

Montaigne's Tupinambá, discussed in Chapter Five, gave the half its social meaning. They called other men their "halves", and they were astonished that in France the needy halves endured their exclusion by the gorged. The anthropology implicit in their word is the foundation of the ethic. If each person is the half of another, then to seize the whole is to destroy oneself, and to keep the half for the other is not charity but a self-preservation that includes the other in the self. Hesiod's maxim, Plato's warnings to kings, Plutarch's table and the Tupinambá's halves converge on a single proposition: the relation is more than the whole, because the whole, consumed, leaves no one to share it with. The world of 802,701 is a world without halves. The Morlocks eat the whole of each Eloi; the Eloi share nothing; the Palace is plundered entire. Only Weena gives part of what she has — a garland, a pocketful of flowers — to someone else. That part is the remainder.

8.3 The Gift Gives Time: Mauss, Derrida and the Anachronic Gift

The economy of the remainder is the gift, and its classical analysis is Marcel Mauss's *Essai sur le don* (1925). Mauss showed that in the societies he studied, from Polynesia to the American Northwest, exchange took the form of obligatory gifts — the obligations to give, to receive and to reciprocate — and that the gift carried something of the giver with it: the Maori called it the *hau* of the thing, the spirit that wishes to return to its source. Bronisław Malinowski's *Argo-*

nauts of the Western Pacific (1922), on which Mauss drew, described the kula ring of the Trobriand Islands, in which necklaces travel in one direction and armshells in the other across hundreds of miles of sea and years of time, never kept for long and valued for their histories of giving. Malinowski's title borrowed the Greek voyage that Wells had borrowed for "The Chronic Argonauts": the voyagers of the gift and the voyagers of time sail under one name. Mauss also observed, in a short note of 1924 titled "Gift, gift", that in the Germanic languages one word meant both gift and poison. Chapter Six found the poisoned remainder in the buried waste of the nuclear age; the ethics of the remainder must distinguish the gift from the *Gift*.

Georges Bataille radicalized Mauss into a cosmology. In *La Part maudite* (1949) he grounded what he called general economy in the sun, which "gives without ever receiving": the energy that reaches the earth is always in excess of what life can use for growth, and every society must spend the surplus, gloriously in luxury, art, eroticism and festival, or catastrophically in war. The potlatch, in which chiefs of the Northwest Coast destroyed their wealth in rivalry, was his example of expenditure as sovereignty. Bataille's economy reads the world of 802,701 with startling precision. The Traveller's sunset theory — that energy in security "takes to art and to eroticism, and then come languor and decay" — is a theory of Bataillean expenditure; the Eloi are the last glorious spending of a surplus, and the Morlocks, who spend it as appetite, its catastrophic form. The terminal beach is the end of the general economy itself, a sun with almost nothing left to give. The flowers belong to the economy of the sun. They are the organs by which a plant receives the sun's unreturnable gift, and Weena's giving of them repeats, in a human relation, the gift on which all life depends.

Gratitude is the second term of the gift, and the ancient world gave it an emblem. Seneca's *De Beneficiis*, the longest classical treatise on giving and receiving, reports Chrysippus's allegory of the three Graces: sisters who dance hand in hand in a ring because a benefit passes from hand to hand and returns to the giver; young, because the memory of benefits should not grow old; maidens, because benefits should be pure; dressed in loose, transparent robes, because benefits should be seen (I.3). Seneca's Graces are the Eloi with a single difference. They are young, girlish, lightly clad and dancing, as the Eloi are; but their dance is the circulation of benefits, and the Eloi's is not. No Eloi saves another from the river. Weena alone completes the figure: she receives a benefit, her rescue, and returns it as a garland made "for me and me alone". The novel's final sentence

gives her return its classical name, gratitude. Shakespeare's Time was "a great-sized monster of ingratitudes", devouring good deeds "as fast as they are made". Weena's gratitude is the deed that Time did not devour.

Jacques Derrida's *Donner le temps* (1991) poses the problem that Weena's flowers solve. A gift, Derrida argued, is the impossible: as soon as giver or receiver recognizes it as a gift, it enters the circle of exchange, obliges a return and ceases to be a gift; the pure gift would have to be given without return, without recognition, almost without memory. Yet the gift is bound to time in a second sense, for it gives time: it demands the interval between gift and return that Pierre Bourdieu identified as the device by which gift economies disguise exchange (*Esquisse d'une théorie de la pratique*, 1972). What the gift gives is the delay. The flowers satisfy Derrida's conditions in the only way they could be satisfied, which is by time travel. They are a gift that cannot be returned, because the giver lived eight hundred thousand years after the receiver and is lost; a gift recognized, but too late for any exchange; a gift that gives time in the most literal sense, carrying a moment of 802,701 into an evening of 1895. Such a gift may be called anachronic. Its giver cannot be repaid, and so the obligation it creates cannot be discharged in kind. It can be discharged only as a way of life.

That is the meaning of the narrator's resolve. He holds a gift from a future that cannot be repaid and a prophecy from the same future that cannot be refuted, and the only counter-gift available to him is to live in a certain manner — "as though it were not so" — in the time that remains. The ethics of the remainder is, in this sense, the ethics of an unrepayable debt: the obligation to givers who are not present, past or future, discharged not by return but by conduct. Every inheritance is of this kind. The dead who gave us a language, a city and a body of knowledge cannot be repaid; the unborn to whom we will give a world cannot repay us. Between them the living hold remainders, and how to hold them is the question of the novel's last page.

8.1
Negentropy and the Flower: Schrödinger and the Thermodynamics of Care

The physics of the remainder was stated half a century after the novel. In *What Is Life?* (1944) Erwin Schrödinger asked how living things maintain their order in

a universe whose entropy tends to a maximum, and answered that an organism keeps itself at a high level of orderliness by "continually sucking orderliness from its environment": "What an organism feeds upon is negative entropy." The metaphor is alimentary, even vampiric, and it confirms the premise of this thesis: life is a metabolism that maintains one order by consuming another. Schrödinger added that the ultimate source of the order on which life feeds is sunlight, captured by plants. The flower is the organ of a being that feeds on light. Ilya Prigogine, whose work on systems far from equilibrium won the Nobel Prize in chemistry in 1977, showed that such systems can generate and sustain structure precisely by dissipating energy, so that life does not defy the second law but rides it, creating local order while hastening the general increase of entropy (*La Nouvelle Alliance*, with Isabelle Stengers, 1979). Life is not exempt from chronophagy. It is the local and temporary form of order that chronophagy makes possible.

Care is the name of the human labour that sustains such local order. Berenice Fisher and Joan Tronto defined it, in a formulation that has become standard, as a species activity that includes everything we do to maintain, continue and repair our world so that we can live in it as well as possible. Maintenance, continuance and repair are negentropic work: they hold back, for a time, the decay to which every body, house, city and relationship is subject. The artist Mierle Laderman Ukeles drew the consequence in her *Maintenance Art Manifesto* of 1969, a proposal for an exhibition to be called "CARE", which asked the question every revolutionary utopia avoids: after the revolution, who is going to pick up the garbage on Monday morning? A decade later, as artist in residence with the New York City Department of Sanitation, Ukeles performed *Touch Sanitation* (1979–80), shaking the hand of each of some eight and a half thousand sanitation workers and saying to each: "Thank you for keeping New York City alive." The gesture is the exact reversal of the Eloi's relation to the Morlocks. The surface seeks out those who maintain it, looks them in the face and thanks them. Gratitude, the word on which the novel ends, is extended to the people of the abyss.

Hannah Arendt's *The Human Condition* (1958) supplies the categories in which to place the flowers. Arendt distinguished labour, the cyclical activity by which the body sustains its metabolism with nature and whose products are consumed almost as soon as they are made; work, the fabrication of the durable things that compose a human world; and action, the deeds and words by which human beings appear to one another, whose only lasting products are the stories

told about them. The Morlocks labour; the Palace of Green Porcelain is work, failing; the Traveller's journey is action, and survives as a story. The flowers belong to none of the categories and to all of them. They are the product of Weena's labour of gathering, durable as a work only by accident, and significant as the trace of an action, a gift that discloses who she was. Arendt also identified the two faculties by which action remedies its own predicaments. Against the irreversibility of what has been done there is forgiveness; against the unpredictability of what is to come there is the faculty of making and keeping promises. The novel confronts both predicaments at their limit — an irreversible history and a future "black and blank" — and its last object is a promise kept across time: the gratitude Weena returned, which the narrator keeps.

The thermodynamics of care can now be stated. Chronophagy is consumption with irreversible loss; care is the work that slows the loss locally and for a time; the remainder is what care leaves behind when the work stops, the trace that persists after the maintainer has gone. A pressed flower is its purest instance: the residue of a living order, held out of the cycle of consumption by an act of keeping, testifying to the care that made and gave it. It does not reverse entropy. It is brown, flat and brittle, and it will crumble in time like the herbarium of the Palace. But it persists long enough to be handed on, and handing on is all that any remainder can do.

8.5
The Carrier Bag and the Pressed Flower: Le Guin, *Flatland* and the Herbarium

The form of the remainder is the container. Ursula K. Le Guin's essay "The Carrier Bag Theory of Fiction" (1986), drawing on Elizabeth Fisher's *Women's Creation* (1975), proposed that the first cultural device was probably not a weapon but a receptacle — a bag, a sling, a net, a gourd — in which gathered food could be carried home and shared. The story of the hunter with his spear, killing the mammoth, is the story our culture has told about itself; the story of the gatherer with her bag is the one it has neglected. Le Guin asked for fiction of the second kind, a novel that would be not an arrow aimed at a climax but a container, a medicine bundle, holding things in a particular relation to one another and to us.

The Time Machine contains both stories, and it lets the first consume itself. The hunter's story is the Traveller's: the iron lever broken from a machine and used as a mace, the matches struck to blind the Morlocks, the camphor lit, the forest fire that kills indiscriminately, the killing of "one's own descendants". Every instrument of that story is used up. The matches are all burned, the camphor is gone, and the fire he set to protect himself and Weena coincides with her loss. The gatherer's story is Weena's: the garland made for one person, the flowers placed in a pocket she has taken for a vase. Its instrument is the pocket itself, converted by her from a receptacle of tools into "an eccentric kind of vase for floral decoration". The pocket is the carrier bag, and its contents are the only things that survive the journey. The novel tells a hero's story and keeps a gatherer's remainder.

The remainder is pressed. To preserve a flower one must take something from it: its water, its volume, its colour, the third dimension in which it lived. The flowers on the Richmond table are "shrivelled now, and brown and flat and brittle". A novel that began by adding a dimension to the world, in the lecture on the fourth dimension, ends with an object that has lost one. Every remainder is a reduction of this kind. The photograph flattens the scene, the book flattens the life, the story told in an evening flattens eight days and eight hundred thousand years; preservation is always paid for in loss. The museum of 802,701 tried to preserve its specimens whole, in jars of spirit, and they have become "a brown dust". The humbler mode of keeping, the pressed flower in a pocket, has outlasted the grand one. The ethics of the remainder accepts that what is kept will be less than what was lived, and keeps it anyway.

The Victorian herbarium was the domestic form of this practice. Emily Dickinson assembled one as a girl in Amherst, some four hundred specimens pressed and labelled on sixty-six pages, now preserved at Harvard, and one of her poems states the economy of the remainder with perfect exactness. "Essential Oils — are wrung — / The Attar from the Rose / Be not expressed by Suns — alone — / It is the gift of Screws —"; and while "The General Rose — decay —", the essence kept "in Lady's Drawer" will "Make Summer — When the Lady lie / In Ceaseless Rosemary —". The essence is a remainder pressed from the flower by force, and it outlasts both the rose and its keeper. Victorian dictionaries of the language of flowers commonly assigned to the mallow the sense of mildness and to the marsh mallow that of beneficence. Whether or not Wells consulted them, his choice of flower is apt: the remainder of the human species is

a mild flower whose name, in the floral code of his century, meant a benefit given.

Flatland returns at the end in a new role. Abbott's Square, visited by a Sphere, could not at first imagine the third dimension, and Wells's Traveller lectured his guests on the fourth. The narrator of the epilogue goes the other way. He does not ask for a further dimension, a vision of the whole of time; he accepts a flat thing in place of the solid, a sign in place of a presence, and finds it enough "for my comfort". The ethics of the remainder is, in this sense, an ethics of Flatland: the acceptance of a reduced dimension in which to live well, after the full dimension of time has been seen and found unlivable.

8.6
Remainders in the Machine Age: *WALL-E, Castle in the Sky, The Platform, The Road, Nausicaä, NieR: Automata*

The fictions of the machine age have rediscovered the remainder with a consistency that suggests a structural necessity. Andrew Stanton's *WALL-E* (Pixar, 2008) is the most complete. Its earth has been abandoned under mountains of refuse, and its last working robot, a trash compactor, spends its days stacking cubes of garbage into towers and collecting treasures from the waste. In its most quietly eloquent gag the robot opens a ring box, throws away the diamond ring and keeps the box: it prefers the container to the jewel, the carrier bag to the treasure. It finds a seedling and plants it in an old boot. Humanity meanwhile lives aboard a starship, grown soft and passive in hover-chairs, fed through straws and governed by an autopilot that suppresses every sign that the earth could live again. The seedling in the boot is that sign, and its arrival brings the passengers to their feet. The closing credits show the history of art beginning again, from cave painting onward, as the humans return to farm. *WALL-E* is *The Time Machine* redeemed: a Morlock that tends rather than eats, Eloi who wake, and a remainder, carried in a container, that turns the story back towards life.

Hayao Miyazaki's *Castle in the Sky* (1986) imagined the same reconciliation in the ruins of a floating city. Laputa, its builders long gone, is tended by a robot that was once a soldier and has become a gardener: it waters the gardens, shelters the nests of birds on its shoulders and lays a flower on a memorial. The arsenal of

Chapter Six has become a garden, and the weapon a carer. The film's villain seeks the city's power as a weapon; its heroine refuses him, recalling that human beings must have roots in the soil; and when the destructive core of Laputa falls away, the great tree at its heart rises into the sky, carrying the garden with it. The remainder of the city is not its power but its tree.

Galder Gaztelu-Urrutia's *The Platform* (2019), discussed in Chapter Four as the purest diagram of the vertical city, supplies the remainder's semiotics. Its hero guards one untouched dessert as a message to the administration above, on the reasoning that only food left uneaten can carry meaning against a system of consumption; then, finding a child at the bottom of the pit, he sends the child up instead. The remainder that was a thing becomes a person, the uneaten dish a living being. Cormac McCarthy's *The Road* (2006) makes the same substitution in a world reduced to ash and cold. A father and son cross a burned continent where the last human beings hunt and farm one another — in one house they find captives kept in a cellar to be eaten piece by piece — and the father tells the boy that they are "carrying the fire". The phrase reverses Wells's method. In *The Time Machine* fire was literal, the Traveller's matches; in *The Road* it has become a figure for the goodness that will not eat others. The speciation of metaphor runs backwards, and the figure, restored, becomes an ethic. The father dies; the boy, the remainder, is taken in by a family; and the novel closes on a remembered image of trout in mountain streams, older than humanity and patterned like maps.

Miyazaki's manga *Nausicaä of the Valley of the Wind* (1982–94) carries the reasoning to its hardest conclusion. Its heroine discovers that the toxic forest overrunning her world is in fact purifying it; that the people of her time were engineered by a vanished civilization to survive in the poison and would die in the clean world being prepared; and that in a sealed crypt the old civilization has kept its arts, its knowledge and the means to replace living humanity with a purified one when the world is ready. Offered, in a hidden garden, a peaceful refuge among the preserved music and books of the old world, she refuses it. She destroys the custodian of the crypt and its planned future, declaring that life is a light that flickers in the darkness. The decision is morally fraught, and Miyazaki presents it as such. Its logic is nonetheless the logic of this thesis: Nausicaä chooses the living, impure, endangered present over a perfected future that would consume it, and she sacrifices an archive that served that future to the life it would have replaced.

NieR: Automata (2017) gives the remainder its purest form as a gift. Its androids fight machine lifeforms on behalf of a humanity that, they learn, has long been extinct, its last trace a server of genetic data on the moon; its machines, meanwhile, imitate the human culture they never knew, founding villages, religions and families in its ruins. At the close of its final ending, after a battle against the scrolling names of the game's makers, fought with the help of messages from players the player will never meet, the game offers a last choice: to delete the player's entire saved game, every hour of progress, in order to help a stranger elsewhere through the same ordeal. The gift is anonymous, unrepayable and costly; it is made from the giver's own archive; and it can be neither acknowledged nor returned. It meets Derrida's conditions for the impossible gift, and it meets them as an action rather than a contemplation. Players who accept hand on their remainder by destroying it.

The six works share a structure that *The Time Machine* founded. Each sets a remainder — a seedling, a tree, a child, a boy, a refused future, an erased archive given away — against a system of consumption, and each makes the remainder the hinge on which the story turns from ending to continuance. None of them restores the world. Each of them hands something on.

8.7
Against Optimization: Jonas, Parfit and the Longtermist Future

The ethics of the remainder must answer the most ambitious contemporary ethics of the future, and it must answer fairly. Its precursor is Hans Jonas's *Das Prinzip Verantwortung* (1979), which argued that technological civilization had extended the reach of human action beyond the horizon of traditional ethics and required a new imperative: act so that the effects of your action are compatible with the permanence of genuine human life on earth. Jonas proposed a heuristics of fear, the deliberate imagination of the worst outcomes in order to discover what must be preserved, and he found the archetype of responsibility in the newborn child, whose mere breathing addresses to the world around it an unconditional demand for care. *The Time Machine* is a heuristics of fear avant la lettre, and Weena — fragile, childlike, drawn from the river — is its archetype of the claim that vulnerability makes on the strong.

Derek Parfit gave the ethics of the future its analytic rigour. In the closing pages of *Reasons and Persons* (1984) he compared three outcomes: peace; a nuclear war that kills ninety-nine per cent of the world's population; and a war that kills everyone. Most people, he observed, would judge the difference between the first and second outcomes to be greater than the difference between the second and third; he argued that the reverse is true, because the third destroys not only the living but the whole of the possible future. From that argument descends the movement now called longtermism. Nick Bostrom's "Astronomical Waste" (2003) argued that every delay in the expansion of civilization beyond the earth forgoes an immense number of potential lives; Toby Ord's *The Precipice* (2020) put the risk of existential catastrophe in the present century at about one in six; William MacAskill's *What We Owe the Future* (2022) argued that positively shaping the long-term future is among the key moral priorities of our time. The case is serious. Future people will matter as much as present people; there may be vastly more of them; and what the present does may decide whether they exist at all. Frank Ramsey had argued as early as 1928 that discounting the welfare of later generations is ethically indefensible and arises from a weakness of the imagination.

The objections are serious too. Critics have argued that expected-value reasoning over astronomical numbers can license almost any sacrifice in the present for a sufficiently large speculative benefit — a version of the problem Bostrom himself, in another context, called Pascal's mugging; that our ignorance of the far future makes such calculations unreliable; that an ethics organized around the potential lives of the future can neglect the injustices suffered by actual people now; and that Parfit's own non-identity problem shows how hard it is to speak of benefiting people whose very existence depends on what we do. Keynes's quip that in the long run we are all dead is not an argument, but it names an intuition that ethics cannot simply dismiss. Edmund Burke's description of society as a partnership "between those who are living, those who are dead, and those who are to be born" (1790), and the principle, widely attributed to Haudenosaunee tradition, of weighing decisions by their effect on the seventh generation, offer older models in which the future is honoured without being made a calculation.

The Time Machine has a position in this debate, and it is not the one usually attributed to it. The novel is often read as a founding text of existential-risk thinking: it imagines the extinction of humanity and asks its reader to fear it. But its ethics is not optimization. The narrator does not propose to maximize the

expected value of the future; he does not know the future at all, for it is "black and blank". He proposes to live, now, in a certain manner, and he holds as his evidence not a forecast but a gift. Chapter Two noted that Christopher Nolan's *Tenet* imagines a posterity that would devour its ancestors to secure its own survival. The danger of an optimizing ethics of the future is that it reverses the direction of chronophagy without abolishing it. Where the present has consumed the future's resources, the future, in the name of its astronomical numbers, may consume the present, treating living people as fuel for the lives to come. The ethics of the remainder refuses both directions of consumption. It does not eat the future, which is the vice of the present's extraction of resources and its warming of the planet; it does not let the future eat the present, which is the vice of optimization. It leaves something uneaten in each direction — a world fit for those who come after, and lives that are not mere means to them — and it holds the two remainders together in the subjunctive, as though the end were not decided. The ethic is bidirectional, as the Traveller's machine was.

The difference can be put in Jonas's terms. The longtermist asks how much future value present action can produce. The ethics of the remainder asks what present action must not consume — of the earth, of other people, of the gifts of the past — if anything worth receiving is to be handed on. The first is a calculus of maximization; the second is a discipline of measure, Hesiod's half carried into an age of planetary consequence. Both take the future seriously. Only the second takes seriously that the future will be received, as Weena's flowers were, by people who will judge it not by its quantity but by what it shows of the hearts of those who kept it.

8.8
Kronos Unseated: Conclusion

This thesis has argued that *The Time Machine* constructs time itself as an alimentary process — chronophagy — and that the construction organizes the novel at every level. It identified four registers of time's appetite, the oikonomic, the zoological, the thermodynamic and the mediatic, and showed that the time machine renders them convertible; it argued that Wells's characteristic procedure is the speciation of metaphor, by which the social figures of Victorian discourse are run through deep time until they become flesh, and proposed that temporal

literalization is the constitutive operation of speculative fiction (Chapter One). It read the machine as both a geometry that spatializes time and a perceptual apparatus born with the cinema (Chapter Two); the Eloi as the entropic pastoral of a Golden Age that resolves Hesiod's two Kronoi into a single regime (Chapter Three); the Morlocks as the speciation of the Victorian residuum and the double the Traveller refuses (Chapter Four); the whole as an experiment in the dissolution of the alimentary code of the human after the extinction of the sacrificial substitute (Chapter Five); the Palace of Green Porcelain as the archive become arsenal, against which the novel sets the gift carried by a witness (Chapter Six); and the terminal beach as the death of the witness, from which the Traveller returns (Chapter Seven). This final chapter has developed from the novel's last two sentences an ethics of the remainder: Pauline in its grammar, Hesiodic in its measure, Maussian in its economy, negentropic in its physics, Le Guinian in its form, and bidirectional in its regard for the future.

Hesiod's *Theogony* supplies the image with which the argument closes. When Rhea bore Zeus, she gave Kronos in his place a stone wrapped in swaddling-clothes, and the devourer swallowed it. In time, overcome by the craft of Gaia and the strength of his son, Kronos brought up his offspring, and the first thing he brought up was the stone he had swallowed last. Zeus set it fast at Pytho under the glens of Parnassus, "to be a sign thenceforth and a marvel to mortal men" (497–500). Kronos was unseated by a substitution: a thing given in the place of a child, which he could swallow but not digest, and which, cast up, became a monument. Centuries later Pausanias saw at Delphi a small stone, anointed every day with oil, that was said to be the one Kronos had swallowed. It is the first remainder of European myth.

Hannah Arendt named the principle the stone protects. Against the predicaments of action she set natality, the fact that human beings are born and that with each birth something new can begin: "Men, though they must die, are not born in order to die but in order to begin." Kronos devours his children to prevent beginnings; his appetite is the refusal of natality. Every remainder this thesis has followed is, in the end, a protection of beginning: Weena drawn from the river, the seed planted in a ruined dome, the child sent up the shaft, the boy carrying the fire, the seedling in a boot, the drift of seeds from Neptune, the new universe kindled around a campfire. The ethics of the remainder unseats Kronos not by defeating time, which cannot be defeated, but by keeping open, at the table where time is eaten, the possibility that something will begin.

The novel's table bears the whole argument. On the night of his return the Traveller measures the last predator on earth against the furniture of the room — "a crab as large as yonder table" — and, his story told, he places on "the little table" two withered flowers. The same table carries the measure of appetite and the gift. Kronos sat at that table in the form of the mutton, the substitute that keeps the human from the Morlock's fare, and in the form of the story, which the guests consumed with their wine and doubted. He is unseated there in the form of the flowers, which no one eats, which pass from a creature of the year 802,701 to a man of 1895 and from him to his friend, and which survive, brown and flat and brittle, to witness what remains when mind and strength have gone. Written some thirteen decades after the novel — at a time when the machines that Wilde hoped would free humanity are being built, when the planet's heat is rising and its species are being consumed, and when the long future has again become a subject of calculation — this thesis ends where Wells's narrator ends: with an object on a table, and a resolve. If the heap must fall, it remains for us to live as though it were not so — to eat well, to keep the half, to thank the hands that maintain the world, to carry what is given, and to set down, at the table of time, something that time does not eat.



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The bibliography lists the works cited in the thesis, divided by kind. Within each division entries are alphabetical by author or, for works without a single author, by title; Section I, the works of Wells, is chronological, and Section X lists games by title. Classical, biblical and medieval texts are cited by their standard divisions, and the translations quoted are identified in the Note on Texts and Citation. Square brackets give dates of composition or first publication where these differ from the edition listed. Films, television, games and performances are listed with the year of first release or performance.

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Keywords

H. G. Wells; The Time Machine*; scientific romance; speculative fiction; chronophagy; speciation of metaphor; social second law; entropic pastoral; ethics of the remainder; deep time; time travel; fourth dimension; Charles Howard Hinton; spacetime; Henri Bergson; thermodynamics; entropy; heat death; Rudolf Clausius; William Thomson (Lord Kelvin); degeneration; E. Ray Lankester; T. H. Huxley;* Evolution and Ethics*; Darwinism; class; two nations; submerged tenth; Eloi; Morlocks; Kronos; Saturn; Golden Age; Hesiod;* Works and Days*; Plato's* Statesman*; Ovid;* tempus edax rerum*; pastoral; Arcadia; utopia; dystopia; William Morris; Oscar Wilde; Max Weber; Nietzsche's last man; cannibalism; anthropophagy; sacrifice; alimentary code; Moloch; vampire capital; biopolitics; anthropological machine; archive; ruin; museum; Walter Benjamin; heterotopia; extinction; Anthropocene; Dipesh Chakrabarty; speculative realism; longtermism; gift; Marcel Mauss; Jacques Derrida; negentropy; carrier bag theory; Ursula K. Le Guin; cognitive estrangement; Darko Suvin; subjunctivity; Samuel R. Delany; reception; transmedial comparatism; cinema; flicker fusion; Robert W. Paul; video games; manga; fin de siècle; dying earth; Last Man;* Metropolis*;* Snowpiercer*;* The Platform.

"It remains for us to live as though it were not so."